

Airy amplitudes for relaxed trees, compacted trees and finite-language automata

Positive amplitudes and expansions to every fixed order,
binary and every fixed arity (OEIS A082161 and relatives)

A merged research report built from six AI-assisted manuscripts
(batch 77, manuscripts 62, 15, 18, 66, 29 and 48)

Manuscripts dated 2 October 2026; merged 2 October 2026

Abstract

Relaxed trees, compacted trees and minimal deterministic automata accepting finite languages form one family of triangular recurrences indexed by the arity, or alphabet size, k . Their counts grow like $(n!)^{k-1}$ times an exponential, times $\exp(cn^{1/3})$ with c a multiple of the largest Airy zero $z = -2.338107410459767\dots$, times a power of n . For $k = 2$ the three counts are OEIS A082161, A254789 and A331120; for $k = 3$ the relaxed counts are A082162, for $k = 4$ A102102. The order of growth was known (Elvey Price, Fang and Wallner; Ghosh Dastidar and Wallner); the existence of a limiting amplitude was open. This report prints six manuscripts as four Parts. Part I (relaxed binary trees) and Part II (compacted binary trees) prove convergence to a positive amplitude and a Poincaré expansion to every fixed order in $n^{-1/3}$, with three explicit logarithmic corrections. Part I uses an exact Jacobi symmetrization; Part II treats the compacted recurrence as a signed delay whose scalar drift cancels, and shows that the $n^{-1/3}$ term cancels in the compacted-to-relaxed ratio. Part III proves the same for every fixed $k \geq 3$ and all three models, with common first two corrections, the first model-dependent defects, eventual decrease of the normalized ratios and a cancellation identity. Its ternary section gives the $k = 3$ base proof, the third coefficient and a second route to the automaton amplitude; its comparison section gives the earliest, Θ -level form of the positive completed-run comparison, with an explicit constant uniform in k and n . Part IV proves the binary automaton theorem, with power $n^{7/8}$, by the method of Part II. Every Part inverts its expansion with Lambert W and separates smooth inverses from integer thresholds. The amplitudes are proved positive but are not evaluated; no transseries, convergent series, uniformity in k or certified decimal is claimed. The report is AI-assisted and unrefereed, and none of it is formalized.

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Part 0. Guide: the family, its normalizations and this merge

This guide was written when the six manuscripts were merged. Everything after it, from Part I on, is the manuscripts' text, except for the passages marked [write, 2 October 2026] and the changes listed in “Conventions of the merge” below.

What the report proves

- **Part I** (source 62, relaxed binary trees, A082161). There is $\gamma > 0$ with $a_n = \gamma n! 4^n e^{3zn^{1/3}} n(1 + \sum_{k \leq M} c_k n^{-k/3} + O_M(n^{-(M+1)/3}))$ for every M , $c_k \in \mathbb{Q}[z]$; logarithmic corrections $\frac{53z^2}{90}, \frac{44z}{27}, \frac{141}{140} - \frac{1304z^3}{42525}$ (equations (F1)–(F3) in Section 1). Also the rational coefficient field, and Lambert- W inverses with discrete threshold brackets.
- **Part II** (source 15; compacted binary trees, A254789). By Theorem 16.1, $c_n = \gamma_c n! 4^n e^{3zn^{1/3}} n^{3/4}(1 + \sum c_k n^{-k/3} + O_M)$ with $\gamma_c > 0$ and logarithmic corrections $\frac{53z^2}{90}, \frac{271z}{216}, \frac{393}{1120} - \frac{1304z^3}{42525}$; exact positive four-state, three-state and renewal models and their symmetry obstructions; the ratio corollary (27.2), in which the $n^{-1/3}$ term of $\log(c_n/r_n)$ cancels; inverses. Appendix A is its self-contained proof of the shared Jacobi estimates.
- **Part III** (sources 18, 66 and 29, every fixed arity $k \geq 3$). Theorem 30.1: positive amplitudes A_k^R, A_k^C, A_k^B and expansions to every fixed order for relaxed, compacted and automaton counts, with the common corrections (30.7)–(30.8). Theorem 30.2: the first model-dependent defects $(q/k)^k/(k-2) \cdot n^{-(k-2)}$ and half of it, and eventual decrease of both normalized ratios. The cancellation identity (35.4) and an inverse. The ternary section (source 66): the $k = 3$ base proof, Theorems 38.1 and 38.2 with the explicit c_3 and scalar coefficients $\mathfrak{s}_3, \dots, \mathfrak{s}_6$, the explicit ternary weight with $1 \leq \omega \leq 3/2$, and the bridge-tail route to the automaton amplitude. The comparison section (source 29): Theorem 47.1, the comparison $P_{k,n} \leq B_n/(2^{n-1}R_n) \leq 1$ with $P_{k,n} = \prod_{j=2}^n (1 - 1/(2j^{k-1}))$ and $P_k \geq (2\sqrt{2}/\pi) \sin(\pi/\sqrt{2})$; a uniform finite-level approximation; a Θ -only inverse enclosure; and the boundary obstruction that the weights of the ternary section and of Lemma 31.1 resolve.
- **Part IV** (source 48, binary finite-language automata, A331120). Theorem 53.1: $b_n = \gamma n! 8^n e^{3zn^{1/3}} n^{7/8}(1 + \sum c_k n^{-k/3} + O_M)$ with $\gamma > 0$ and logarithmic corrections $\frac{53z^2}{90}, \frac{623z}{432}, \frac{3497}{4480} - \frac{1304z^3}{42525}$; a nonnegative renewal representation; a selected smooth inverse.

One family, indexed by the arity

In the coordinates of Part III (Section 31), fix $k \geq 2$, put $q = k - 1$, and for $m \geq 1$, $x \geq qm$ let

$$\begin{aligned} r(x, m) &= r(x, m-1) + (m+1)r(x-1, m), \\ c(x, m) &= c(x, m-1) + (m+1)c(x-1, m) - (m-1)c(x-k, m-1), \\ b(x, m) &= 2b(x, m-1) + (m+1)b(x-1, m) - mb(x-k, m-1), \end{aligned}$$

with base rows $r(x, 0) = c(x, 0) = b(x, 0) = 1$ for $x \geq 0$, the automaton auxiliary value $b(-1, 0) = 1$, and all other missing values zero. The relaxed, compacted and automaton counts are $R_n = r(qn, n)$,

$C_n = c(qn, n)$ and $B_n = b(qn, n)$, where n counts internal nodes of the trees and transient states of the automata (each automaton has one further, rejecting, sink). At $k = 2$ the three recurrences are, letter for letter, Part I's $r_{n,m}$, Part II's (16.1) and Part IV's (53.1). The initial terms below were recomputed from these recurrences when the report was written.

k	model	OEIS	$n = 0, 1, 2, \dots$	Part
2	relaxed	A082161	1, 1, 3, 16, 127, 1363, 18628, 311250, ...	I
2	compacted	A254789	1, 1, 3, 15, 111, 1119, 14487, 230943, ...	II
2	automata	A331120	1, 1, 6, 60, 900, 18480, 487560, 15824880, ...	IV
3	relaxed	A082162 ^a	1, 1, 7, 139, 5711, 408354, 45605881, ...	III
3	compacted	—	1, 1, 7, 133, 5299, 371329, 40898599, ...	III
3	automata	— ^b	1, 1, 14, 532, 42644, 6011320, 1330452032, ...	III
4	relaxed	A102102	1, 1, 15, 1000, 189035, 79278446, ...	III
4	automata	— ^b	1, 1, 30, 3900, 1460700, 1220162880, ...	III

^aThe entry starts at $n = 1$; the sources adjoin $R_0 = 1$. The array A128249 lists the relaxed counts of all arities (Section 30).

^bSource 29's bounded search verified no OEIS identifier for automata over $k \geq 3$ letters; A082162 counts relaxed ternary trees, not ternary automata.

Where the method changes. After normalization, the compacted and automaton recurrences are the relaxed one plus a delayed term (Part III, equation (35.1)) whose relative size per step is $O_k(i^{-(k-1)})$ (Section 35.2); its cumulative effect on the logarithm is of order $n^{-(k-2)}$ (Theorem 30.2). For $k \geq 3$ this is summable, the completed-run products P_C , P_B of (35.2) are positive, and the three models share the scale $\mathcal{G}_{k,n} = (n!)^q (k^k/q^q)^n e^{\gamma n^{1/3}} n^{(2k-1)/3}$ (automata with an extra 2^n). For $k = 2$ the delay is of order N^{-1} and not summable: it changes the power of n instead of adding a correction. Parts II and IV cancel the resulting scalar drift, $1/(4N)$ and $1/(8N)$, against the factors $N^{-1/4}$ and $N^{-1/8}$, which gives $n^{3/4}$ and $n^{7/8}$ in place of the relaxed n . Source 29 notes the same threshold from the other side: at $k = 2$ its lower product tends to zero like $n^{-1/2}$. Source 48's remark on the exclusion family $(\alpha m + \beta)$, with heuristic power $1 - \alpha/4$, is a heuristic it states without proof.

The binary and fixed-arity coefficients agree where they should. With $k = 2$, $q = 1$, Part III's general formulas (30.7)–(30.8) give $c_{2,1} = 53z^2/90$ and $c_{2,2} = z(2809z^3 + 26400)/16200$, exactly Part I's c_1, c_2 in (F3), and with $k = 3$ they give the c_1, c_2 of Theorem 38.2. Both reductions were checked symbolically (difference zero). Source 18's proof note `18-arity-all-orders-proof.md` observes the $k = 2$ reduction of $c_{k,1}$ itself. Part III claims nothing for $k = 2$, and must not be read as doing so: at $k = 2$ the compacted and automaton second coefficients differ from the relaxed one ($\frac{271z}{216}$ and $\frac{623z}{432}$ against $\frac{44z}{27}$ in the logarithm; the differences $-\frac{3}{8}z$ and $-\frac{3}{16}z$ are in ratio two, as Theorem 30.2 has it for $k \geq 3$, while the n^{-1} differences $-\frac{21}{32}$ and $-\frac{29}{128}$ are not).

Normalization dictionary

Notion	Parts I, II, IV (sources 62, 15, 48) and source 29	Part III (sources 18, 66)
Largest Airy zero	$z = -2.338107410459767\dots$	a (also a_1 ; a_m the m th zero)
Scaled zero	$a := 2^{-1/3}z$ (not the zero itself; source 29 has none)	$\lambda := a/B$, $B = (2/q)^{1/3}$; at $k = 2$, λ is Parts I, II, IV's a
Airy profile	$F(x) = \text{Ai}(z + 2^{1/3}x)$, $F'' = 2(x + a)F$	$f(x) = \text{Ai}(a + Bx)$, $f'' = (2/q)(x + \lambda)f$; source 66: $B = 1$, $f(x) = \text{Ai}(a + x)$
Time, height	$N = n + m$, $j = n - m$; endpoint $N = 2n$, $j = 0$	$i = x + m$, $j = x - qm$; endpoint $i = kn$, $j = 0$
Normalized array	$d_{N,j} = r/((N + j)/2)!$ (I); $e_{N,j}$ (II, IV)	$d_i(j) = q^{2x}r(x, m)/x!$; at $q = 1$ this is Part I's $d_{N,j}$
Stretched exponent	$3zn^{1/3}$	$\gamma n^{1/3}$, $\gamma = 3(kq/2)^{1/3}a$; source 29: β_k
Power of n	n (I), $n^{3/4}$ (II), $n^{7/8}$ (IV)	n^η , $\eta = (2k - 1)/3$ (source 29: α_k)
Amplitude	γ (I, IV), γ_c and γ_r (II)	A_k^R , A_k^C , A_k^B (18); C_R , C_B (66)
Index n	internal nodes (I, II); transient states (IV, 29)	internal nodes (R_n , C_n); transient states (B_n)

Source 29 also counts by total states $N = n + 1$ (its Remark on total states), which shifts the power of n to $(2 - k)/3$. Source 48's array $c_{n,m} = b_{n,m}/2^m$ is a normalized automaton array, not Part II's compacted count.

One polynomial–Airy identity. Every Part solves its formal profiles with the same identity. For $L = \partial_x^2 - cx - \ell$ and polynomials P, Q ,

$$L(PF + QF') = (P'' + 2(cx + \ell)Q' + cQ)F + (2P' + Q'')F',$$

which reduces $L(PF + QF') = RF + SF'$ to a triangular polynomial equation for Q . Parts I, II and IV use $c = 2$, $\ell = 2a$ (equation (F28) and (22.5)); source 66 uses $c = 1$, $\ell = a$ ((43.3)); source 18's (34.2) is the case $c = 2/q$, $\ell = 2\lambda/q$ after division by $q/2$. The batch-77 report on tree-child networks, [SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/a213863-tree-child-networks](#), proves the same identity with general c, ℓ for d -combining tree-child networks and uses the same zero-limit projection bootstrap; the two reports neither cite nor depend on each other.

Notation: one letter, several meanings

No symbol of any source was renamed: each Part, and each source section of Part III, keeps its source's letters. The sources use one letter for several notions and one notion under several letters. The table fixes the meaning in each Part and prints the tempting false reading beside the true one.

Symbol	Meaning and where	Do not read it as
z, a	See the dictionary above. In Part I, a_n is the counting sequence and a_1^{Ai} the Airy zero.	Parts I, II, IV's a is <i>not</i> the Airy zero; Part III's a is.

Symbol	Meaning and where	Do not read it as
γ	Parts I and IV: the positive amplitude. Part II: γ_c , with γ_r the relaxed amplitude of Part I. Part III: $\gamma = 3(kq/2)^{1/3}a < 0$, the coefficient of $n^{1/3}$ (source 66: $\gamma = 3 \cdot 3^{1/3}a$). Part I's (F23): γ_E is Euler's constant.	Part III's γ is not an amplitude.
k, q	Part III: the arity $k \geq 3$ and $q = k - 1$. Parts I, II, IV: k is an order index ($c_k n^{-k/3}$) or a run length ($w_m(k)$, $W_m(k)$); q is $p \log x + C$ in the inverse sections, q_0 a limiting projection, q an exponent in the bootstraps. Source 66: p_j, q_j are row probabilities.	In Parts I, II, IV, k is never an arity.
$c, \mathbf{c}$	Part II: $c_{n,m}, c_n$ compacted counts. Part IV: $c_{n,m} = b_{n,m}/2^m$. Part III: $c(x, m)$, C_n compacted k -ary counts, $c_i(t)$ column sums, $c_{k,r}^X$ coefficients. Relative coefficients: c_k (I), $\mathbf{c}_k$ (II, IV), c_r (66).	Part IV's $c_{n,m}$ is not a compacted count.
b, B	$b_{N,j}$ Jacobi edge weights and $b_N = \langle g_N, B_N g_{N-3} \rangle$ (II, IV), B_N the normalized delay (II, IV). Part IV: $b_{n,m}$, b_n automaton counts. Part III: $b(x, m)$, B_n automaton counts, $B = (2/q)^{1/3}$. Source 66: b_i a scalar projection, B a polynomial in $Af + Bf'$ and the weighted adjoint $B = \omega^{-1}T^*\omega$. Parts I, II: $B = 53z^2/90$ in the inverse sections.	Part IV uses b three ways, as its source did.
λ	Parts I, II, IV: λ_N , the top eigenvalue of S_N (≈ 2). Part III: $\lambda = a/B$. Source 29: $\lambda_k = C_k^{1/q}/e$.	Not the same quantity in any two of these.
R, P	Part III: R_n relaxed counts; Parts II, IV: $R_N = P_N N^{-1/4}$, resp. $P_N N^{-1/8}$; Part I: $R(a, \varepsilon)$, R_m . P_N spectral product (I, II, IV); P_C, P_B, P_i (18); P_3, P_i (66); $P_{k,n}, P_k$ (29); $p = 3/2$ (I) and $p = 5/4$ (II) in the inverses, p also the residual order.	
C, A	C : matching constant A_∞/q_0 (I; C_* in II, IV), $\log(\gamma\sqrt{2\pi})$ in Part I's inverse, compacted counts C_n (III), amplitudes C_R, C_B (66), growth constant $C_k = 2k^k/q^q$ (29). A : amplitudes A_k^X and growth constant k^k/q^q (III), $3z$ in the inverses (I, II), completed-run arrays $A(n, m)$, $A(x, m)$ (IV, 29), A_∞ limiting projections.	Source 29's C_k is a growth constant, not an amplitude.
η, α, β	$\eta = (2k - 1)/3$ (18; α_k in 29); η_N (II, IV); η_i (66); $\eta_k = \alpha_k + q/2$ (29). α_N, α_i projection errors; $\beta(x, \varepsilon)$ delay coefficients (II, IV); $\beta = (7k - 6)/6$, β^X (18); β_k (29); β_i (66); (α, β) exclusion parameters (48).	
ω, h	ω : a primitive cube root of unity (I, II, IV); the weight ω_j (III). h : h_N parity vector, h_k amplitude coefficients, $h(n)$ inverse model (I, II); $h_j = j + 1$ (III).	
$\mathcal{G}, \mathfrak{s}, \sigma$	$\mathcal{G}_k(a, x)$ normalized profiles (I); $\mathcal{G}_{k,n}$ the common scale (III). s_m ratio coefficients (I, II, IV); $\sigma(\varepsilon)$ scalar series and $\sigma_{i,m}$ singular values (III); source 66's $\mathfrak{s}(\varepsilon)$ is Part III's σ at $k = 3$.	
D, Q, L, U, W	D_N gauge, Q_N contraction (I, II, IV); D_i delay and Q_i kernel (III); $D = q(w + 1)$ slope (III); D_N total-state count (29). L_N delay (II, IV); L_q (III); $L = \log y$ (III). U upward shift (II, IV); $U_i(j)$ up coefficients (III, 29). W_0 Lambert; W_N transverse error (I, II, IV); W_m run weights (IV, 29, 66).	

Symbol	Meaning and where	Do not read it as
Thresholds	$N(X)$ (I), $N_c(Y)$ (II), $N(y)$ (18), $\nu(y)$ (66), $I_k(X)$ (29), “the first integer index with $b_n \geq y$ ” (IV): the least n with count at least the target.	Not the inverse of a smooth model.

Sources and provenance

The report was built from six manuscripts, all delivered in the arrival commit 096ee7b87 of 2 October 2026 as batch 77. Five were placed in this directory by commit 34f1acd4b (cluster P3; archive 29 was handed over to that cluster by cluster P4) and source 48 by commit aa7345800 (cluster P2). All six are dated 2 October 2026. The author line is empty in sources 62, 15, 29 and 48 and reads “Research report” in sources 18 and 66; no tool or person is named. Each calls itself an independently checked research report and disclaims publication and peer review.

Part	Source	Archive (.zip) / main file	Lines	pp.	Pin	Labels
I	62	relaxed-binary-trees-reproducibility / relaxe d-binary-trees.tex	1221	23	none	air:rb:
II	15	compacted-binary-trees-reproducibility / comp acted-binary-trees.tex	1274	21	none ^a	air:cb:
III	18	fixed-arity-airy-package / fixed-arity-airy- expansions.tex	515	12	none	air:ka:
III	66	ternary-airy-amplitudes-result / ternary-airy- amplitudes.tex	656	14	none	air:t3:
III	29	larger-alphabet-automata-report / larger-alpha abet-automata.tex	344	6	none ^a	air:ka:al:
IV	48	oeis-dfa-finite-language-report / dfa-finite- languages.tex	967	16	a68b926cf	air:dfa:

^aRecords a search of the repository, not a continuation pin; see below.

Pins. No source continues a repository path. Source 48’s overlap receipt cites repository files at commit a68b926cf. Source 15’s provenance note (shipped as 15-compacted-src-literature-and-provenance.md) records a repository search at commit 89daa64f that found no A254789 material, and source 29’s raw search receipt (not shipped; see the README) refers to commit 4b874cea0. All three commits are ancestors of the placement commits. Those searches were right when made: before this batch the repository had no report on any of these sequences. This report is now that material; the sentences in the Parts that report the empty searches are kept, with a note.

Where the merge had to choose.

1. *Base and order.* Source 62 is the base: source 15 cites it, source 18’s formulas reduce to it at $k = 2$, and it treats the binary problem that Elvey Price, Fang and Wallner left open. Part II follows because its ratio corollary uses Part I. Part III collects every fixed $k \geq 3$. Part IV is last because its source was placed by a later commit, into the Part number reserved for it; mathematically it is the automaton twin of Part II.
2. *One Jacobi appendix.* Source 15’s appendix “Self-contained Jacobi estimates” is printed once, as Appendix A. Source 48’s appendix is the same text with one word changed (“DFA theorem” for “compacted theorem”) and is not reprinted. Source 62’s own Sections 2–5, which prove the

same estimates in the base text’s own words, are kept; the appendix is a second, self-contained write-up of them.

3. *Part IV prints what is specific to automata.* Source 48’s Sections 3–8 and 10 follow Part II line by line. Five passages are word for word Part II’s: the definitions and Jacobi lemma of Section 3, the finite-memory lemma, the formal-equation paragraph of Section 7, the bootstrap of Section 8.2 and the rational-field paragraph of Section 10. They are replaced by pointers. Every display in which the automaton differs is printed: the renewal model, the delay coefficient with its $m = 1$ term, the seeds, the drift $1/(8N)$, the exponents, the scalar and logarithmic coefficients, the amplitude formula and the inverse.
4. *Source 66 inside Part III.* Source 66 is the $k = 3$ case of source 18 and the base proof that 18 generalizes: 18’s proof notes say so and cite 66’s proof file by hash (SHA-256 `18cdbc4e...`, the file shipped as `66-ternary-amplitude-proof-candidate.md`). Its two theorems are kept in its section, marked as the $k = 3$ instance of Theorems 30.1 and 30.2, because the proof printed after them is that base proof. Its own additions are named there.
5. *Source 29 inside Part III.* The positive completed-run comparison appears three times in the sources: as source 29’s main theorem (every $k \geq 3$, explicit finite- n constant, Θ only), as step (35.2) of source 18’s proof (every $k \geq 3$, infinite product) and as step (44.1) of source 66’s ($k = 3$). It is stated as a theorem once, Theorem 47.1, credited to source 29 as its earliest form; the two proof steps carry pointers to it. Source 29’s other results are printed in full.
6. *Nothing else is omitted* except the title blocks, the tables of contents and the separate bibliographies (merged at the end). Each Part, and each source section of Part III, opens with its source’s abstract as delivered.

How the Parts depend on one another

- Part I is self-contained apart from the Elvey Price–Fang–Wallner lower bound [1], used for positivity.
- Part II’s main theorem is self-contained through Appendix A and the same lower bound. Only its ratio corollary (Section 27) uses Part I’s (F2).
- Part III uses neither Part I nor Part II. Its only external input is the relaxed lower bound of Ghosh Dastidar and Wallner [3]. Its general proof extends the ternary proof of source 66; the comparison of source 29 is a Θ -level statement that the general theorem upgrades.
- Part IV uses Part II’s Section 18, its finite-memory Lemma 20.1, its bootstrap and Appendix A, together with the relaxed array of Part I (for domination) and the Elvey Price–Fang–Wallner automaton lower bound [2].

Status, scope and non-claims

The six manuscripts are AI-assisted and unrefereed; each disclaims publication and peer review, and their audits are research verification. Their finite computations (exact recurrences, symbolic coefficient recursions, floating diagnostics) check algebra and indexing and are not proofs of the

analytic steps. Nothing in this report is formalized in Lean or Rocq, and placing it in this repository confers no formal status.

The union of the sources’ non-claims, all kept in the Parts:

- No amplitude is evaluated or enclosed. The decimals 166.9506 and 166.9515 (Part I), 173.12105 (Part II) and the normalized ratios of Part IV are evidence, not enclosures; Part III gives no amplitude values at all.
- Every expansion is a Poincaré expansion to each fixed order. No convergence of the series, Gevrey bound, optimal truncation, exponentially small term or transseries is claimed, and no canonical real interpolation of an integer sequence.
- Part III is not uniform in k and proves eventual, not everywhere, decrease of the normalized ratios.
- Inverses: smooth-model inverses, interpolation inverses and integer thresholds are kept apart; unspecified O -constants give eventual statements, not certified finite thresholds.
- The growth orders are due to Elvey Price, Fang and Wallner [1, 2] and to Ghosh Dastidar and Wallner [3]; the sources claim only the amplitudes, the expansions and what follows from them. Their literature and repository checks are bounded searches, not priority claims.
- Source 66’s “no general-alphabet theorem” and “no compacted ternary result”, source 29’s “no limiting amplitude”, source 62’s “nothing on compacted trees” and source 15’s higher-arity question are statements about each source alone; other Parts of this report now supply those results, as notes in the Parts say.

A correction of the literature that this report has not verified. Sources 18 and 66 state that the proof of the high-path lemma of [3] (Lemma 14 of arXiv v1, Lemma 15 of the published version) contains a false pointwise bound $U_i(j) \leq q$ ($U_i(j) \leq 2$ for $k = 3$; source 66 gives $U_4(1) = 8/3$). For the coefficient $U_i(j)$ as defined in Part III the inequality does fail near the boundary. Whether the published text prints it was not checked against the paper when this report was written. Neither source uses the bound: source 18 avoids it, and source 66 repairs the lemma’s conclusion.

Instances of results elsewhere in the repository; neighbouring reports

Every inverse section is an instance of results proved in the repository’s transseries volume, `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/`. No novelty is claimed for these methods.

- The Lambert anchors $x \log(4x/e) = T$ (Parts I, II), $X(\log X + \log 8 - 1) = Y$ (Part IV) and $qn_0 \log n_0 + (\log A - q)n_0 = L$ (Part III, sources 18, 66, 29) solve the dominant block exactly, as in theorem `p0:thm:lambert-core`.
- The explicit corrections of (I10)–(I11), (26.5), (36.1), (45.1), (50.4) and (63.2) are reversions of a perturbed core, as in theorem `p0:thm:perturbed-inversion`.
- The threshold brackets (I18), (26.9) and (50.5), and the rules $N = \lceil x \rceil$ of sources 18 and 66, are the exact-rounding and separation steps of theorem `p0:thm:staircase`. Its order-theoretic parts are formalized as `Fabius.staircase_ceil` and `Fabius.staircase_separation` in

`Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`; these declarations verify the general rounding step only, nothing specific to this report.

Neighbouring reports. The tree-child network report named above shares the polynomial–Airy lemma family. The batch-77 report `SetTheory/Cardinals/docs/reports/automata-and-formal-languages/accessible-and-strong-automata` counts other automata (accessible and strongly connected deterministic automata, in the Korshunov–Liskovets regime of inverse-power corrections); it has no stretched exponential and shares no lemma with this report. The Airy law of `a290268-unbounded-deficits`, in the same collection directory, is an additive fold from an integral representation, unrelated to the Airy zero here.

Conventions of the merge

- Every label carries the report prefix `air:` and the sub-prefix of the table above; guide labels are `air:guide:`, Part labels `air:part:`, the appendix `air:app:jacobi`. Source 62’s section labels were generated from its section titles and keep those names after the prefix. The new section labels added to sources 15, 18, 66, 29 and 48 have the form `air:(sub-prefix)sec:....`
- Sections are numbered through the report. Source 62’s Section k is Section k (Sections 1–15). Source 15’s Section k is Section $k + 15$; source 18’s is $k + 29$; source 66’s is $k + 37$; source 29’s is $k + 46$; source 48’s is $k + 52$. Theorems keep their position within their section, so source 15’s Theorem 1.1 is Theorem 16.1. Equations are numbered within sections; source 62’s tagged equations (F1)–(F36), (G1)–(G5) and (I1)–(I19) keep their tags. Source 62’s references “§ k ” are now references to its sections, which kept their numbers.
- Citations are merged into one bibliography. The companion-article entry of source 15 (**relaxed**) is replaced by references to Part I; its annotation said that the companion was used only for the relaxed all-orders input of source 15’s Section 12 (Section 27 here), that all shared Jacobi estimates are proved in the appendix, and that no publication or peer-review status is implied. Sources 18 and 66 cite Ghosh Dastidar and Wallner as DW, the others as **gdw**; source 29’s **binary** is source 48’s **dfa**. Source 62’s numbered list of references is merged into the bibliography, its annotations kept.
- Sentences in the Parts about “the accompanying archive”, “the package”, “**replay.py**”, “the SHA-256 manifest”, “the README”, “**dependencies/relaxed/**” or “this standalone article” describe the source’s delivered archive, which survives in commit 096ee7b87. Which of its files this report ships, and under which names, is listed in the README. PDFs, checksum manifests, the members’ manuscripts and READMEs and a few duplicate or empty files are not shipped.

Part I

Relaxed binary trees

Source. Batch-77 manuscript 62, archive `relaxed-binary-trees-reproducibility.zip`, main file `article/relaxed-binary-trees.tex` (1,221 lines; 23-page PDF), titled “Asymptotic amplitude and all orders for relaxed binary trees”, dated 2 October 2026, empty author line. The base of this report: its text is printed in full, with label prefix `air:rb:`. Its numbered list of references is merged into the bibliography at the end.

Abstract of the source.

The number of relaxed binary trees is known to have order $n!4^n \exp(3zn^{1/3})n$, where z is the largest negative Airy zero. We give a spectral argument for convergence to a positive limiting amplitude and for a full expansion in powers of $n^{-1/3}$. An exact time-dependent gauge turns the counting recurrence into a symmetric Jacobi operator followed by a diagonal contraction. A finite Airy quasimode, the continuum spectral gap, and summable ground-mode drift establish norm tracking. A Catalan kernel estimate recovers the boundary coordinate. Polynomial–Airy approximate solutions and a zero-limit projection bootstrap then transfer every finite formal order to the exact sequence. The known lower bound is used to prove strict positivity of the amplitude. We also derive Lambert W inverse expansions and distinguish them from certified discrete threshold decisions. The accompanying reproducibility archive checks the exact algebra and numerical evidence; its amplitude estimates are not rigorous enclosures.

1 Statement and scope

Let a_n be OEIS A082161, the number of relaxed binary trees with n internal nodes. Its defining triangular recurrence is

$$r_{n,m} = r_{n,m-1} + (m+1)r_{n-1,m}, \quad r_{n,0} = 1, \quad r_{n,m} = 0 \ (m > n), \quad a_n = r_{n,n}.$$

Let $z = a_1^{\text{Ai}} = -2.338107410459767\dots$ be the largest real zero of Ai. (This Airy zero is denoted by z throughout, to avoid confusing it with the counting sequence.) Set

$$a = 2^{-1/3}z, \quad F(x) = \text{Ai}(z + 2^{1/3}x).$$

Theorem. There is a constant $\gamma > 0$ such that, for each integer $M \geq 0$,

$$a_n = \gamma n!4^n e^{3zn^{1/3}} n \left(1 + \sum_{k=1}^M c_k n^{-k/3} + O_M(n^{-(M+1)/3}) \right). \quad (\text{F1})$$

The coefficients belong to $\mathbb{Q}[z]$ and are determined by the polynomial–Airy recursion in §8. The first logarithmic coefficients are

$$\log \frac{a_n}{\gamma n!4^n e^{3zn^{1/3}} n} = \frac{53z^2}{90} n^{-1/3} + \frac{44z}{27} n^{-2/3} + \left(\frac{141}{140} - \frac{1304z^3}{42525} \right) n^{-1} + O(n^{-4/3}). \quad (\text{F2})$$

Consequently

$$\begin{aligned} c_1 &= 53z^2/90, \\ c_2 &= z(2809z^3 + 26400)/16200, \\ c_3 &= (1042139z^6 + 28444320z^3 + 30836700)/30618000. \end{aligned} \tag{F3}$$

Numerically these are 3.2193061325653844, 1.3717168742447035, and -5.306490818002809 . No rigorous numerical enclosure or elementary evaluation of γ is asserted.

The counting recurrence, the Θ estimate, and the conjectural limiting amplitude and all-orders shape are due to Andrew Elvey Price, Wenjie Fang, and Michael Wallner, Compacted binary trees admit a stretched exponential [1], especially Theorem 1.1 and §3.4. Their paper explicitly leaves existence of the amplitude open and reports the fit 166.95208957. The contribution here is a convergence/stability argument for the relaxed recurrence. It does not extend without further work to compacted trees, whose recurrence is different. It proves a Poincaré expansion to every finite algebraic order, not an exponentially improved transseries or a canonical interpolation between integers.

2 Exact recurrence and contractive symmetrization

[write, 2 October 2026] Sections 2–5 prove the gauge, gap, quasimode and projection estimates that Parts II and IV also use. Source 15 proves the same estimates again, self-contained and with the explicit matrices of Part II, in its appendix, printed as Appendix A; source 48 ships a verbatim copy of that appendix. Both write-ups are kept: this one is the base text, the appendix the version the later Parts cite.

Put

$$d_{N,j} = \frac{r_{(N+j)/2, (N-j)/2}}{((N+j)/2)!}$$

when $0 \leq j \leq N$ and $j \equiv N \pmod{2}$, and zero otherwise. Direct substitution gives

$$d_{0,j} = \mathbf{1}_{j=0}, \quad d_{N,-1} = 0, \quad d_{N,j} = \frac{N-j+2}{N+j} d_{N-1,j-1} + d_{N-1,j+1}. \tag{F4}$$

In particular $a_n = n!d_{2n,0}$. Values outside the admissible support are interpreted as zero.

For $N \geq 1$, define on $0 \leq j \leq N+1$

$$D_N(j)^2 = \frac{\Gamma(N+2)\Gamma(N+1)}{\Gamma(N-j+2)\Gamma(N+j+1)}, \quad D_N(0) = 1,$$

and $v_N(j) = d_{N,j}/D_N(j)$. All vectors are padded with zeros in $\ell^2(\mathbb{Z}_{\geq 0})$. Define S_N to be the symmetric Jacobi matrix on $0, \dots, N+1$, padded by zero, with edge weights

$$b_{N,j} = \sqrt{\frac{N-j+2}{N+j}}, \quad 1 \leq j \leq N+1,$$

and define a diagonal contraction by

$$Q_N(j) = \sqrt{1 - \frac{j(j-1)}{N(N+1)}} \quad (0 \leq j \leq N+1), \quad Q_N(j) = 0 \quad (j > N+1).$$

The factorial identities

$$\frac{D_N(j)}{D_N(j-1)} = b_{N,j}, \quad \frac{D_{N-1}(j)}{D_N(j)} = Q_N(j)$$

on the support where the second ratio is needed establish the exact identity

$$v_N = S_N Q_N v_{N-1}. \quad (\text{F5})$$

Both b and Q lie in $[0, 1]$. If J denotes the half-line adjacency matrix, then $0 \leq S_N Q_N \leq J$ entrywise. If λ_N is the largest eigenvalue of S_N , then $\|S_N Q_N\| \leq \lambda_N$. The finite Jacobi matrix is irreducible, so its top eigenvector ψ_N can be taken positive and of norm one. Its spectrum is symmetric about zero because it exchanges the two parity subspaces. In particular its operator norm is λ_N .

3 The coarse Airy limit and a uniform spectral gap

Write $\varepsilon = N^{-1/3}$ and $x_j = (j+1)\varepsilon$. For vectors supported on the finite interval $0, \dots, N+1$, the quadratic form is exactly

$$\begin{aligned} \langle v, (2I - S_N)v \rangle &= \sum_{j=0}^N b_{N,j+1} |v_{j+1} - v_j|^2 + |v_0|^2 \\ &\quad + \sum_{j=1}^N (2 - b_{N,j} - b_{N,j+1}) |v_j|^2 + (2 - b_{N,N+1}) |v_{N+1}|^2. \end{aligned} \quad (\text{F6})$$

Here $|v_0|^2$ is the first Dirichlet-gradient interval, with fictitious value $v_{-1} = 0$; no nonphysical edge coefficient $b_{N,0}$ is introduced.

For $1 \leq j \leq N+1$,

$$1 - b_{N,j} = \frac{2(j-1)}{(N+j)(1+b_{N,j})} \geq \frac{j-1}{N+j}.$$

It follows from (F6) that, for universal positive constants c, C ,

$$\varepsilon^{-2} \langle v, (2I - S_N)v \rangle \geq c \sum_{j=0}^{N+1} x_j |v_j|^2 - C\varepsilon \|v\|^2. \quad (\text{F7})$$

On each fixed x window the edge weights tend uniformly to one, and

$$\varepsilon^{-2} (2 - b_{N,j} - b_{N,j+1}) \longrightarrow 2x_j$$

uniformly there. Interpolate linearly with value $\varepsilon^{-1/2}v_j$ at x_j and zero at zero. Add one zero node immediately after the final finite node and extend by zero thereafter; the endpoint term in (F6) controls this final interpolation interval. The gradient terms in (F6) give local H^1 bounds for any sequence with bounded scaled energy. Equation (F7) gives uniformly small L^2 tails outside a large fixed window. Local compactness, followed by this tail bound, gives strong L^2 compactness. The boundary interval enforces the trace zero at zero. The resulting liminf form is

$$\int_0^\infty (|f'(x)|^2 + 2x|f(x)|^2) dx.$$

For completeness, convergence of norms in this interpolation follows first on a fixed window from the local gradient bound and the Riemann-sum identity, and then globally by (F7). Sampling smooth compactly supported functions with zero boundary value gives the reverse, limsup bound. Applying the elementary min-max principle in both directions therefore gives convergence of each fixed low eigenvalue of $\varepsilon^{-2}(2I - S_N)$ to that of

$$\mathcal{H} = -\frac{d^2}{dx^2} + 2x, \quad f(0) = 0.$$

The latter eigenvalues are $\mu_k = -2^{2/3}z_k$, where $z_1 = z > z_2 > \dots$ are the negative Airy zeros. Thus, if $\lambda_{2,N}$ is the second-largest positive eigenvalue for large N ,

$$\lambda_N = 2 + 2a\varepsilon^2 + o(\varepsilon^2), \quad \lambda_N - \lambda_{2,N} \sim 2^{2/3}(z - z_2)\varepsilon^2. \quad (\text{F8})$$

As a map between opposite parity subspaces, S_N has singular values equal to its positive eigenvalues (and possibly additional zeros). Hence, after deleting its top singular direction, its norm divided by λ_N is at most $1 - cN^{-2/3}$ for all sufficiently large N . This parity formulation is essential: the negative eigenvalue $-\lambda_N$ is not a second slow direction within the occupied parity space.

4 Quantitative estimates from a finite quasimode

The Airy equation reads $F'' = 2(x + a)F$. Define

$$\begin{aligned} F_2(x) &= \frac{2(-a + 2x)}{15}F(x) + \frac{x(2a + x)}{15}F'(x), \\ F_3(x) &= \frac{F(x) - xF'(x)}{3}, \\ f_N(x) &= F(x) + \varepsilon^2 F_2(x) + \varepsilon^3 F_3(x), \\ \hat{\lambda}_N &= 2 + 2a\varepsilon^2 + 3\varepsilon^3 + \frac{13a^2}{15}\varepsilon^4 + \frac{5a}{3}\varepsilon^5. \end{aligned} \quad (\text{F9})$$

Choose a fixed smooth cutoff χ , equal to one on $[0, 1]$ and zero on $[2, \infty)$, and sample $\chi(j/N^{1/2})f_N(x_j)$. Let $\hat{f}_N$ denote its ℓ^2 normalization. Changing or omitting the cutoff in local calculations produces an error smaller than every inverse power of N .

Here is an explicit way to check and bound the quasimode residual. At a sample point the two edge coefficients are

$$b_-(x, \varepsilon) = \sqrt{\frac{1 - x\varepsilon^2 + 3\varepsilon^3}{1 + x\varepsilon^2 - \varepsilon^3}}, \quad b_+(x, \varepsilon) = \sqrt{\frac{1 - x\varepsilon^2 + 2\varepsilon^3}{1 + x\varepsilon^2}}.$$

Expand $b_-f_N(x-\varepsilon) + b_+f_N(x+\varepsilon) - \hat{\lambda}_N f_N(x)$ through degree five in ε , reducing every derivative using $F'' = 2(x+a)F$. Every coefficient vanishes. At the left boundary the omitted sample is zero because $F(0) = F_2(0) = F_3(0) = 0$. On the support of the cutoff, $x\varepsilon^2 = O(N^{-1/2})$; Taylor remainders are bounded by $C\varepsilon^6$ times a fixed polynomial in x multiplying Airy decay. All such polynomial–Airy envelopes are square-integrable. Derivatives of the cutoff contribute only $\exp(-cN^{1/4})$ times a power of N . Since the unnormalized sampled norm is asymptotic to $\varepsilon^{-1/2}\|F\|_{L^2}$, this proves

$$\|(S_N - \hat{\lambda}_N)\hat{f}_N\| = O(\varepsilon^6). \quad (\text{F10})$$

The Airy gap (F8) identifies the nearby eigenvalue with λ_N ; projecting the residual on the complement of ψ_N then gives

$$\lambda_N = \hat{\lambda}_N + O(\varepsilon^6), \quad \|\psi_N - \hat{f}_N\| = O(\varepsilon^4). \quad (\text{F11})$$

The sign is fixed by the positive leading Airy profile. Several useful consequences follow without a separate quantitative localization theorem:

$$\begin{aligned} \|\psi_N - \psi_{N-1}\| &= O(N^{-1}), \\ \|(I - Q_N)\psi_N\| &= O(N^{-4/3}), \\ \psi_N(0) &= \sqrt{2} N^{-1/2}(1 + O(N^{-1/3})). \end{aligned} \quad (\text{F12})$$

For the first, the explicitly normalized samples in (F9) have ℓ^2 derivative $O(N^{-1})$ with respect to continuous N ; the two approximation errors in (F11) are smaller. For the second, use $1 - \sqrt{1-t} \leq t$ for $0 \leq t \leq 1$, the finite polynomial moments of the sampled Airy profile, and (F11). For the endpoint, the pointwise error is at most the ℓ^2 error, and

$$I := \int_0^\infty F(x)^2 dx = 2^{-1/3} \text{Ai}'(z)^2, \quad \frac{F'(0)}{\sqrt{I}} = \sqrt{2}.$$

A Riemann-sum estimate with an integrable derivative gives a relative $O(\varepsilon)$ normalization error; the endpoint Taylor error is $O(\varepsilon^2)$ and the normalized quasimode error divided by the leading endpoint is $O(\varepsilon^{5/2})$. These identities therefore verify the quantitative endpoint estimate in (F12). Standard Airy decay needed above follows, for example, directly from the decaying solution of the Airy ODE; only its polynomially weighted integrability and exponential tail are used.

5 Convergence of the scalar amplitude

Set

$$P_N = \prod_{i=1}^N \lambda_i, \quad u_N = v_N/P_N, \quad M_N = S_N Q_N / \lambda_N.$$

Then $u_N = M_N u_{N-1}$, $\|M_N\| \leq 1$, and $\|u_N\| \leq 1$. Let E_p be projection onto height parity p . The positive eigenvector has equal squared mass on the two parities: this follows by taking the scalar product of each half of $S_N \psi_N = \lambda_N \psi_N$ with that half. Therefore

$$g_N = \sqrt{2} E_{N \bmod 2} \psi_N, \quad h_N = \sqrt{2} E_{(N-1) \bmod 2} \psi_N$$

are unit vectors, with $S_N h_N = \lambda_N g_N$. By (F12),

$$\|h_N - g_{N-1}\| = O(N^{-1}), \quad \|(I - Q_N)h_N\| + \|(I - Q_N)g_{N-1}\| = O(N^{-4/3}). \quad (\text{F13})$$

The second estimate for g_{N-1} uses the first estimate only for its approximation by explicit Airy samples, rather than the crude $O(N^{-1})$ difference: direct use of (F11) at $N-1$ gives the stated $O(N^{-4/3})$ estimate with Q_N . It also follows by observing that $Q_N \geq Q_{N-1}$ pointwise on the old support and treating its negligible eigenvector tail.

Decompose

$$u_N = A_N g_N + w_N, \quad w_N \perp g_N.$$

Write $\Pi_N = I - |g_N\rangle\langle g_N|$ on the occupied parity subspace. The singular gap from §3 gives, for every vector y on the preceding parity,

$$\|\Pi_N M_N y\| \leq (1 - cN^{-2/3})\|Q_N y\| \leq (1 - cN^{-2/3})\|y\|. \quad (\text{F14})$$

Indeed $\Pi_N S_N = S_N(I - |h_N\rangle\langle h_N|)$ on that parity; this projection removes the top singular direction, regardless of whether y is perpendicular to h_N . Also, (F13) implies

$$\|M_N g_{N-1} - g_N\| = O(N^{-1}).$$

Applying (F14) to w_{N-1} and the latter estimate to the scalar term yields

$$\|w_N\| \leq (1 - cN^{-2/3})\|w_{N-1}\| + CN^{-1}. \quad (\text{F15})$$

Iteration gives $\|w_N\| = O(N^{-1/3})$. One general estimate used here and later is

$$\sum_{k=N_0}^N k^{-r} \prod_{i=k+1}^N (1 - ci^{-2/3}) = O_r(N^{2/3-r}). \quad (\text{F16})$$

To prove it, split at $N/2$. The earlier half is exponentially small in $N^{1/3}$ times a polynomial; on the later half bound the product by $\exp(-c'(N-k)N^{-2/3})$ and sum a geometric series. A finite initial segment makes no difference.

The scalar equation is

$$A_N = t_N A_{N-1} + \langle k_N, w_{N-1} \rangle, \quad t_N = \langle Q_N h_N, g_{N-1} \rangle, \quad k_N = Q_N h_N - g_{N-1}. \quad (\text{F17})$$

The unit-vector identity $\langle h, g \rangle = 1 - \|h - g\|^2/2$ and (F13) show

$$t_N = 1 + O(N^{-4/3}), \quad \|k_N\| = O(N^{-1}). \quad (\text{F18})$$

Since $|A_N| \leq 1$, (F15), (F17), and (F18) give

$$A_N - A_{N-1} = O(N^{-4/3}).$$

Thus

$$A_N = A_\infty + O(N^{-1/3}). \quad (\text{F19})$$

Positivity gives $A_N \geq 0$ and hence $A_\infty \geq 0$; strict positivity is established in §7.

6 A boundary smoothing estimate

Norm tracking alone cannot read a coordinate whose leading size is $N^{-1/2}$. A short final block of the original positive kernel provides the missing smoothing.

Let $U(N, k) = M_N \cdots M_{k+1}$. For $0 \leq \ell \leq N^{2/3}$, positivity and $S_i Q_i \leq J$ give

$$\|e_0^T U(N, N - \ell)\|_2 \leq \frac{\|e_0^T J^\ell\|_2}{\lambda_{N-\ell+1} \cdots \lambda_N}.$$

The squared numerator is the number of nonnegative walks of length 2ℓ returning to zero:

$$\|e_0^T J^\ell\|_2^2 = (J^{2\ell})_{0,0} = \frac{1}{\ell+1} \binom{2\ell}{\ell} \leq C 4^\ell (\ell+1)^{-3/2}.$$

Equation (F11) implies

$$\frac{2^\ell}{\lambda_{N-\ell+1} \cdots \lambda_N} \leq \exp(C\ell N^{-2/3}) \leq C'$$

over the same range. Therefore

$$\|e_0^T U(N, N - \ell)\|_2 \leq C(\ell+1)^{-3/4}. \quad (\text{F20})$$

The exact error recurrence for w_N is

$$w_N = M_N w_{N-1} + f_N^{\text{err}}, \quad f_N^{\text{err}} = A_{N-1} M_N g_{N-1} - A_N g_N, \quad \|f_N^{\text{err}}\| = O(N^{-1}).$$

Use Duhamel over $L = \lfloor N^{2/3} \rfloor$ steps. Since all times in this block are comparable with N , (F15) and (F20) imply

$$\begin{aligned} |w_N(0)| &\leq CL^{-3/4} \|w_{N-L}\| + CN^{-1} \sum_{\ell=0}^{L-1} (\ell+1)^{-3/4} \\ &= O(N^{-5/6}). \end{aligned} \quad (\text{F21})$$

For even N , (F12) gives $g_N(0) = 2N^{-1/2}(1 + O(N^{-1/3}))$.

7 The limiting constant and its strict positivity

From (F11),

$$\log \lambda_N = \log 2 + aN^{-2/3} + \frac{3}{2}N^{-1} + O(N^{-4/3}).$$

Summation gives

$$P_N = \kappa 2^N e^{3aN^{1/3}} N^{3/2} (1 + O(N^{-1/3})), \quad \kappa > 0. \quad (\text{F22})$$

If desired, a convergent expression is

$$\log \kappa = a\zeta(2/3) + \frac{3}{2}\gamma_E + \sum_{N=1}^{\infty} \left[\log(\lambda_N/2) - aN^{-2/3} - \frac{3}{2N} \right]. \quad (\text{F23})$$

Here γ_E is Euler's constant, not the counting amplitude. This is an existence formula; it does not by itself give efficient certified numerical evaluation.

Equations (F19)–(F22), for even N , now give

$$d_{N,0} = 2\kappa A_{\infty} 2^N e^{3aN^{1/3}} N + O\left(2^N e^{3aN^{1/3}} N^{2/3}\right). \quad (\text{F24})$$

The known lower half of the Elvey Price–Fang–Wallner Θ theorem states that $d_{2n,0}$ is bounded below by a positive constant times $4^n e^{3zn^{1/3}} n$ for all sufficiently large n . If $A_{\infty} = 0$, (F24) contradicts that bound. Hence $A_{\infty} > 0$ and

$$\gamma = 4\kappa A_{\infty} > 0. \quad (\text{F25})$$

Taking $N = 2n$ in (F24) proves (F1) with $M = 0$. This completes the amplitude-convergence argument, using the explicit prior lower-bound theorem and the elementary Airy facts used above.

8 Formal profiles at every finite order

Work now with the original recurrence (F4), rather than the symmetric gauge. Seek

$$Z_{N,j} = H_N \sum_{k=0}^K \varepsilon^k G_k(x_j), \quad G_0 = F, \quad G_k(0) = G'_k(0) = 0 \quad (k \geq 1),$$

with

$$\frac{H_N}{H_{N-1}} \sim 2 \left(1 + \sum_{m \geq 2} s_m \varepsilon^m \right). \quad (\text{F26})$$

At the previous time the two arguments and powers are

$$y_{\pm} = (x \pm \varepsilon)(1 - \varepsilon^3)^{-1/3}, \quad \varepsilon_{N-1}^k = \varepsilon^k (1 - \varepsilon^3)^{-k/3},$$

and the downward coefficient is

$$\frac{1 - x\varepsilon^2 + 3\varepsilon^3}{1 + x\varepsilon^2 - \varepsilon^3}.$$

After the leading orders, the coefficient at degree m has the form

$$\mathcal{L}G_{m-2} - 2s_m F + R_m(x)F + S_m(x)F', \quad \mathcal{L} = \frac{d^2}{dx^2} - 2(x+a), \quad (\text{F27})$$

where R_m, S_m are known polynomials over $\mathbb{Q}[a]$, formed from earlier orders. In particular $s_2 = a$. This form follows by Taylor expansion: the new profile first appears through the second-order spatial operator; its lower, zeroth-order terms cancel between the two sides of (F4). Time-change terms involving this new profile enter only at later degree.

To solve $\mathcal{L}(PF + QF') = RF + SF'$, compute

$$\mathcal{L}(PF + QF') = (P'' + 4(x+a)Q' + 2Q)F + (2P' + Q'')F'.$$

Eliminating $P' = (S - Q'')/2$ reduces the problem to

$$-\frac{1}{2}Q''' + 4(x+a)Q' + 2Q = R - \frac{1}{2}S'. \quad (\text{F28})$$

On polynomials of degree at most d , the operator on the left is upper triangular by degree with nonzero diagonal entries $4j + 2$, $0 \leq j \leq d$; hence it is invertible. The new scalar s_m adds $2s_m$ to the forcing R , thus adds exactly s_m to the constant term of Q . Choose it uniquely to enforce $Q(0) = 0$, which is the boundary condition. Integrating P' leaves one free constant; choose it so $P(0) + Q'(0) = 0$, which enforces the derivative normalization. This proves existence and uniqueness at every formal order within the stated normalization, not merely consistency of finitely many guessed coefficients.

[write, 2 October 2026] This polynomial–Airy identity, with its triangular reduction, is used by every Part of this report and by the tree-child network report; see “One polynomial–Airy identity” in Part 0.

The first scalar coefficients are

$$s_2 = a, \quad s_3 = \frac{4}{3}, \quad s_4 = \frac{29a^2}{270}, \quad s_5 = -\frac{23a}{81}, \quad s_6 = \frac{74385 - 8783a^3}{85050}. \quad (\text{F29})$$

For example,

$$G_1 = -\frac{2}{3}x^2F, \quad G_2 = \frac{2(2a + 15x^4 + 111x)}{135}F + \frac{x(-8a + 51x)}{270}F'.$$

The amplitude factor can be chosen as a genuine smooth function of N with finite logarithmic expansion

$$\log H_N = N \log 2 + 3aN^{1/3} + \frac{4}{3} \log N + \sum_{k=1}^J h_k N^{-k/3}. \quad (\text{F30})$$

Expanding its exact one-step ratio matches any prescribed finite number of coefficients in (F26). At degree ε^{k+3} , the new h_k has coefficient $-k/3$, so the construction is again triangular and nonsingular. Crucially the leading multiplicative constant in (F30) is fixed to one for every truncation. This avoids an artificial truncation-dependent amplitude.

9 Approximate solutions in the normalized norm

Fix any $p > 1$. Take enough orders in §8, multiply the sampled profile by the cutoff $\chi(j/N^{1/2})$, and restrict to the occupied parity. Call the result Z_N . Choose the truncation so that its local formal residual starts at a power $N^{-p'}$ with $p' \geq p$. Define

$$z_N = D_N^{-1} Z_N / P_N.$$

Then

$$\|z_N\| = O(1), \quad r_N := z_N - M_N z_{N-1} = O(N^{-p}) \quad \text{in } \ell^2. \quad (\text{F31})$$

Here are the needed global bounds. On $j \leq 2N^{1/2}$, the product formula for D_N gives $D_N^{-1} \leq C$ uniformly: $-\log D_N(j) \leq Cj^2/N \leq C'$. On any fixed scaled window $D_N^{-1} = 1 + O(N^{-1/3}(1+x)^2)$. Taylor remainders in the rational coefficients and the time dilation are bounded by N^{-p} times a fixed polynomial–Airy envelope on the cutoff support. Cutoff errors decay like $\exp(-cN^{1/4})$ times a polynomial, also after multiplication by D_N^{-1} . Finally,

$$H_N / P_N \sim \kappa^{-1} N^{-1/6},$$

while a parity-restricted sampled profile has norm asymptotic to $N^{1/6} \sqrt{I/2}$. These bounds prove (F31) by direct summation. The same reasoning proves

$$\langle g_N, z_N \rangle \longrightarrow q_0 := \kappa^{-1} \sqrt{I/2} > 0. \quad (\text{F32})$$

Indeed, after the natural sample normalization, both profiles converge to the same positive Airy function. Neither (F31) nor (F32) needs a pointwise estimate on the unknown solution.

10 A scalar bootstrap to every finite order

Set

$$C = A_\infty / q_0, \quad e_N = u_N - C z_N.$$

Then $e_N = M_N e_{N-1} - C r_N$ and the ground projection tends to zero. Decompose

$$e_N = \alpha_N g_N + W_N, \quad W_N \perp g_N, \quad \alpha_N \longrightarrow 0.$$

The exact same estimates (F14), (F17), and (F18) now give

$$\begin{aligned}\|W_N\| &\leq (1 - cN^{-2/3})\|W_{N-1}\| + CN^{-1}|\alpha_{N-1}| + CN^{-p}, \\ |\alpha_N - \alpha_{N-1}| &\leq CN^{-4/3}|\alpha_{N-1}| + CN^{-1}\|W_{N-1}\| + CN^{-p}.\end{aligned}\tag{F33}$$

Suppose initially that $\alpha_N = O(N^{-q})$, where $q \geq 0$. Formula (F16) gives

$$\|W_N\| = O(N^{-q-1/3} + N^{2/3-p}).$$

Substitute this into the second inequality of (F33). Its right-hand side is summable because $p > 1$; summing from $N + 1$ to infinity and using the zero limiting projection gives

$$|\alpha_N| = O(N^{-q-1/3} + N^{1-p}).\tag{F34}$$

The smaller tail $O(N^{2/3-p})$ has been absorbed by $O(N^{1-p})$. Start with $q = 0$, which holds since u_N, z_N are bounded. Finitely many repetitions of (F34) reach

$$\alpha_N = O(N^{1-p}), \quad \|W_N\| = O(N^{2/3-p}), \quad \|e_N\| = O(N^{1-p}).\tag{F35}$$

There is no boundary-at-infinity assumption beyond the already proved, explicit limit (F32) and the choice of C . No inverse operator with a small gap is silently assumed uniformly bounded.

Apply Duhamel directly to the inhomogeneous e recurrence on the final block $L = \lfloor N^{2/3} \rfloor$. By (F20), (F31), and (F35),

$$\begin{aligned}|e_N(0)| &\leq CL^{-3/4}N^{1-p} + CN^{-p} \sum_{\ell=0}^{L-1} (\ell+1)^{-3/4} \\ &= O(N^{1/2-p}) + O(N^{1/6-p}) = O(N^{1/2-p}).\end{aligned}\tag{F36}$$

The main endpoint is of size $N^{-1/2}$, so the relative endpoint error is $O(N^{1-p})$. For the remainder at order M , take $p \geq 1 + (M + 1)/3$ and enough formal terms. Since p can be taken as large as desired, every prescribed finite algebraic order of the explicit approximation transfers to the exact sequence. All truncations have the same q_0 , hence the same C and the same $\gamma = 2CF'(0) = 4\kappa A_\infty$. This proves (F1).

11 Coefficient extraction and computational checks

Solving the scalar ratio matching (F30) using (F29) gives

$$h_1 = 53a^2/45, \quad h_2 = 79a/27, \quad h_3 = 143/210 - 5216a^3/42525.$$

The normalized endpoint profile is

$$\frac{\sum_k \varepsilon^k G_k(\varepsilon)}{F'(0)\varepsilon} = 1 + \frac{a}{3}\varepsilon^2 + \frac{4}{3}\varepsilon^3 + O(\varepsilon^4).$$

Consequently the logarithmic endpoint corrections in time N are

$$\frac{53a^2}{45}N^{-1/3} + \frac{88a}{27}N^{-2/3} + \left(\frac{141}{70} - \frac{5216a^3}{42525}\right)N^{-1}.$$

Substituting $N = 2n$ and $a = 2^{-1/3}z$ yields (F2), and exponentiation gives (F3).

12 The rational coefficient field

The recursion immediately puts every coefficient at time N in $\mathbb{Q}[a]$, where $a = 2^{-1/3}z$. The substitution $N = 2n$ initially gives the weaker field $\mathbb{Q}(2^{1/3})[z]$. To justify the stronger field $\mathbb{Q}[z]$ asserted in the theorem, one needs the following mod-3 grading.

Let ϕ_a be the formal power-series solution of

$$\phi_a'' = 2(x+a)\phi_a, \quad \phi_a(0) = 0, \quad \phi_a'(0) = 1.$$

Every coefficient of ϕ_a belongs to $\mathbb{Q}[a]$. At the special Airy parameter used in the proof, this is exactly $F(x)/F'(0)$. For any primitive cube root of unity ω , uniqueness of the initial-value solution gives

$$\phi_{\omega a}(\omega x) = \omega \phi_a(x), \quad \phi_{\omega a}'(\omega x) = \phi_a'(x). \quad (\text{G1})$$

Indeed, the left side of the first identity, as a function of x , obeys the same ODE as ϕ_a because $\omega^3 = 1$, and its derivative at zero is ω .

Divide every formal profile in §8 by $F'(0)$ and denote the resulting formal sum by

$$\Phi(a, \varepsilon, x) = \sum_{k \geq 0} \varepsilon^k \mathcal{G}_k(a, x), \quad \mathcal{G}_0 = \phi_a.$$

Its normalization is $\Phi(a, \varepsilon, 0) = 0$ and $\partial_x \Phi(a, \varepsilon, 0) = 1$. Denote the half-ratio of the amplitude factors by

$$R(a, \varepsilon) = 1 + \sum_{m \geq 2} s_m(a) \varepsilon^m.$$

In the formal recurrence, the rational coefficient depends only on $x\varepsilon^2$ and ε^3 ; the dilation depends only on ε^3 ; and the shifted arguments are $(x \pm \varepsilon)(1 - \varepsilon^3)^{-1/3}$. Consequently the transformation

$$(a, \varepsilon, x) \longmapsto (\omega a, \omega \varepsilon, \omega x)$$

preserves the recurrence. The transformed profile

$$\tilde{\Phi}(a, \varepsilon, x) = \omega^{-1} \Phi(\omega a, \omega \varepsilon, \omega x)$$

has the same leading solution and the same boundary and derivative normalizations, by (G1). The order-by-order polynomial recursion in §8 is unique under exactly these normalizations. Therefore

$$R(\omega a, \omega \varepsilon) = R(a, \varepsilon), \quad \Phi(\omega a, \omega \varepsilon, \omega x) = \omega \Phi(a, \varepsilon, x). \quad (\text{G2})$$

In particular,

$$s_m(\omega a) = \omega^{-m} s_m(a). \quad (\text{G3})$$

The uniquely determined coefficients $h_k(a)$ of the finite logarithmic amplitude satisfy the same grading,

$$h_k(\omega a) = \omega^{-k} h_k(a). \quad (\text{G4})$$

One way to see (G4) without manipulating a logarithm branch is to use its ratio equation. The leading ratio contributions involve $3a/\varepsilon$, multiplied by a series in ε^3 , and $\log(1 - \varepsilon^3)$. They are invariant under the transformation. The term involving $h_k \varepsilon^k$ first enters at degree $k+3$ with nonzero coefficient $-k/3$, so the unique recursive solution must have the indicated grading.

The normalized endpoint factor is

$$E(a, \varepsilon) = \frac{\Phi(a, \varepsilon, \varepsilon)}{\varepsilon}.$$

Equation (G2) shows $E(\omega a, \omega \varepsilon) = E(a, \varepsilon)$. The constant term of E is one. Thus every coefficient of both E and $\log E$ has the same mod-3 grading. Combining it with (G4), the time- N logarithmic and relative correction coefficients $b_k(a)$ obey

$$b_k(\omega a) = \omega^{-k} b_k(a). \quad (\text{G5})$$

Since b_k is a polynomial with rational coefficients, every nonzero monomial a^r in it satisfies

$$r + k \equiv 0 \pmod{3}.$$

Finally its contribution after changing variables is

$$a^r N^{-k/3} = 2^{-(r+k)/3} z^r n^{-k/3}.$$

The exponent $(r+k)/3$ is an integer, so the factor of two is rational. This proves that every forward correction coefficient at time n lies in $\mathbb{Q}[z]$, rather than merely in $\mathbb{Q}(2^{1/3})[z]$.

13 Asymptotic inversion and discrete thresholds

[**write, 2 October 2026**] The Lambert anchor, the explicit corrections (I10)–(I11) and the brackets (I18) are instances of theorems `p0:thm:lambert-core`, `p0:thm:perturbed-inversion` and `p0:thm:staircase` of the repository’s transseries volume; see Part 0, “Instances of results elsewhere in the repository”. No novelty is claimed for the inversion mechanics.

13.1 Input, normalization, and scope

Let $a_n = A082161(n)$, and let $z = a_1^{\text{Ai}} = -2.338107410459767\dots$ denote the first negative zero of the Airy function. To avoid confusing this zero with the first sequence term, this report uses z throughout. The forward theorem provides a fixed $\gamma > 0$ satisfying

$$\log a_n = \log \gamma + \log \Gamma(n+1) + n \log 4 + 3zn^{1/3} + \log n + Bn^{-1/3} + Dn^{-2/3} + L_3n^{-1} + O(n^{-4/3}), \quad (\text{I1})$$

where

$$B = \frac{53z^2}{90}, \quad D = \frac{44z}{27}, \quad L_3 = \frac{141}{140} - \frac{1304z^3}{42525}. \quad (\text{I2})$$

The error in (I1) is initially only an assertion at integer n . All conclusions about an actual sequence below use only that assertion, not unproved differentiability of its remainder.

There are three distinct inversion objects:

1. the inverse of an explicit smooth asymptotic model;
2. the inverse of a chosen interpolation of the integer sequence;
3. the discrete threshold $N(X) = \min\{n : a_n \geq X\}$.

They have different error statements. In particular, no real-valued asymptotic expansion, by itself, proves a single unqualified floor/ceiling rule at thresholds arbitrarily close to sequence values.

13.2 The factorial contributes a crucial logarithm

Stirling's formula gives

$$\log \Gamma(n+1) = n \log n - n + \frac{1}{2} \log n + \frac{1}{2} \log(2\pi) + \frac{1}{12n} + O(n^{-3}).$$

Consequently, set

$$A = 3z, \quad p = \frac{3}{2}, \quad C = \log(\gamma\sqrt{2\pi}), \quad E = L_3 + \frac{1}{12} = \frac{229}{210} - \frac{1304z^3}{42525}, \quad (\text{I3})$$

and define

$$h(n) = n \log(4n/e) + An^{1/3} + p \log n + C + Bn^{-1/3} + Dn^{-2/3} + En^{-1}. \quad (\text{I4})$$

Then (I1) is $\log a_n = h(n) + O(n^{-4/3})$. In particular, the coefficient of $\log n$ is **3/2**, not 1. Numerically,

$$\begin{aligned} A &= -7.014322231379301\dots, & B &= 3.219306132565383\dots, \\ D &= -3.810249113341843\dots, & E &= 1.482422558319535\dots \end{aligned}$$

An equally useful model retains the factorial exactly:

$$h_\Gamma(n) = \log \gamma + \log \Gamma(n+1) + n \log 4 + An^{1/3} + \log n + Bn^{-1/3} + Dn^{-2/3} + L_3/n. \quad (\text{I5})$$

Its difference from (I4) is $-1/(360n^3) + O(n^{-5})$, below the precision needed here. Its inverse differs from the inverse of (I4) by $O(n^{-3}/\log n)$.

For (I4),

$$h'(n) = \log(4n) + \frac{A}{3}n^{-2/3} + \frac{p}{n} - \frac{B}{3}n^{-4/3} - \frac{2D}{3}n^{-5/3} - En^{-2} \sim \log(4n) > 0, \quad (\text{I6})$$

and $h''(n) \sim 1/n$. Thus h has a unique increasing large-argument inverse.

13.3 Lambert W anchor and the first Airy correction

Put

$$T = \log X, \quad x = \frac{T}{W_0(4T/e)}, \quad s = \log(4x) = 1 + W_0(4T/e), \quad q = p \log x + C. \quad (\text{I7})$$

Here W_0 is the principal real Lambert W branch and X is sufficiently large. The identity

$$x \log(4x/e) = T$$

holds exactly. The leading correction needed to undo the negative Airy term is positive:

$$h^{-1}(T) = x - \frac{A}{s}x^{1/3} - \frac{q}{s} + O\left(\frac{x^{-1/3}}{s}\right). \quad (\text{I8})$$

Equivalently, the displayed first terms are

$$x - \frac{3z}{\log(4x)}x^{1/3} - \frac{3}{2} + \frac{\frac{3}{2}\log 4 - \log(\gamma\sqrt{2\pi})}{\log(4x)}. \quad (\text{I9})$$

This shows the hierarchy clearly: the Airy correction grows like $x^{1/3}/\log x$, the factorial/prefactor combination gives a constant shift $-3/2$, and the unknown amplitude first enters at order $1/\log x$.

Equation (I8), including its stated error scale for suitable interpolations or the discrete bracketing interpretation below, already follows from the weaker core hypothesis

$$a_n = \gamma n! 4^n e^{3zn^{1/3}} n(1 + O(n^{-1/3})).$$

It does not require the all-orders construction.

An alternative anchor explicitly absorbs the unknown amplitude: set $y = T - C$, $x = y/W_0(4y/e)$, and $q = p \log x$. Every formula below remains valid with this replacement. The unshifted version (I7) makes the dependence on γ easier to see.

13.4 Explicit inverse through the third fractional order

Define

$$\begin{aligned}
u &= -\frac{A}{s}, \\
v &= -\frac{q}{s}, \\
w &= -\frac{B}{s} + \frac{A^2}{3s^2} - \frac{A^2}{2s^3}, \\
r &= -\frac{D}{s} + \frac{A(p+q/3)}{s^2} - \frac{Aq}{s^3}, \\
k &= -\frac{E}{s} + \frac{pq}{s^2} - \frac{AB+q^2/2}{s^3} + \frac{A^3}{3s^4} - \frac{A^3}{2s^5}.
\end{aligned} \tag{I10}$$

Then the inverse of the explicit smooth model (I4) satisfies

$$\boxed{h^{-1}(T) = x + ux^{1/3} + v + wx^{-1/3} + rx^{-2/3} + kx^{-1} + O\left(\frac{x^{-4/3}}{\log(4x)}\right)}. \tag{I11}$$

The error order in (I11) is also the uncertainty imposed by the forward remainder in (I1). It is therefore the natural stopping point using the three displayed forward logarithmic coefficients.

13.4.1 Derivation

Write $t = x^{-1/3}$ and

$$\delta = u/t + v + wt + rt^2 + kt^3, \quad n = x + \delta.$$

The leading function $f(n) = n \log(4n/e)$ has expansion

$$f(x + \delta) - f(x) = s\delta + \frac{\delta^2}{2x} - \frac{\delta^3}{6x^2} + \cdots.$$

For the perturbations, use

$$\begin{aligned}
A(x + \delta)^{1/3} &= A/t + (A/3)ut + (A/3)vt^2 + ((A/3)w - (A/9)u^2)t^3 + O(t^4), \\
p \log(x + \delta) + C &= q + put^2 + pvt^3 + O(t^4), \\
B(x + \delta)^{-1/3} &= Bt - (B/3)ut^3 + O(t^4), \\
D(x + \delta)^{-2/3} &= Dt^2 + O(t^4), \\
E(x + \delta)^{-1} &= Et^3 + O(t^5).
\end{aligned}$$

The required equations, in order, are

$$\begin{aligned}
su + A &= 0, \\
sv + q &= 0, \\
sw + u^2/2 + Au/3 + B &= 0, \\
sr + uv + Av/3 + pu + D &= 0, \\
sk + v^2/2 + (u + A/3)w - u^3/6 - Au^2/9 + pv - Bu/3 + E &= 0.
\end{aligned} \tag{I12}$$

Solving (I12) gives (I10); the substantial cancellation in k is exact. SymPy verifies every coefficient from t^{-1} through t^3 is identically zero with A, B, D, E, p, q, s left symbolic.

Although q depends on x , this calculation is a Taylor expansion about the fixed anchor x . Its contribution to derivatives is already included in $p \log(x + \delta)$; one must not separately differentiate q within (I12).

For the remainder, $q = O(s)$, $u = O(1/s)$, $v = O(1)$, and $w, r, k = O(1/s)$. Hence $\delta = O(x^{1/3}/s)$ and $\delta/x = O(x^{-2/3}/s)$. All omitted terms in the residual $h(x + \delta) - T$ are $O(x^{-4/3})$. Since the local derivative is asymptotic to s , the mean-value theorem yields (I11). No derivative bound on the unknown sequence remainder was used.

13.5 Two Newton steps are a simpler practical inverse

For computation it is usually preferable to avoid manually evaluating (I10). Starting from the Lambert anchor $x_0 = x$, take

$$x_{j+1} = x_j - \frac{h(x_j) - T}{h'(x_j)}, \quad j = 0, 1. \quad (\text{I13})$$

If $n_* = h^{-1}(T)$, then

$$\begin{aligned} x_0 - n_* &= O(x^{1/3}/s), \\ x_1 - n_* &= O(x^{-1/3}/s^3), \\ x_2 - n_* &= O(x^{-5/3}/s^7). \end{aligned} \quad (\text{I14})$$

Indeed, throughout the relevant interval, h' is comparable to s and $h'' = O(1/x)$, so one Newton step squares the error and divides it by xs . The last bound is smaller than the forward uncertainty $O(x^{-4/3}/s)$. Thus two Newton steps solve the available asymptotic inverse more accurately than its supplied coefficients justify.

The same procedure works using h_Γ from (I5), with

$$h'_\Gamma(n) = \psi(n+1) + \log 4 + (A/3)n^{-2/3} + n^{-1} - (B/3)n^{-4/3} - (2D/3)n^{-5/3} - L_3n^{-2},$$

where ψ is the digamma function. Work with $T = \log X$ rather than forming very large X .

More generally, if a smooth model has forward logarithmic error $O(n^{-\rho})$, its inverse uncertainty is $O(x^{-\rho}/s)$. Additional Newton steps can make model inversion error smaller than any prescribed finite power. Starting from (I7), after j steps the bound is

$$x_j - n_* = O\left(x^{1-2^{j+1}/3} s^{1-2^{j+1}}\right).$$

13.6 What “smooth inverse” means for an integer sequence

The asymptotic assertion (I1) alone does not specify a canonical interpolation between integers. It is incorrect simply to differentiate its $O(n^{-4/3})$ remainder as though a differentiable real-variable asymptotic theorem had been proved.

There is, however, a precise conditional statement:

- If an eventually increasing interpolation $L(t)$ satisfies $L(n) = \log a_n$ and $L(t) = h(t) + O(t^{-4/3})$ uniformly for real t , then

$$L^{-1}(T) = h^{-1}(T) + O(x^{-4/3}/s). \quad (\text{I15})$$

The argument is value bracketing using monotonicity and the slope of h ; no derivative estimate for the remainder of L is necessary.

- Such smooth increasing interpolations **exist** under (I1). For example, put $\varepsilon_n = \log a_n - h(n)$ and choose a fixed smooth bump ϕ , supported in $(-1/3, 1/3)$, with $\phi(0) = 1$. For large real t , set

$$L(t) = h(t) + \sum_n \varepsilon_n \phi(t - n).$$

The supports are disjoint, the perturbation and its first derivative are both $O(t^{-4/3})$, and therefore $L'(t) > 0$ eventually. This supplies a smooth interpolation meeting (I15), but it is an arbitrary choice, not an intrinsic continuous extension of the combinatorial sequence.

- The common interpolation that is piecewise linear in $\log a_n$ has a weaker generic comparison. Because $h''(t) \sim 1/t$, its interpolation error relative to h is $O(1/t)$, larger than the forward remainder. Its inverse satisfies only

$$L_{\text{linear}}^{-1}(T) = h^{-1}(T) + O(1/(x \log x)). \quad (\text{I16})$$

- An arbitrary monotone interpolation can differ by order 1 in its inverse. Integer samples alone do not force the sharper real-valued estimate.

Every increasing interpolation through the exact sequence nevertheless gives the same discrete threshold after taking a ceiling, with $N(X) = \text{ceil}(L^{-1}(\log X))$ for large X . Thus interpolation distinctions do not obstruct the direct discrete bracketing below.

13.7 Discrete threshold bracketing and the rounding barrier

The forward hypothesis itself implies eventual strict monotonicity. In fact,

$$\log a_{n+1} - \log a_n = \log(4n) + O(n^{-2/3}) > 0$$

under (I1). Under just the weaker core hypothesis, the error can instead be written $O(n^{-1/3})$, still sufficient. As a_n is unbounded, finitely many earlier terms do not affect $N(X)$ for sufficiently large X .

Suppose that an actual forward bound is available:

$$|\log a_n - h(n)| \leq K n^{-4/3} \quad (n \geq n_0). \quad (\text{I17})$$

Let $r_* = h^{-1}(\log X)$ and $K_* = \max(K, 1)$. For all sufficiently large r_* , one valid, deliberately conservative choice is

$$\eta = \frac{8K_* r_*^{-4/3}}{\log(4r_*)}.$$

Then

$$\boxed{\lceil r_* - \eta \rceil \leq N(X) \leq \lceil r_* + \eta \rceil.} \quad (\text{I18})$$

To verify (I18), test the integer immediately below $\text{ceil}(r_* - \eta)$ and the integer $\text{ceil}(r_* + \eta)$. They lie within a bounded distance of r_* ; there h' is at least half $\log(4r_*)$, while the allowed forward error is at most twice $Kr_*^{-4/3}$. The displacement η leaves a strict margin on either side. Eventual monotonicity gives the result.

If the explicit approximation (I11) is used in place of r_* , enlarge η by a bound for its error, which has the same order. If the two-step Newton approximation is used, its much smaller error from (I14) can be added instead.

Important qualifications:

1. Once $2\eta < 1$, (I18) leaves either one integer or two adjacent candidates.
2. If $\text{dist}(r_*, \mathbb{Z}) > \eta$, both ceilings agree and $N(X) = \text{ceil}(r_*)$.
3. If r_* is within η of an integer, the next omitted term or exact sequence evaluation can decide which adjacent candidate is correct. The supplied asymptotic does not always decide it.
4. One may write $N(X) = \text{ceil}(r_* + O(r_*^{-4/3}/\log r_*))$ only in the existential/bracketing sense of (I18). It does **not** mean $N(X) = \text{ceil}(r_*) + o(1)$ uniformly in X .
5. An unquantified O bound proves the existence of (I18) with some constant and starting point. It does not yield a numerically certified enclosure for a particular finite X . That requires effective forward constants, a certified amplitude, and numerical error control.

Under the weaker core theorem, the corresponding bracket radius is $O(r_*^{-1/3}/\log r_*)$, still tending to zero. Therefore the core theorem alone already localizes the threshold to at most two adjacent integers asymptotically.

13.8 Unknown gamma and numerical amplitude estimates

The constant γ is a fixed positive parameter, not a free constant that can be silently dropped after leading order. Changing γ to $\tilde{\gamma}$ shifts the smooth inverse, to first order, by

$$-\frac{\log(\tilde{\gamma}/\gamma)}{\log(4n)}. \quad (\text{I19})$$

A bounded multiplicative amplitude uncertainty thus causes an inverse uncertainty $O(1/\log n)$. This is $o(1)$, but is much larger than the algebraic error in (I11), and can still change the discrete answer at near-integer thresholds.

If certified information gives

$$|\log \gamma - \log \tilde{\gamma}| \leq \varepsilon,$$

the forward error budget for a model using $\tilde{\gamma}$ becomes $\varepsilon + Kn^{-4/3}$. The threshold radius is then

$$O\left(\frac{\varepsilon + r_*^{-4/3}}{\log(4r_*)}\right).$$

An uncertified fitted decimal for γ does not supply such a bound. In particular, the supplied approximate value 166.95063854245 is suitable for illustrative computations, not for asserting rigorous threshold decisions.

13.8.1 An amplitude-free locator and its existing provenance

If γ is unknown, one can still use

$$r_0 = x - \frac{3z}{\log(4x)}x^{1/3} - \frac{3}{2}, \quad x = \frac{\log X}{W_0(4 \log X/e)}.$$

The real inverse uncertainty is $O(1/\log x) = o(1)$. In discrete form, for some fixed constant K and sufficiently large X ,

$$\left\lceil r_0 - \frac{K}{\log x} \right\rceil \leq N(X) \leq \left\lceil r_0 + \frac{K}{\log x} \right\rceil.$$

This coarse result already follows from the published Elvey Price–Fang–Wallner Θ theorem: its logarithm has an $O(1)$ remainder, which is divided by the slope $\sim \log x$ on inversion. It should therefore not be presented as a consequence unique to the amplitude proof. The amplitude convergence identifies the fixed $\log \gamma$ term; the all-orders theorem provides the finer algebraic inverse corrections. As before, existence of an unspecified constant K is not a numerical finite-threshold certificate.

14 Reproducible computation and numerical limitations

The accompanying source archive is relocatable and includes every script needed to generate finite-order coefficients, verify the algebraic identities used above, and reproduce the reported numerical evidence. It requires Python with SymPy, NumPy, SciPy, and mpmath. The exact versions used in the delivered replay are recorded in its environment report.

The main replay command is `python replay.py --max-n 20000`. It verifies the SHA-256 manifest of its inputs before executing the checks, writes every result beneath `output/`, and reports any failure. The coefficient command `python coefficients.py --order 8 --output output/coefficients.json` constructs all scalar and profile terms through the requested order. Raising the order invokes the same finite polynomial algorithm, with no lookup table or fitted coefficients.

The coefficient construction is a finite, exact procedure:

1. Represent a profile by a polynomial pair (P, Q) for $PF + QF'$ and differentiate using $F'' = 2(x + a)F$
2. Expand the rational recurrence, the two shifted arguments, and the previous-time powers to the next required degree in ε
3. Collect the known forcing in the two components F and F'

4. Solve the triangular polynomial equation (F28), choose the scalar so that $Q(0) = 0$, and integrate for P with $P(0) + Q'(0) = 0$
5. Verify the resulting residual coefficient, both boundary conditions, and the mod-three grading exactly
6. Match the logarithmic one-step ratio and expand the endpoint factor to obtain forward logarithmic and relative coefficients

The archive also checks the original triangular recurrence against the transformed recurrence with exact rational arithmetic, verifies the squared gauge identities, verifies the finite frozen quasimode, and checks the inverse cancellations with symbolic parameters. Its numerical tests use the full finite support of the forward recurrence, rather than an unquantified height cutoff.

At $n = 20000$, the normalized raw ratio is 187.01754328008607. Dividing it by the first three relative corrections gives 166.95063854245257; removing the first three logarithmic corrections instead gives 166.95152887373968. These values are consistent with the empirical amplitude 166.95208957 reported by Elvey Price, Fang, and Wallner. They are not certified enclosures, and the last displayed digits should not be interpreted as established digits of γ .

The exact symbolic checks establish the displayed cancellation identities. Floating-point checks establish only reproducibility of numerical evidence. Neither kind of computation replaces the analytic compactness, stability, or endpoint arguments above. The theorem's constant is the convergent expression (F25), not a fitted decimal. In particular, certified finite-threshold decisions require effective forward constants, a certified amplitude interval, and controlled numerical evaluation.

15 Further research questions

Certified amplitude evaluation. Can the convergent spectral-product and projection representation be supplemented with explicit tail constants to give an efficient interval enclosure of γ ? The present proof gives existence and algebraic rates, but its compactness-based spectral gap has not been made numerically effective. A constructive lower gap, certified Airy residuals, and explicit endpoint bounds would make this a concrete certification problem.

Compacted binary trees. The corresponding compacted-tree recurrence has a subtraction term. Its established leading scale has the same stretched exponential and a different power of n . Extending the present positive-kernel smoothing argument requires an appropriate positive lift or a different stability estimate. The relaxed-tree proof does not automatically establish the compacted amplitude.

[write, 2 October 2026] Answered by the same batch: Part II (source 15) proves the compacted amplitude and every fixed order, Theorem 16.1. It does not use a positive lift: it treats the subtraction as a globally bounded signed delay and cancels its scalar drift. The relaxed-tree proof indeed does not by itself establish the compacted amplitude, as said above.

Higher arity. The exact gauge and parity decomposition exploit a nearest-neighbor binary recurrence. For relaxed k -ary structures, one can ask whether a comparable slowly varying transfer operator admits a coercive continuum limit and summable ground-mode drift. The specific gauge, gap, and boundary exponent would need to be derived anew.

[write, 2 October 2026] Answered for every fixed arity $k \geq 3$ by the same batch: Part III (sources 18 and 66) proves positive amplitudes and every fixed order for relaxed k -ary trees,

and for compacted trees and automata, starting from the Θ theorem of Ghosh Dastidar and Wallner [3]. As this question anticipates, the gauge, gap and boundary exponent are derived anew: a fixed weighted norm matching forward and adjoint boundary harmonics replaces the exact Jacobi symmetrization. Uniformity in k remains open (Section 37).

Beyond algebraic orders. The construction proves a Poincaré expansion at every fixed order. It gives no optimal-truncation theorem, no Gevrey estimate on the coefficient growth, and no exponentially small correction. Those questions require quantitative control uniform in the expansion order, together with an analysis of higher Airy modes and possible additional asymptotic sectors.

Intrinsic interpolation and inverse certification. The combinatorial sequence has no canonical real interpolation supplied by its recurrence. The threshold brackets in §13 avoid assuming one. It would be useful to combine interval evaluation of nearby exact terms with the asymptotic locator, yielding a certified algorithm that resolves the occasional two-candidate ambiguity.

Closing note of source 62. The proposed convergence argument and its scope are distinguished from the existing result in §1. No assertion of an exhaustive global priority search or of peer-reviewed acceptance is made.

Part II

Compacted binary trees

Source. Batch-77 manuscript 15, archive `compacted-binary-trees-reproducibility.zip`, main file `article/compacted-binary-trees.tex` (1,274 lines; 21-page PDF), titled “Asymptotic amplitude and all orders for compacted binary trees”, dated 2 October 2026, empty author line. Printed in full with label prefix `air:cb::`; its appendix is Appendix A of this report. Its “companion relaxed-tree theorem” is Part I.

Abstract of the source.

The number of compacted binary trees with n internal nodes is known to have order $n!4^n \exp(3zn^{1/3})n^{3/4}$, where z is the largest negative Airy zero. We prove convergence to a positive limiting amplitude and a Poincaré expansion to every finite order in $n^{-1/3}$. The exact counting recurrence is a signed three-step perturbation of the relaxed-tree Jacobi evolution. A global $O(N^{-1})$ delay bound, cancellation of its scalar drift, and a finite-memory tracking argument establish amplitude convergence. Polynomial–Airy profiles and a zero-limit bootstrap then transfer arbitrary formal orders to the exact sequence without assuming positivity of the signed propagator or an endpoint heat-kernel estimate. The published lower bound supplies strict positivity. We also give exact positive finite-state and renewal models, explain their symmetry obstructions, derive explicit Lambert W inverse corrections, and state the precise discrete threshold interpretation. A self-contained appendix proves the shared Jacobi estimates. Numerical amplitude values are evidence, not certified enclosures.

16 Statement and scope

Let $c_{n,m}$ be defined for $0 \leq m \leq n$ by

$$c_{n,0} = 1, \quad c_{n,m} = c_{n,m-1} + (m+1)c_{n-1,m} - (m-1)c_{n-2,m-1}, \quad (16.1)$$

with invalid cells zero, and put $c_n = c_{n,n}$. The mathematical index counts internal nodes, so $c_0 = c_1 = 1$ and

$$(c_n)_{n=0}^8 = (1, 1, 3, 15, 111, 1119, 14487, 230943, 4395855).$$

This is the compacted binary-tree sequence associated with OEIS A254789 [7]. Proposition 2.11 and Definition 2.12 of Elvey Price, Fang and Wallner [1] give the recurrence and the equinumerous decorated-path model used below. Their Theorem 1.1 proves

$$c_n = \Theta\left(n!4^n e^{3zn^{1/3}} n^{3/4}\right), \quad (16.2)$$

where

$$z = -2.338107410459767\dots, \quad \text{Ai}(z) = 0, \quad a = 2^{-1/3}z, \quad F(x) = \text{Ai}(z + 2^{1/3}x).$$

Here z is the largest real Airy zero and $F(x) > 0$ for $x > 0$. The asymptotic equivalent and full expansion are conjectural in Section 3.4 of that paper; they do not follow merely from its two-sided estimate.

Theorem 16.1. *There is a constant $\gamma_c > 0$ such that, for every fixed nonnegative integer M ,*

$$c_n = \gamma_c n!4^n e^{3zn^{1/3}} n^{3/4} \left(1 + \sum_{k=1}^M \mathbf{c}_k n^{-k/3} + O_M(n^{-(M+1)/3})\right), \quad \mathbf{c}_k \in \mathbb{Q}[z]. \quad (16.3)$$

The amplitude is the convergent spectral-projection expression in (24.1). The first three logarithmic corrections are

$$\begin{aligned} \log \frac{c_n}{\gamma_c n!4^n e^{3zn^{1/3}} n^{3/4}} &= \frac{53z^2}{90} n^{-1/3} + \frac{271z}{216} n^{-2/3} \\ &+ \left(\frac{393}{1120} - \frac{1304z^3}{42525} \right) n^{-1} + O(n^{-4/3}). \end{aligned} \quad (16.4)$$

This is a Poincaré expansion: the error statement applies at each fixed finite order. It asserts neither an exponentially improved transseries nor a canonical real interpolation of the integer sequence. No elementary evaluation or rigorous numerical enclosure of γ_c is claimed. The fitted decimal 173.12670485 in [1] is consistent with the computations accompanying this article, but is not part of the theorem.

The argument is an independently checked research proof. This description is not a claim of publication or peer review. A bounded search of primary literature and the ProveIt repository found no prior resolution of this unrestricted compacted amplitude problem; that search is not a guarantee of priority or exhaustive novelty. The later k -ary work [3] and the bounded-right-height problem [4] have different stated conclusions or restrictions.

[write, 2 October 2026] The repository search (recorded in 15-compactd-src-literature-and-provenance.md at commit 89daa64f) was right when made. The repository's A254789 material is now this report, which the same batch delivered together with the relaxed, fixed-arity and automaton companions.

17 Exact positive path models

17.1 Four states and the signed recurrence

A path from $(0, 0)$ to (n, m) stays in $0 \leq m \leq n$, using horizontal steps H and vertical steps V . An H step at ordinate m has one of the labels $1, \dots, m+1$. The forbidden pattern is HHV in which the two horizontal labels are equal and greater than one. A trailing equal pair is allowed until a V step follows it.

At a path prefix define four states:

- Z : no trailing H , so the prefix is empty or ends in V ;
- A : the last H label is 1;
- B : the last H label exceeds 1, without a trailing equal non-1 pair;
- C : the trailing two H labels are equal and exceed 1.

Write $\mathbf{F}(n, m) = (Z, A, B, C)^\top$. The exact matrices are

$$H_m^{(4)} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 \\ m & m & m-1 & m-1 \\ 0 & 0 & 1 & 1 \end{pmatrix}, \quad V^{(4)} = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (17.1)$$

with

$$\mathbf{F}(n, m) = H_m^{(4)} \mathbf{F}(n-1, m) + V^{(4)} \mathbf{F}(n, m-1), \quad \mathbf{F}(0, 0) = (1, 0, 0, 0)^\top.$$

Invalid cells are zero. At $m = 0$, states B, C are unreachable. Replacing $m-1$ by $\max(m-1, 0)$ gives globally nonnegative formal matrices without changing the reachable evolution.

To verify the transitions, a new label 1 always enters A . After a prefix in Z or A , each of the m non-1 labels enters B . After a prefix in B or C , precisely one label enters C and the other $m-1$ enter B . A V accepts exactly Z, A, B . In particular, a prefix in C is not discarded prematurely: another H replaces the trailing pair.

If $c = Z + A + B + C$, these transitions imply

$$\begin{aligned} A(n, m) &= c_{n-1, m}, & B(n, m) + C(n, m) &= m c_{n-1, m}, \\ C(n, m) &= m c_{n-2, m}, & Z(n, m) &= c_{n, m-1} - (m-1) c_{n-2, m-1} \end{aligned} \quad (17.2)$$

for the last identity when $m \geq 1$. Adding them gives (16.1). At $n = m > 0$ the last step cannot be H , so $c_n = Z(n, n)$.

17.2 A three-state local kernel

States Z and A have identical outgoing transitions and can be merged into $E = Z + A$. In the state order (E, B, C) ,

$$H_m = \begin{pmatrix} 1 & 1 & 1 \\ m & m-1 & m-1 \\ 0 & 1 & 1 \end{pmatrix}, \quad V = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \mathbf{e} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \mathbf{t} = (1, 1, 1). \quad (17.3)$$

The same convention at $m = 0$ applies. For every fixed $m > 0$, the reachability matrix $(\mathbf{e}, H_m \mathbf{e}, H_m^2 \mathbf{e})$ has determinant m^2 , while the matrix with rows $\mathbf{t}, \mathbf{t}V, \mathbf{t}VH_m$ has determinant -1 . Their Hankel product has determinant $-m^2$. Thus this fixed-altitude linear suffix-counting representation has dimension three; a two-state linear representation of that same problem is impossible. This statement does not exclude coordinate-dependent or nonlocal encodings.

With physical time $N = n + m$, height $j = n - m$, and factorial normalization

$$\mathbf{f}_N(j) = \frac{\mathbf{F}_3((N+j)/2, (N-j)/2)}{((N+j)/2)!},$$

the positive nearest-neighbor kernel is

$$\mathbf{f}_N(j) = \frac{2}{N+j} H_{(N-j)/2} \mathbf{f}_{N-1}(j-1) + V \mathbf{f}_{N-1}(j+1). \quad (17.4)$$

Only valid parity and support are used; its initial vector is $\mathbf{f}_0(0) = \mathbf{e}$.

17.3 A scalar positive renewal representation

A horizontal run of length k at altitude m , completed by a V , has admissible weight

$$w_m(0) = 1, \quad w_m(1) = m + 1, \quad w_m(k) = (m^2 + m + 1)(m + 1)^{k-2} \quad (k \geq 2). \quad (17.5)$$

Indeed among the $(m + 1)^k$ labelings, exactly $m(m + 1)^{k-2}$ have a forbidden final pair; earlier pairs do not immediately precede a V . Let $a(n, m)$ count paths ending in V , with $a(0, 0) = 1$ and all other $a(n, 0) = 0$. Then

$$\begin{aligned} a(n, m+1) &= \sum_{k=0}^{n-m} w_m(k) a(n-k, m), & n \geq m+1, \\ c_{n,m} &= \sum_{k=0}^{n-m} (m+1)^k a(n-k, m), & a(n, n) = c_n. \end{aligned} \quad (17.6)$$

The completed-run generating function is

$$W_m(t) = \sum_{k \geq 0} w_m(k) t^k = \frac{1 - mt^2}{1 - (m+1)t}.$$

At the vertical-step clock, distance $h = n - m$ changes to $h' = h + k - 1$. For $b_m(h) = a(m + h, m)/(m + h)!$, the transfer weight from h to h' is

$$w_m(k) \frac{(m+h)!}{(m+h+k)!}, \quad k \geq 0, \quad h' \geq 0.$$

This kernel is scalar and positive, but has arbitrarily large upward jumps and downward jumps of size at most one.

17.4 Why the positive models are not symmetric Jacobi reductions

Several exact obstructions delimit direct symmetrization:

1. For $m > 0$, $\text{rank } H_m = 2$ and

$$H_m^2 = (m+1, m^2, m)^\top (1, 1, 1).$$

Its characteristic polynomial is $\lambda^2(\lambda - m - 1)$, but the zero eigenspace has dimension one. Its nontrivial zero Jordan block prevents similarity to a real symmetric matrix, even under a general invertible change of basis. In particular, no positive-definite common metric makes both H_m and V self-adjoint.

2. In the height/state transfer there is a positive directed edge $B_h \rightarrow E_{h+1}$ and no reverse V edge. Positive diagonal conjugation preserves such zero patterns. The renewal kernel similarly has a $+2$ jump without a -2 jump.
3. Freeze $m = 1$, use horizontal coefficient $1/2$ and vertical coefficient 1 , and truncate heights to $0, 1, 2, 3$. The resulting 12 by 12 matrix has characteristic polynomial

$$\frac{\lambda^6}{4}(4\lambda^6 - 12\lambda^4 + 6\lambda^2 - 1).$$

The cubic in λ^2 has discriminant -432 , so this frozen model has nonreal eigenvalues and cannot be similar to a self-adjoint matrix. This is an obstruction for the stated frozen example, not every time-dependent compacted transfer.

4. A self-adjoint moment representation $w(k) = \langle v, S^k v \rangle$ would have nonnegative first Hankel minor, whereas

$$w_m(0)w_m(2) - w_m(1)^2 = -m < 0 \quad (m > 0).$$

These observations do not rule out dilations or asymptotic effective self-adjoint descriptions. They motivate the signed-delay route below, which requires neither a symmetric lift nor a new singular gap for the three-state kernel.

18 The shared Jacobi evolution

All vectors below are zero-padded elements of $\ell^2(\mathbb{N}_0)$. For $N \geq 1$ define

$$D_N(j)^2 = \frac{\Gamma(N+2)\Gamma(N+1)}{\Gamma(N-j+2)\Gamma(N+j+1)}, \quad 0 \leq j \leq N+1, \quad D_N(0) = 1. \quad (18.1)$$

Let S_N be the symmetric Jacobi matrix on $0, \dots, N+1$ with edge weights

$$b_{N,j} = \sqrt{\frac{N-j+2}{N+j}} \quad (1 \leq j \leq N+1), \quad Q_N(j) = \sqrt{1 - \frac{j(j-1)}{N(N+1)}}. \quad (18.2)$$

Both are zero outside that finite interval. Let λ_N be the largest eigenvalue of S_N , and ψ_N its positive unit eigenvector. If E_p projects onto height parity p , put

$$\begin{aligned} g_N &= \sqrt{2}E_{N \bmod 2}\psi_N, & h_N &= \sqrt{2}E_{(N-1) \bmod 2}\psi_N, \\ P_N &= \prod_{i=1}^N \lambda_i, & M_N &= \frac{S_N Q_N}{\lambda_N}, & \Pi_N &= I - |g_N\rangle\langle g_N| \end{aligned} \quad (18.3)$$

on the occupied parity subspace. Both g_N and h_N are unit vectors.

Lemma 18.1. *There are positive constants c, κ such that, for all sufficiently large N and any y on the preceding parity,*

$$\begin{aligned}
\|M_N\| &\leq 1, & \|\Pi_N M_N y\| &\leq (1 - cN^{-2/3}) \|y\|, \\
\|M_N g_{N-1} - g_N\| &= O(N^{-1}), & \|M_N^* g_N - g_{N-1}\| &= O(N^{-1}), \\
t_N := \langle g_N, M_N g_{N-1} \rangle &= 1 + O(N^{-4/3}), \\
\lambda_N &= 2 + 2aN^{-2/3} + 3N^{-1} + O(N^{-4/3}), \\
P_N &= \kappa 2^N e^{3aN^{1/3}} N^{3/2} (1 + O(N^{-1/3})).
\end{aligned} \tag{18.4}$$

Moreover ψ_N is within $O(N^{-4/3})$ in ℓ^2 of the normalization of cutoff samples of

$$\begin{aligned}
F(x) + N^{-2/3} \left[\frac{2(-a+2x)}{15} F(x) + \frac{x(2a+x)}{15} F'(x) \right] \\
+ N^{-1} \frac{F(x) - xF'(x)}{3}, \quad x = (j+1)N^{-1/3},
\end{aligned} \tag{18.5}$$

with cutoff $\chi(j/N^{1/2})$. These normalized samples have time derivative $O(N^{-1})$, one-site shift difference $O(N^{-1/3})$, and bounded polynomial moments on the Airy scale.

A complete proof is given in Appendix A. The gap in (18.4) is a parity-restricted singular-value gap. It is not a second spectral-radius estimate on the whole space, where $-\lambda_N$ is also an eigenvalue.

The relaxed counts $r_{n,m}$ satisfy

$$r_{n,0} = 1, \quad r_{n,m} = r_{n,m-1} + (m+1)r_{n-1,m},$$

with invalid cells zero. They count the same decorated paths without the forbidden HHV restriction. Thus $0 \leq c_{n,m} \leq r_{n,m}$. The common gauge turns the relaxed recurrence into $v_N^{(r)} = S_N Q_N v_{N-1}^{(r)}$; consequently $\|v_N^{(r)} / P_N\| \leq 1$. Only this domination and the explicit Jacobi estimates, not an automatic extension of a relaxed amplitude theorem, will be used to prove Theorem 16.1.

19 A globally bounded signed delay

Set

$$e_{N,j} = \frac{c_{(N+j)/2, (N-j)/2}}{((N+j)/2)!}, \quad v_N(j) = \frac{e_{N,j}}{D_N(j)}$$

on $0 \leq j \leq N$, $j \equiv N \pmod{2}$, and zero otherwise. The scalar form of (17.4), equivalently direct substitution in (16.1), is

$$e_{N,j} = \frac{N-j+2}{N+j} e_{N-1,j-1} + e_{N-1,j+1} - \frac{2(N-j-2)}{(N+j)(N+j-2)} e_{N-3,j-1}. \tag{19.1}$$

The delayed term is used only for a valid source $(N-3, j-1)$; it is zero at $j=0$ and outside that support. In particular its coefficient at $j=N-2$ is zero. This convention excludes the spurious negative rational coefficient at the nonphysical source for $j=N$.

For all sufficiently large N , the exact gauge recurrence is

$$v_N = S_N Q_N v_{N-1} - L_N v_{N-3}, \quad (L_N f)(j) = \ell_{N,j} f(j-1), \tag{19.2}$$

where on $1 \leq j \leq N-2$

$$\ell_{N,j} = \frac{2(N-j-2)}{(N+j)(N+j-2)} \frac{D_{N-3}(j-1)}{D_N(j)}, \quad (19.3)$$

and it is zero elsewhere. Cancellation of factorials gives

$$\left(\frac{D_{N-3}(j-1)}{D_N(j)} \right)^2 = \frac{(N-j+1)(N-j)(N+j)(N+j-1)(N+j-2)(N+j-3)}{(N+1)N^2(N-1)^2(N-2)}. \quad (19.4)$$

For $N \geq 6$ this ratio is uniformly bounded on the entire physical support; the rational prefactor in (19.3) is $O(N^{-1})$. Hence

$$\|L_N\| \leq CN^{-1}. \quad (19.5)$$

This is a global operator bound, not an estimate limited to the Airy window.

For $1 \leq j \leq 2N^{1/2}$ on the physical support, expanding (19.3)–(19.4) gives

$$\ell_{N,j} = \frac{2}{N} + O\left(\frac{j+1}{N^2}\right).$$

Let U be the unilateral upward shift: $(Uf)(0) = 0$ and $(Uf)(j) = f(j-1)$. The finite profile in Lemma 18.1, its Airy moments, and (19.5) imply

$$\left\| L_N g_{N-3} - \frac{2}{N} U g_{N-3} \right\| = O(N^{-5/3}), \quad \|U g_{N-3} - g_N\| = O(N^{-1/3}). \quad (19.6)$$

For the first estimate the typical height is $O(N^{1/3})$, and the $O(N^{-4/3})$ quasimode error is multiplied by $O(N^{-1})$. Cutoff tails are exponentially small. The second estimate uses a one-site sample shift and an $O(N^{-1})$ time change. The shifted parity is exactly the parity occupied at time N .

20 Cancellation of the scalar drift

Put

$$R_N = P_N N^{-1/4}, \quad u_N = v_N / R_N, \quad (20.1)$$

and begin the following recurrences at any fixed sufficiently large N . Then

$$u_N = \eta_N M_N u_{N-1} - B_N u_{N-3}, \quad \eta_N = \left(\frac{N}{N-1} \right)^{1/4}, \quad B_N = \frac{R_{N-3}}{R_N} L_N. \quad (20.2)$$

Since $R_{N-3}/R_N = 1/8 + O(N^{-2/3})$, Lemma 18.1 and (19.6) yield

$$\begin{aligned} \eta_N &= 1 + \frac{1}{4N} + O(N^{-2}), \quad \|B_N\| = O(N^{-1}), \\ \left\| B_N g_{N-3} - \frac{1}{4N} g_N \right\| &= O(N^{-4/3}), \\ b_N := \langle g_N, B_N g_{N-3} \rangle &= \frac{1}{4N} + O(N^{-4/3}), \\ \eta_N t_N - 1 - b_N &= O(N^{-4/3}). \end{aligned} \quad (20.3)$$

The last line cancels the nonsummable order- N^{-1} drift. It explains the additional factor $N^{-1/4}$ relative to the relaxed spectral normalization.

Lemma 20.1 (Finite-memory bounds). *Suppose nonnegative x_N satisfy, eventually,*

$$x_N \leq (1 - cN^{-2/3})x_{N-1} + CN^{-1}x_{N-3} + CN^{-r}. \quad (20.4)$$

For every fixed real r , $x_N = O(N^{2/3-r})$. A sum of power forcings has the sum of the corresponding bounds. If a polynomially bounded sequence δ_N satisfies

$$|\delta_N| \leq CN^{-1}(|\delta_{N-1}| + |\delta_{N-2}|) + O(N^{-r}), \quad (20.5)$$

then $\delta_N = O(N^{-r})$.

Proof. For the first statement use the barrier $KN^{2/3-r}$. Its difference from the two homogeneous terms on the right of (20.4) is

$$KcN^{-r} + O(KN^{2/3-r-1}).$$

The error is smaller by $N^{-1/3}$ regardless of the sign of $2/3 - r$. Choose a sufficiently large starting time and then K large enough to dominate both the forcing and the finite initial triple. Induction proves the estimate. Adding barriers handles sums of powers. For the second statement, repeatedly substitute the polynomial bound in (20.5); each substitution reduces its exponent by one until the forcing dominates. Equivalently the barrier KN^{-r} gives an eventual induction, since C/N tends to zero. $\square$

21 Convergence of the scalar amplitude

The positive gauge preserves $0 \leq v_N \leq v_N^{(r)}$ entrywise. The relaxed contraction therefore gives the initial bound

$$\|u_N\| \leq N^{1/4}. \quad (21.1)$$

Decompose

$$u_N = A_N g_N + w_N, \quad w_N \perp g_N.$$

Applying Lemma 18.1 to the transverse part of (20.2), and bounding the delayed whole vector using (21.1), gives

$$\|w_N\| \leq (1 - c'N^{-2/3})\|w_{N-1}\| + O(N^{-3/4}).$$

The factor $\eta_N = 1 + O(N^{-1})$ is absorbed in the gap. Lemma 20.1 then gives

$$\|w_N\| = O(N^{-1/12}). \quad (21.2)$$

No cancellation or positivity of the delayed evolution is needed here.

Projecting the recurrence gives

$$A_N = \eta_N t_N A_{N-1} - b_N A_{N-3} + \xi_N, \quad |\xi_N| \leq CN^{-1}(\|w_{N-1}\| + \|w_{N-3}\|). \quad (21.3)$$

The one-step cross term uses $\|M_N^* g_N - g_{N-1}\| = O(N^{-1})$ and orthogonality; the delayed cross term uses only $\|B_N\| = O(N^{-1})$. Set $\delta_N = A_N - A_{N-1}$. Then

$$\delta_N = b_N(\delta_{N-1} + \delta_{N-2}) + (\eta_N t_N - 1 - b_N)A_{N-1} + \xi_N. \quad (21.4)$$

Equations (20.3), (21.1) and (21.2) give (20.5) with $r = 13/12$. Thus $\delta_N = O(N^{-13/12})$ is summable, and A_N has a finite limit A_∞ . Since u_N and g_N are nonnegative, $A_\infty \geq 0$.

Now A_N is bounded and $w_N \rightarrow 0$, so u_N is bounded. Repeating the transverse and scalar estimates improves them to

$$A_N = A_\infty + O(N^{-1/3}), \quad \|w_N\| = O(N^{-1/3}), \quad \|u_N\| = O(1). \quad (21.5)$$

There is not yet an endpoint conclusion: the normalized ground endpoint itself is only of order $N^{-1/2}$.

22 Formal profiles at every finite order

Let $\varepsilon = N^{-1/3}$ and $x = (j+1)\varepsilon$. Seek an approximate solution to the ungauged recurrence in the form

$$Z_{N,j} = H_N \Phi_N(x), \quad \Phi_N(x) = \sum_{k=0}^K \varepsilon^k G_k(x), \quad G_0 = F, \quad (22.1)$$

with $G_k(0) = G'_k(0) = 0$ for $k \geq 1$, and formal ratio

$$\frac{H_N}{H_{N-1}} \sim 2 \left(1 + \sum_{m \geq 2} s_m \varepsilon^m \right). \quad (22.2)$$

The one-step arguments are $(x \pm \varepsilon)(1 - \varepsilon^3)^{-1/3}$, and the delayed argument is $(x - \varepsilon)(1 - 3\varepsilon^3)^{-1/3}$. The horizontal and delayed coefficients in (19.1) become

$$\begin{aligned} \mathcal{A}(x, \varepsilon) &= \frac{1 - x\varepsilon^2 + 3\varepsilon^3}{1 + x\varepsilon^2 - \varepsilon^3}, \\ \beta(x, \varepsilon) &= \frac{2\varepsilon^3(1 - x\varepsilon^2 - \varepsilon^3)}{(1 + x\varepsilon^2 - \varepsilon^3)(1 + x\varepsilon^2 - 3\varepsilon^3)}. \end{aligned} \quad (22.3)$$

After division by H_{N-1} , the delayed term has the extra factor

$$\frac{H_{N-3}}{H_{N-1}} = \left(\frac{H_{N-1}}{H_{N-2}} \frac{H_{N-2}}{H_{N-3}} \right)^{-1},$$

whose leading constant is $1/4$. As β starts at degree three, neither the new profile nor the new scalar in those previous ratios changes the leading Airy inversion at the current degree.

At formal degree m , the equation is

$$\mathcal{L}G_{m-2} - 2s_m F + R_m(x)F + S_m(x)F' = 0, \quad \mathcal{L} = \partial_x^2 - 2(x+a), \quad (22.4)$$

where $R_m, S_m \in \mathbb{Q}[a][x]$ are already known; in particular $s_2 = a$. For a polynomial pair P, Q ,

$$\mathcal{L}(PF + QF') = (P'' + 4(x+a)Q' + 2Q)F + (2P' + Q'')F'.$$

Thus solving $\mathcal{L}G = RF + SF'$ reduces to

$$-\frac{1}{2}Q''' + 4(x+a)Q' + 2Q = R - \frac{1}{2}S'. \quad (22.5)$$

On polynomials of degree at most d , its matrix is triangular with nonzero diagonal $4j+2$, $0 \leq j \leq d$. It is therefore invertible over $\mathbb{Q}[a]$. Changing s_m shifts the constant term of Q by s_m , so it uniquely enforces $Q(0) = 0$. The integration constant of P then enforces $P(0) + Q'(0) = 0$. Since $F(0) = 0$

and $F'(0) \neq 0$, these are exactly the value and derivative normalizations of G . This proves existence and uniqueness at every finite formal order.

The initial scalar coefficients are

$$\begin{aligned} s_2 &= a, & s_3 &= \frac{13}{12}, & s_4 &= \frac{29a^2}{270}, \\ s_5 &= -\frac{11a}{324}, & s_6 &= \frac{2054835 - 140528a^3}{1360800}. \end{aligned} \quad (22.6)$$

A genuine finite logarithmic amplitude may consequently be chosen as

$$\log H_N = N \log 2 + 3aN^{1/3} + \frac{13}{12} \log N + \sum_{k=1}^J h_k N^{-k/3}. \quad (22.7)$$

At degree $k+3$ in the one-step ratio, the new h_k has coefficient $-k/3$, so any finite list of ratio coefficients can be matched uniquely. For every truncation the leading multiplicative constant in H_N is one. Initially,

$$h_1 = \frac{53a^2}{45}, \quad h_2 = \frac{235a}{108}, \quad h_3 = -\frac{641}{1680} - \frac{5216a^3}{42525}. \quad (22.8)$$

23 Global residuals and the zero-limit bootstrap

23.1 Approximate solutions in the normalized norm

Fix any $p > 1$. Take sufficiently many terms of (22.1) and (22.7), multiply by the cutoff $\chi(j/N^{1/2})$, restrict to occupied parity, and define

$$z_N = D_N^{-1} Z_N / R_N.$$

Then

$$\|z_N\| = O(1), \quad r_N := z_N - \eta_N M_N z_{N-1} + B_N z_{N-3}, \quad \|r_N\| = O(N^{-p}), \quad (23.1)$$

and

$$\langle g_N, z_N \rangle \longrightarrow q_0 = \kappa^{-1} \sqrt{I/2} > 0, \quad I = \int_0^\infty F(x)^2 dx. \quad (23.2)$$

Here is the required global justification. On $j \leq 2N^{1/2}$ the factorial product for D_N has $\log D_N^{-1} = O(j^2/N)$, so D_N^{-1} is uniformly bounded there. Taylor remainders in (22.3), the time dilations and the finite amplitude ratios are bounded by an arbitrarily prescribed power of $N^{-1/3}$ times a fixed polynomial–Airy envelope. All such envelopes are square-integrable. The coefficient denominators stay uniformly away from zero. Cutoff derivative errors, and differences between one- and three-step cutoff supports, are exponentially small in $N^{1/4}$, up to powers of N . The bounded gauge preserves this property. At the left boundary all fictitious samples vanish exactly, because every $G_k(0) = 0$; the upper moving boundary lies outside the cutoff for large N . These facts give the norm residual after summation.

For the normalization, $H_N/R_N \sim \kappa^{-1} N^{-1/6}$, and a parity-restricted Airy sample has norm asymptotic to $N^{1/6} \sqrt{I/2}$. Its direction tends to g_N , proving (23.2). Only the leading asymptotic of P_N has been used: the original approximate recurrence is divided by the exact R_N , so an all-orders expansion of the relaxed spectral product is unnecessary.

23.2 A bootstrap with vanishing limiting projection

Set

$$C_* = A_\infty/q_0, \quad E_N = u_N - C_* z_N = \alpha_N g_N + W_N, \quad W_N \perp g_N. \quad (23.3)$$

Equations (21.5) and (23.2) imply $\alpha_N \rightarrow 0$. The error recurrence is

$$E_N = \eta_N M_N E_{N-1} - B_N E_{N-3} - C_* r_N.$$

The same projection estimates as before imply

$$\begin{aligned} \|W_N\| &\leq (1 - cN^{-2/3}) \|W_{N-1}\| + CN^{-1} \|W_{N-3}\| \\ &\quad + CN^{-1}(|\alpha_{N-1}| + |\alpha_{N-3}|) + CN^{-p}, \\ |\delta_N| &\leq CN^{-1}(|\delta_{N-1}| + |\delta_{N-2}|) + CN^{-4/3}|\alpha_{N-1}| \\ &\quad + CN^{-1}(\|W_{N-1}\| + \|W_{N-3}\|) + CN^{-p}, \end{aligned} \quad (23.4)$$

where $\delta_N = \alpha_N - \alpha_{N-1}$. If $\alpha_N = O(N^{-q})$, $q \geq 0$, Lemma 20.1 gives

$$\|W_N\| = O(N^{-q-1/3} + N^{2/3-p}), \quad \delta_N = O(N^{-q-4/3} + N^{-p}). \quad (23.5)$$

The latter is summable. Crucially, summing the differences to infinity uses the known zero limit and yields

$$\alpha_N = O(N^{-q-1/3} + N^{1-p}). \quad (23.6)$$

Begin with $q = 0$, since u_N and z_N are bounded, and iterate finitely many times. The version of Lemma 20.1 for sums of powers handles the residual floor throughout. Thus

$$\|E_N\| = O(N^{1-p}). \quad (23.7)$$

Neither positivity nor contractivity of the signed delayed propagator is assumed. The argument uses the gap for the unperturbed Jacobi evolution, smallness of the delay, and the zero limiting scalar projection.

24 The endpoint and the common positive amplitude

Point evaluation has norm one on ℓ^2 , so $|E_N(0)| \leq \|E_N\| = O(N^{1-p})$. For even N , the leading normalized approximate endpoint is of order $N^{-1/2}$, because

$$H_N/R_N \sim \kappa^{-1}N^{-1/6}, \quad \Phi_N(\varepsilon) \sim F'(0)\varepsilon.$$

We may choose p arbitrarily large. For the relative remainder in (16.3), it is enough to take $p \geq 3/2 + (M+1)/3$ and retain enough formal terms. Thus (23.7) supplies every fixed finite relative endpoint order; no endpoint smoothing estimate for the signed kernel is needed.

If $A_\infty = 0$, then $C_* = 0$. Equation (23.7) with $p > 3/2$ would imply $u_N(0) = o(N^{-1/2})$. On the other hand, the known lower bound (16.2), divided by R_{2n} , implies $u_{2n}(0) \geq cN^{-1/2}$ for large even $N = 2n$. This contradiction proves $A_\infty > 0$.

Since $D_N(0) = 1$ and $c_n = n!e_{2n,0}$, conversion from N to $2n$ gives

$$\gamma_c = 2^{3/4}C_*F'(0) = 2^{7/4}\kappa A_\infty > 0, \quad A_\infty = \lim_{N \rightarrow \infty} \left\langle g_N, v_N / (P_N N^{-1/4}) \right\rangle. \quad (24.1)$$

We used the Airy identity $F'(0)/\sqrt{I} = \sqrt{2}$, proved in the appendix. Every truncation has the same q_0 and the same leading normalization of H_N . Hence C_* and γ_c are independent of the expansion order. This completes the analytic part of Theorem 16.1.

25 Explicit coefficients and their rational field

The normalized endpoint factor begins

$$\frac{\Phi_N(\varepsilon)}{F'(0)\varepsilon} = 1 + \frac{a}{3}\varepsilon^2 + \frac{13}{12}\varepsilon^3 + O(\varepsilon^4).$$

Combining this with (22.8), then setting $N = 2n$, gives (16.4). Exponentiation gives

$$\begin{aligned} \mathfrak{c}_1 &= \frac{53z^2}{90}, \\ \mathfrak{c}_2 &= \frac{z(2809z^3 + 20325)}{16200}, \\ \mathfrak{c}_3 &= \frac{2084278z^6 + 43365690z^3 + 21487275}{61236000}. \end{aligned} \tag{25.1}$$

These coefficients were independently checked by a direct polynomial linear-system solve, separate from the triangular recursion.

The formal construction first gives coefficients in $\mathbb{Q}[a]$. To obtain $\mathbb{Q}[z]$ after the change of time scale, let ω be a primitive cube root of unity. For general complex a , define ϕ_a by the entire initial-value problem

$$\phi_a''(x) = 2(x + a)\phi_a(x), \quad \phi_a(0) = 0, \quad \phi_a'(0) = 1.$$

At the physical parameter $a = 2^{-1/3}z$ it equals $F/F'(0)$. At other parameters it need not be a decaying Airy function. Uniqueness of this initial-value problem gives

$$\phi_{\omega a}(\omega x) = \omega \phi_a(x).$$

The full formal recurrence, including its delayed dilation and both previous ratios, is invariant under $(a, \varepsilon, x) \mapsto (\omega a, \omega \varepsilon, \omega x)$. Uniqueness of (22.4)–(22.5) therefore implies

$$s_m(\omega a) = \omega^{-m} s_m(a).$$

The finite logarithm and normalized endpoint inherit the same grading: every monomial a^r at order k has $r + k$ divisible by three. Therefore

$$a^r N^{-k/3} = 2^{-(r+k)/3} z^r n^{-k/3}$$

has a rational power of two as its multiplier. All final relative coefficients lie in $\mathbb{Q}[z]$, as asserted.

26 Inverse asymptotics and discrete thresholds

[write, 2 October 2026] The anchor (26.3), the corrections (26.5) and the bracket (26.9) are instances of theorems `p0:thm:lambert-core`, `p0:thm:perturbed-inversion` and `p0:thm:staircase` of the repository's transseries volume (Part 0). No novelty is claimed for the inversion mechanics.

26.1 The smooth logarithmic model

Define

$$\begin{aligned} A &= 3z, & p &= \frac{5}{4}, & \mathcal{C} &= \log(\gamma_c \sqrt{2\pi}), \\ B &= \frac{53z^2}{90}, & D &= \frac{271z}{216}, & E &= \frac{1459}{3360} - \frac{1304z^3}{42525}. \end{aligned} \quad (26.1)$$

Here the p in this section is the fixed exponent $5/4$, rather than the freely chosen residual order above. Set

$$h(t) = t \log(4t/e) + At^{1/3} + p \log t + \mathcal{C} + Bt^{-1/3} + Dt^{-2/3} + Et^{-1}. \quad (26.2)$$

Stirling's formula and Theorem 16.1 give $\log c_n = h(n) + O(n^{-4/3})$. The factorial contributes both $\frac{1}{2} \log n$ and $1/(12n)$; omitting either changes the inverse. As $t \rightarrow \infty$, $h'(t) \sim \log(4t) > 0$ and $h''(t) \sim 1/t$.

26.2 The Lambert anchor and explicit corrections

For $T = \log Y$ put

$$x = \frac{T}{W_0(4T/e)}, \quad s = \log(4x), \quad q = p \log x + \mathcal{C}. \quad (26.3)$$

Then $x \log(4x/e) = T$. With

$$\begin{aligned} u &= -\frac{A}{s}, & v &= -\frac{q}{s}, \\ w &= -\frac{B}{s} + \frac{A^2}{3s^2} - \frac{A^2}{2s^3}, \\ r &= -\frac{D}{s} + \frac{A(p+q/3)}{s^2} - \frac{Aq}{s^3}, \\ k &= -\frac{E}{s} + \frac{pq}{s^2} - \frac{AB + q^2/2}{s^3} + \frac{A^3}{3s^4} - \frac{A^3}{2s^5}, \end{aligned} \quad (26.4)$$

the inverse of the explicit smooth model satisfies

$$h^{-1}(T) = x + ux^{1/3} + v + wx^{-1/3} + rx^{-2/3} + kx^{-1} + O(x^{-4/3}/s). \quad (26.5)$$

In particular,

$$h^{-1}(T) = x - \frac{3z}{s}x^{1/3} - \frac{5}{4} + \frac{(5/4)\log 4 - \mathcal{C}}{s} + O(x^{-1/3}/s). \quad (26.6)$$

The Airy displacement is positive because $z < 0$. The prefactor/factorial shift is $-5/4$, and the amplitude first appears at order $1/\log x$.

For verification, write $t = x^{-1/3}$ and $\delta = u/t + v + wt + rt^2 + kt^3$. The successive coefficients of $h(x + \delta) - T$, from t^{-1} through t^3 , are

$$\begin{aligned} su + A, & \quad sv + q, \\ sw + u^2/2 + Au/3 + B, \\ sr + uv + Av/3 + pu + D, \\ sk + v^2/2 + (u + A/3)w - u^3/6 - Au^2/9 + pv - Bu/3 + E. \end{aligned} \quad (26.7)$$

All vanish under (26.4). Since $q = O(s)$, $u = O(s^{-1})$, $v = O(1)$ and $w, r, k = O(s^{-1})$, the remaining residual is $O(x^{-4/3})$. Division by the local derivative $\sim s$ proves (26.5).

26.3 Newton inversion and higher finite orders

Starting at $x_0 = x$, use

$$x_{j+1} = x_j - \frac{h(x_j) - T}{h'(x_j)}.$$

One Newton step squares the error and multiplies it by $O((xs)^{-1})$. The initial error is $O(x^{1/3}/s)$, so in general

$$x_j - h^{-1}(T) = O\left(x^{1-2^{j+1}/3} s^{1-2^{j+1}}\right).$$

In particular two steps give $O(x^{-5/3}s^{-7})$, smaller than the forward uncertainty $O(x^{-4/3}/s)$. One may retain $\log \Gamma(t+1)$ instead of truncating Stirling, with the corresponding changes to (26.2). Higher forward orders produce higher smooth models. A logarithmic forward error $O(n^{-\rho})$ becomes an inverse uncertainty $O(x^{-\rho}/\log x)$; enough Newton steps make numerical inversion error smaller than any prescribed finite forward order.

These statements invert explicit smooth models. They never differentiate an unknown remainder of the integer sequence, and do not define an intrinsic real interpolation of the combinatorial counts.

26.4 The exact discrete interpretation

Define

$$N_c(Y) = \min\{n : c_n \geq Y\}.$$

The forward expansion implies eventual strict monotonicity, since

$$\log c_{n+1} - \log c_n = \log(4n) + O(n^{-2/3}) > 0.$$

Suppose $|\log c_n - h(n)| \leq Kn^{-4/3}$ for $n \geq n_0$, and let $r_* = h^{-1}(\log Y)$. For all sufficiently large Y , put

$$\eta = \frac{8 \max(K, 1) r_*^{-4/3}}{\log(4r_*)}. \quad (26.8)$$

Then

$$\lceil r_* - \eta \rceil \leq N_c(Y) \leq \lceil r_* + \eta \rceil. \quad (26.9)$$

Indeed evaluate the integer immediately below the lower ceiling and the integer at the upper ceiling. Near r_* the derivative of the model is at least $\frac{1}{2} \log(4r_*)$, and the forward error at most $2Kr_*^{-4/3}$. The stated displacement gives a strict margin in both directions. Eventual monotonicity completes the argument; any finite initial segment is irrelevant for sufficiently large Y .

When $2\eta < 1$, at most two adjacent integers remain. If $\text{dist}(r_*, \mathbb{Z}) > \eta$, both ceilings agree and give the exact threshold. Replacing r_* by (26.5) or a Newton iterate requires enlarging η by the model-inversion error. A bare ceiling of a truncated asymptotic is not uniformly valid near integer boundaries.

An unspecified asymptotic K proves an eventual theorem, not a certified finite- Y algorithm. Certification also needs effective forward constants, a rigorous amplitude enclosure, and controlled numerical evaluation errors. The fitted decimal for γ_c provides none of these by itself.

26.5 What already follows from the published bounds

The two-sided estimate (16.2) already implies, for some $K' > 0$ and large Y ,

$$\left[x - \frac{3z}{s}x^{1/3} - \frac{5}{4} - \frac{K'}{\log x} \right] \leq N_c(Y) \leq \left[x - \frac{3z}{s}x^{1/3} - \frac{5}{4} + \frac{K'}{\log x} \right]. \quad (26.10)$$

The logarithms of the unknown multiplicative constants are $O(1)$, which becomes $O(1/\log x)$ after inversion. Thus the coarse localization is not new to the amplitude theorem. The new result identifies the fixed amplitude term and permits the finer algebraic corrections and arbitrarily narrow eventual brackets.

Replacing γ_c by $\tilde{\gamma}$ changes the smooth inverse, to first order, by

$$-\frac{\log(\tilde{\gamma}/\gamma_c)}{\log(4n)}.$$

A fixed bounded amplitude error tends to zero in the inverse, yet still exceeds the algebraic precision in (26.5) and can change a threshold near an integer.

27 Cancellation in the compacted to relaxed ratio

The independently checked companion relaxed-tree theorem (Part I, (F1)–(F2); source 15 cites the companion article here) has positive amplitude γ_r (the γ of Part I) and the logarithmic expansion

$$\begin{aligned} \log \frac{r_n}{\gamma_r n! 4^n e^{3zn^{1/3}} n} &= \frac{53z^2}{90} n^{-1/3} + \frac{44z}{27} n^{-2/3} \\ &+ \left(\frac{141}{140} - \frac{1304z^3}{42525} \right) n^{-1} + O(n^{-4/3}). \end{aligned} \quad (27.1)$$

Using that companion theorem, subtraction from (16.4) gives the algebraic corollary

$$\log \frac{c_n}{r_n} = \log \frac{\gamma_c}{\gamma_r} - \frac{1}{4} \log n - \frac{3z}{8} n^{-2/3} - \frac{21}{32} n^{-1} + O(n^{-4/3}), \quad (27.2)$$

and hence

$$\frac{c_n}{r_n} = \frac{\gamma_c}{\gamma_r} n^{-1/4} \left(1 - \frac{3z}{8} n^{-2/3} - \frac{21}{32} n^{-1} + O(n^{-4/3}) \right). \quad (27.3)$$

The $n^{-1/3}$ correction cancels exactly. Since $z < 0$, the first surviving correction is positive. No certified numerical value of the amplitude ratio is supplied.

For the corresponding explicit smooth logarithmic models, let $n_r = h_r^{-1}(T)$ and $n_c = h_c^{-1}(T)$, using their true amplitudes. Subtracting the models and applying the mean-value theorem yields

$$n_c - n_r = \frac{\frac{1}{4} \log n_r - \log(\gamma_c/\gamma_r)}{\log(4n_r)} + O(n_r^{-2/3}/\log n_r). \quad (27.4)$$

This real-model inverse gap tends to $1/4$. Integer threshold differences are subject to rounding and need not converge. The exact thresholds satisfy

$$N_r(Y) \leq N_c(Y) \leq N_r(Y) + 1 \quad (27.5)$$

for sufficiently large Y : use $c_n \leq r_n$ and eventual $c_{n+1} > r_n$. The latter follows already from the published two-sided estimates, since c_{n+1}/r_n grows at least as a positive multiple of $n^{3/4}$. Thus (27.5) is also a consequence of the known bounds, rather than a new amplitude-level conclusion.

28 Reproducibility and numerical limitations

The accompanying reproducibility archive contains the exact finite-state checks, coefficient construction, independent coefficient calculation, gauge and delay checks, inverse algebra, and exact forward recurrence. The computations require Python with SymPy, mpmath, NumPy and SciPy. Run `python replay.py` from the archive root. It verifies its SHA-256 manifest, runs all seven checks under `output/mathematical-replay/`, compares independent coefficients and the exact-state fixture, and checks that its inputs remain unchanged. The option `-manifest-only` checks input integrity without running the mathematical checks. The README records the environment and dependencies; complete approved relaxed proof sources and the coefficient-field supplement are included under `dependencies/relaxed/`. The article source can be rebuilt with `bash build_pdf.sh` and a TeX distribution providing pdfLaTeX and the packages listed in its preamble. In the archive this writes only beneath `output/article/`; beside a standalone article it writes to the local build directory and replaces the adjacent PDF.

[**write, 2 October 2026**] This paragraph describes source 15's delivered archive (commit 096ee7b87). Its `dependencies/relaxed/` files are earlier versions of Part I's text, of its review (shipped as `62-relaxed-verif-mathematical-review.md`) and of its Section 12; they are not shipped. The build script and the manuscript are not shipped either (this report is built as its README says), and the delivered README wrote the build command as `basharticle/build_pdf.sh`. The README of this report lists the shipped files.

The coefficient algorithm has no fitted input:

1. Represent each profile by the polynomial pair (P, Q) for $PF + QF'$.
2. Expand the rational coefficients, time dilations and previous ratios to the required degree, reducing derivatives with $F'' = 2(x + a)F$.
3. Solve the triangular polynomial equation (22.5), choose the scalar to enforce $Q(0) = 0$, and choose the remaining integration constant to enforce $P(0) + Q'(0) = 0$.
4. Match the finite logarithmic amplitude and expand the endpoint factor; then convert $N = 2n$ and $a = 2^{-1/3}z$.
5. Verify the residual coefficient, both boundary conditions and the mod-three grading exactly. An independent linear-system implementation checks the displayed initial coefficients.

Exact computation finds agreement among the signed, four-state and three-state recurrences on all 5151 cells $0 \leq m \leq n \leq 100$. The scalar renewal gives the same diagonal and all V -ending entries. Direct, nonmemoized enumeration of 709506 accepted labeled prefixes with at most seven horizontal steps agrees state by state with the four-state kernel; 13317 attempted vertical moves are rejected. These are finite checks of models whose equality was proved above.

The forward computation uses exact integer compacted counts through $n = 2500$. Removing the first three logarithmic corrections at that point gives an amplitude estimate of approximately 173.121045803661366, consistent with the published fit 173.12670485. Neither number is a rigorous enclosure, and the final displayed digits are not established digits of γ_c .

Symbolic checks verify finite algebraic identities; numerical checks verify the reproducibility of numerical evidence. Neither replaces the compactness, stability, or endpoint arguments. The amplitude in the theorem is (24.1), not a decimal fit.

29 Further questions

Certified amplitude evaluation. An effective version of the Airy gap and explicit residual and tail constants could turn (24.1) into an interval algorithm. The present argument establishes convergence and algebraic rates without furnishing all constants needed for a finite numerical certificate.

Positive effective operators. The exact three-state and renewal representations remain useful for probabilistic or nonselfadjoint analysis. The obstructions above specify what a direct symmetrization cannot do; they do not preclude a controlled asymptotic elimination of fast internal modes.

Higher arity and beyond algebraic orders. Higher-arity compacted recurrences require their own transfer analysis. The present fixed-order construction gives no bound uniform in the expansion order, no Gevrey estimate, and no exponentially small correction. Those questions require additional information on coefficient growth and higher Airy modes.

[write, 2 October 2026] For every fixed arity $k \geq 3$ the compacted recurrence is treated in Part III (source 18), as a finite-memory perturbation with a summable delay; the binary case, where the delay is not summable, is the one treated here. The questions of uniformity in the order, Gevrey bounds and exponentially small corrections remain open for every k .

Certified inverse thresholds. Combining interval evaluation of the amplitude with effective forward error bounds and exact neighboring counts would resolve the occasional two-integer ambiguity in (26.9).

Part III

Fixed arity $k \geq 3$

Source. Three manuscripts. (1) Batch-77 manuscript 18, archive `fixed-arity-airy-package.zip`, main file `report/fixed-arity-airy-expansions.tex` (515 lines; 12-page PDF), titled “Airy expansions for fixed arity trees and finite language automata”, author line “Research report”, dated 2 October 2026: Sections 30–37, label prefix `air:ka:.` (2) Batch-77 manuscript 66, archive `ternary-airy-amplitudes-result.zip`: the ternary section, Sections 38–46, label prefix `air:t3:.` (3) Batch-77 manuscript 29, archive `larger-alphabet-automata-report.zip`, handed over from cluster P4: the comparison section, Sections 47–52, label prefix `air:ka:al:.` All three are printed in full.

[write, 2 October 2026] Source 18’s leading proof extends source 66’s ternary proof. Its detailed proof note (shipped as `18-arity-fixed-arity-proof.md`) says it “extends the independently approved ternary fixed-weight proof” and names as its base a producer workspace path, `/workspace/shared/ternary-airy-amplitude-research/amplitude-proof-candidate.md`, which is not shipped under that name. The file is source 66’s `amplitude-proof-candidate.md`, shipped as `66-ternary-amplitude-proof-candidate.md`; source 18’s independent analytic audit pins it by SHA-256 (`18cdbc4e...`). Source 18’s all-orders proof note (`18-arity-all-orders-proof.md`) says that the structure of its error bootstrap is that of source 66’s `all-orders-reduction.md` (shipped as `66-ternary-all-orders-reduction.md`). The ternary section below prints source 66 as that base.

Abstract of the source.

For every fixed integer arity $k \geq 3$, we obtain positive asymptotic amplitudes and expansions to every fixed finite order for relaxed trees, compacted trees, and minimal deterministic finite automata accepting finite languages. The first two relative corrections are explicit and common to all three models. We determine the first model-dependent logarithmic ratio corrections, prove eventual decrease of the normalized ratios, and give an inverse-count expansion with the necessary interpolation qualification. The method combines exact boundary harmonics, a fixed weighted singular Airy limit, polynomial–Airy recursion, and a finite-memory perturbation argument. The leading amplitudes are positive but are not evaluated explicitly. No assertion is uniform as k grows, and no convergence or exponentially accurate completion of the infinite formal expansion is claimed.

30 Results and indexing

Fix $k \geq 3$, write $q = k - 1$, and let $a = a_1 < 0$ be the largest zero of Ai. Set

$$B = (2/q)^{1/3}, \quad \lambda = a/B, \quad \gamma = 3(kq/2)^{1/3}a, \quad \eta = (2k - 1)/3, \quad (30.1)$$

and define the common relaxed scale

$$\mathcal{G}_{k,n} = (n!)^q \left(\frac{k^k}{q^q} \right)^n e^{\gamma n^{1/3}} n^\eta. \quad (30.2)$$

Let $R_n = r(qn, n)$ and $C_n = c(qn, n)$ count relaxed and compacted k -ary trees, respectively. Let $B_n = b(qn, n)$ use the source DFA normalization: n transient states, together with one rejecting sink. The auxiliary source value specified below is part of this normalization.

For ternary relaxed trees, A082162 has offset 1 and begins 1, 7, 139, 5711, $\dots$; our additional convention is $R_0 = 1$. For arity four, A102102 has offset 0 and begins 1, 1, 15, 1000, 189035, $\dots$. The array A128249 uses alphabet size equal to its column index plus one, so arity k corresponds to column $k - 1$; its sequence is read antidiagonally. These identifiers describe relaxed counts, not the distinct minimal DFA counts.

Theorem 30.1 (All finite orders). *There are constants $A_k^R, A_k^C, A_k^B > 0$ and real coefficients $c_{k,r}^X$, $X \in \{R, C, B\}$, such that for each fixed $M \geq 0$,*

$$R_n = A_k^R \mathcal{G}_{k,n} \left[1 + \sum_{r=1}^M c_{k,r}^R n^{-r/3} + O_{k,M}(n^{-(M+1)/3}) \right], \quad (30.3)$$

$$C_n = A_k^C \mathcal{G}_{k,n} \left[1 + \sum_{r=1}^M c_{k,r}^C n^{-r/3} + O_{k,M}(n^{-(M+1)/3}) \right], \quad (30.4)$$

$$B_n = A_k^B 2^n \mathcal{G}_{k,n} \left[1 + \sum_{r=1}^M c_{k,r}^B n^{-r/3} + O_{k,M}(n^{-(M+1)/3}) \right]. \quad (30.5)$$

Each model has one and the same amplitude for every M . All three share their first two corrections. With

$$h_1 = \frac{\lambda^2(3q^2 + 27q + 23)}{45q}, \quad (30.6)$$

these are

$$c_{k,1}^X = k^{-1/3} h_1, \quad (30.7)$$

$$c_{k,2}^X = k^{-2/3} \left[\frac{h_1^2}{2} + \frac{\lambda(4q^3 + 321q^2 + 429q + 126)}{270q} \right]. \quad (30.8)$$

The published theorem of Dastidar and Wallner [3] supplies the relaxed Θ scale, including its positive lower bound. The equivalences, finite-order expansions, and comparisons here are additional arguments; they are not attributed to that published theorem. Finite-order expansion means a separate controlled remainder for every fixed M , not convergence of the resulting infinite series.

[**write, 2 October 2026**] At $k = 2$ the formulas (30.7)–(30.8) reduce exactly to the relaxed binary coefficients c_1, c_2 of Part I, (F3) (checked symbolically when this report was written; source 18’s proof note `18-arity-all-orders-proof.md` observes it for $c_{k,1}$). The theorem itself is claimed only for $k \geq 3$, and at $k = 2$ it fails as stated for the compacted and automaton models: their powers are $n^{3/4}$ (Part II) and $n^{7/8}$ (Part IV), not $n^{(2k-1)/3} = n$, and their second corrections differ from the relaxed one. See “Where the method changes” in Part 0.

Theorem 30.2 (First model-dependent defects). *The normalized limits*

$$\rho_C = \frac{A_k^C}{A_k^R} = \lim \frac{C_n}{R_n} > 0, \quad \rho_B = \frac{2A_k^B}{A_k^R} = \lim \frac{B_n}{2^{n-1}R_n} > 0$$

obey

$$\log \frac{C_n}{\rho_C R_n} = \frac{(q/k)^k}{k-2} n^{-(k-2)} + O_k(n^{-(k-2)-1/3}), \quad (30.9)$$

$$\log \frac{B_n}{\rho_B 2^{n-1} R_n} = \frac{(q/k)^k}{2(k-2)} n^{-(k-2)} + O_k(n^{-(k-2)-1/3}). \quad (30.10)$$

Thus all coefficients with $r < 3(k-2)$ agree with the relaxed ones. At $r = 3(k-2)$ the compacted coefficient increases by $(q/k)^k/(k-2)$ and the DFA coefficient by half that amount. Both normalized ratios approach their limits from above and are eventually strictly decreasing.

For $k = 3$, the common corrections are

$$c_1 = \frac{89a^2}{90 \cdot 3^{1/3}}, \quad c_2 = 3^{-2/3} \left(\frac{7921a^4}{16200} + \frac{115a}{27} \right),$$

and the logarithmic ratio defects begin with $8/(27n)$ for compacted trees and $4/(27n)$ for DFAs.

[**write, 2 October 2026**] These are the c_1, c_2 of source 66’s Theorem 38.2 (symbolic difference zero). Source 66 also gives the relaxed c_3 at $k = 3$, which this section does not.

31 Exact arrays and boundary harmonics

The source recurrences, for $m \geq 1$ and $x \geq qm$, are

$$r(x, m) = r(x, m-1) + (m+1)r(x-1, m), \quad (31.1)$$

$$c(x, m) = c(x, m-1) + (m+1)c(x-1, m) - (m-1)c(x-k, m-1), \quad (31.2)$$

$$b(x, m) = 2b(x, m-1) + (m+1)b(x-1, m) - mb(x-k, m-1). \quad (31.3)$$

Their ordinary base rows are $r(x, 0) = c(x, 0) = b(x, 0) = 1$ for $x \geq 0$. Values below the physical diagonal are zero, apart from the DFA auxiliary $b(-1, 0) = 1$. Other missing values are zero. In particular,

$$b(q, 1) = 1, \quad b(x, 1) = 2^{x-q+1} - 1 \quad (x \geq q).$$

The last identity follows from $b(x, 1) = 2b(x-1, 1) + 1$ after its first site. The compacted first row agrees with the relaxed row because $m-1 = 0$ there.

For the relaxed model introduce

$$i = x + m, \quad j = x - qm, \quad x = (qi + j)/k, \quad d_i(j) = \frac{q^{2x}}{x!} r(x, m).$$

Then $d_0(0) = 1$ and, on $J_i = \{0 \leq j \leq i : j \equiv i \pmod{k}\}$,

$$(T_i u)(j) = U_i(j)u(j-1) + u(j+q), \quad U_i(j) = \frac{q^2(i-j+k)}{qi+j} = q - \frac{qk(j-q)}{qi+j}. \quad (31.4)$$

Missing inputs are zero. The endpoint conversion is exact:

$$R_n = \frac{(qn)!}{q^{2qn}} d_{kn}(0). \quad (31.5)$$

Let T denote the infinite critical map with up coefficient q and down coefficient 1. Set $h_j = j+1$ and define l_j by

$$\sum_{j \geq 0} l_j z^j = \frac{qk/2}{q - kz + z^k}. \quad (31.6)$$

Then $Th = kh$ and $T^*l = kl$, including the bottom equations, with $l_j = 0$ at negative sites. The exact fixed norm is

$$\|u\|_\omega^2 = \sum_j \omega_j |u_j|^2, \quad \omega_j = \frac{l_j}{j+1}, \quad \pi_j = l_j(j+1).$$

The weighted adjoint also satisfies $T^\dagger h = kh$.

Lemma 31.1 (Positive fixed weight). *For fixed k , the weight is positive and bounded above and below. More precisely,*

$$l_j = j + \frac{q+2}{3} + E_j, \quad \omega_j = 1 + \frac{q-1}{3(j+1)} + \frac{E_j}{j+1},$$

where every fixed-order finite difference of E_j decays exponentially. Hence $|\Delta^m(\omega_j - 1)| \leq C_{k,m}(j+1)^{-m-1}$.

Proof. Factor

$$q - kz + z^k = (1 - z) \left(q - \sum_{m=1}^q z^m \right) = (1 - z)^2 \sum_{r=0}^{q-1} (q - r) z^r.$$

The middle factor has no zero in $|z| < 1$; equality in the triangle inequality on the unit circle forces $z = 1$. That zero is simple, so all zeros of the last polynomial lie outside the unit disk. Its value and derivative at 1 are $qk/2$ and $qk(q-1)/6$, which give the displayed expansion. Positivity follows from the renewal generating function

$$\frac{k/2}{(1 - z)(1 - q^{-1} \sum_{m=1}^q z^m)}.$$

Indeed if the increments are uniform on $\{1, \dots, q\}$ and τ_j is their first passage time above j , then $l_j = (k/2)\mathbb{E}\tau_j = \mathbb{E}S_{\tau_j}$. Here $\tau_j \leq j + 1$, so the stopped-sum identity is a finite sum of independent-increment expectations. Thus $j + 1 \leq l_j \leq j + q$; in particular $1 \leq \omega_j \leq q$. $\square$

32 Positive forms and the singular Airy limit

For $g_j = u_j/h_j$, exact conditional variance of the h -Doob kernel gives

$$k^2 \|u\|_\omega^2 - \|Tu\|_\omega^2 = \sum_{\substack{j \geq 1 \\ \text{output phase}}} q\omega_j j(j+k) |g_{j-1} - g_{j+q}|^2. \quad (32.1)$$

On an input phase r , put $H_m = r + km + 1$. Direct telescoping gives

$$\sum_{m \geq 0} H_m H_{m+1} \left| \frac{u_{m+1}}{H_{m+1}} - \frac{u_m}{H_m} \right|^2 = \sum_{m \geq 0} |u_{m+1} - u_m|^2 + \frac{k|u_0|^2}{r+1}. \quad (32.2)$$

Both formulas include the terminal edge to zero for a finite-support critical vector.

For the physical transfer define

$$\alpha_i = 1 + \frac{q^2 - 1}{qi + 1}, \quad Q_i(j, t) = \frac{T_i(j, t)h_t}{k\alpha_i h_j}, \quad r_i(j) = \sum_t Q_i(j, t), \quad c_i(t) = \pi_t^{-1} \sum_j \pi_j Q_i(j, t).$$

Active up edges have $j \geq 1$, and $U_i(j) \leq q\alpha_i$. Thus $r_i, c_i \leq 1$. Write $v_i(j) = q - U_i(j)$. Every physical input has its up output, so

$$1 - c_i(t) = \frac{\alpha_i - 1 + v_i(t+1)l_{t+1}/(kl_t)}{\alpha_i}. \quad (32.3)$$

At the extra bottom sites $0 \leq t \leq q-1$, this is exactly

$$1 - c_i(t) = \frac{kt(qi + q)}{\alpha_i(qi + 1)(qi + t + 1)}.$$

For all physical $t \geq 1$ it is at least $c_k t/i$. The negative values of v_i at some bottom sites are therefore accounted for, not discarded.

With $p = Q_i(j, j-1)$ and $b = Q_i(j, j+q)$, assigning zero probability to missing inputs, the positive identity is

$$\begin{aligned} \frac{k^2 \alpha_i^2 \|u\|_\omega^2 - \|T_i u\|_\omega^2}{k^2 \alpha_i^2} &= \sum_t \pi_t (1 - c_i(t)) |g_t|^2 \\ &\quad + \sum_j \pi_j p b |g_{j-1} - g_{j+q}|^2 \\ &\quad + \sum_j \pi_j (1 - r_i(j)) (p |g_{j-1}|^2 + b |g_{j+q}|^2). \end{aligned} \quad (32.4)$$

This follows from the two-point variance identity. All three terms are nonnegative, and replacing α_i^2 by 1 changes the form only by $O_k(i^{-1}) \|u\|^2$.

Proposition 32.1 (Compact singular Airy limit). *For each fixed m , the m th largest singular value in the fixed weighted phase norms satisfies*

$$\sigma_{i,m} = k \left[1 + \frac{a_m}{B} i^{-2/3} + o_k(i^{-2/3}) \right],$$

where a_m are the Airy zeros in decreasing order. The leading right singular vector converges, under mesh interpolation, to the positive normalized $\text{Ai}(a_1 + Bx)$.

Proof. Put $\varepsilon = i^{-1/3}$ and divide the unshifted defect by $2k^2\varepsilon^2$. For $x = \varepsilon(j+1)$, take $u_j = \sqrt{k}\varepsilon^{3/2}h_jG(x)$ and $F = xG$. Riemann sums in (32.4) give

$$\|u\|^2 \longrightarrow \int |F|^2, \quad \frac{k^2 \|u\|^2 - \|T_i u\|^2}{2k^2\varepsilon^2} \longrightarrow \int \left(\frac{q}{2} |F'|^2 + x |F|^2 \right).$$

The variance supplies $q/2$ times the radial energy; column and row losses each supply half the potential because $v_i(j) = kx\varepsilon^2 + O_k(\varepsilon^3)$ away from zero.

Bounded forms give the tail bound $\sum_{\varepsilon j \geq R} \omega_j |u_j|^2 \leq C_k/R$. To control the bottom, first localize with a cutoff supported in $x \leq 2R$ and equal to one on $x \leq R$. This costs $O_{k,R}(\varepsilon^2) \|u\|^2$ in the unscaled defect. On its support the variance coefficients are comparable to the critical ones. Now (32.2) applies, including its zero terminal edge, since the localized vector is far below the physical top. It bounds $\sum |\Delta u|^2 + |u_r|^2$ by $C_{k,R}\varepsilon^2$. The interpolation of $u/\sqrt{k\varepsilon}$, extended to zero at $x = 0$, has a bounded local H^1 norm and zero trace. Rellich compactness, the negligible mass near zero from this bound, and the tail estimate give strong L^2 compactness.

For the liminf retain the nonnegative terms on $\delta \leq x \leq R$, pass by weak lower semicontinuity and strong L^2 , then let $\delta \downarrow 0$ and $R \uparrow \infty$. The limiting radial domain is the Dirichlet form domain: the identity $\int x^2 |(F/x)'|^2 = \int |F'|^2$ holds by cutoff approximation, with Hardy's inequality controlling zero. Compactly supported smooth functions give recovery sequences. Min-max therefore gives convergence of each fixed low eigenvalue to that of $-(q/2)D^2 + x$ on the half-line with zero boundary. Those eigenvalues are $-a_m/B$. Simplicity gives ground-state convergence. $\square$

33 Evolving profiles and a positive amplitude

Let $f(x) = \text{Ai}(a + Bx)$ and

$$p(x) = -\frac{q+3}{6q}x^2 - \frac{(q-1)\lambda}{3q}x, \quad \beta = \frac{7k-6}{6}.$$

Fix a sufficiently large starting time I , so all scalar factors and profiles below are defined and positive where required; the earlier evolution is absorbed into finite initial data. Choose a smooth cutoff equal to one up to $x = \log i$ and zero after $2 \log i$, and let

$$F_i(j) = \chi_i(x)[f(x) + \varepsilon p(x)f(x)], \quad N_i = \|F_i\|_\omega, \quad \psi_i = F_i/N_i,$$

$$t_i = k(1 + \lambda i^{-2/3} + \beta i^{-1}), \quad s_i = t_i N_i / N_{i-1}.$$

The exact previous-time dilation is $(i-1)^{-1/3} = \varepsilon(1 + \varepsilon^3/3 + O(\varepsilon^6))$. Writing $g = pf$, the forward expansion is

$$\begin{aligned} T_i F_{i-1} &= kf + k\varepsilon g + k\varepsilon^2((q/2)f'' - xf) \\ &\quad + k\varepsilon^3 \left[\frac{q(q-1)}{6} f''' + \frac{4}{3} x f' + kf + \frac{q}{2} g'' - xg \right] + O_k(\varepsilon^4 W(x)). \end{aligned}$$

Here W is a square-integrable polynomial Airy envelope allowing $|f'|$ as well as $|f|$; it need not vanish at zero. Substitution of $f'' = (2/q)(x + \lambda)f$ and p cancels the f' coefficient and gives βf . Thus

$$\|T_i \psi_{i-1}/s_i - \psi_i\| = O_k(i^{-4/3}), \quad \left\| T_i^\dagger \psi_i/s_i - \psi_{i-1} \right\| = O_k(i^{-1}). \quad (33.1)$$

For clarity, the adjoint estimate retains the exact cancellation $T^\dagger h = Th$. At $t \geq q$ the difference $T^\dagger - T$ has Taylor moments $M_0 = O(h^{-3})$, $M_1, M_2 = O(h^{-2})$, and $hM_0 + M_1 = 0$. Subtracting $f'(0)\varepsilon h$ leaves a profile and derivatives bounded by $C_k \varepsilon^3(h^3 + h^2)$ and its derivative analogues near zero. The finitely many bottom sites use the same exact linear cancellation directly. In the bulk, Airy bounds apply. This proves the second estimate in (33.1); separately estimating the two weight errors would not suffice.

Euler–Maclaurin on each phase, using the exact algebraic-plus-exponential decomposition of ω , yields

$$N_{r,\varepsilon}^2 = C_{-1}\varepsilon^{-1} + C_0 + C_1\varepsilon + O_k(\varepsilon^2), \quad C_{-1} = \frac{1}{k} \int f^2 > 0,$$

with the three written constants independent of phase. The squared profile and its first derivative vanish at zero; the algebraic-weight quotient also vanishes there. Hence phase dependence first enters at absolute order ε^2 . In particular,

$$N_i/N_{i-1} = 1 + O_k(i^{-1}), \quad N_i \asymp_k i^{1/6}, \quad \psi_i(0) \asymp_k i^{-1/2}.$$

Normalize by $A_i = T_i/s_i$ and decompose $y_i = a_i \psi_i + z_i$, $z_i \perp \psi_i$. The residuals give off-diagonal blocks of sizes $O(i^{-1})$ and $O(i^{-4/3})$, and a central entry $1 + O(i^{-4/3})$. Proposition 32.1 gives the stable block bound $1 - c_k i^{-2/3}$: on $\psi_{i-1}^\perp$ the squared Rayleigh quotient is at most $\sigma_{i,2}^2 + (\sigma_{i,1}^2 - \sigma_{i,2}^2)o(1)$.

The quantity $|a_i| + K i^{-1/3} \|z_i\|$ is bounded for sufficiently large K . Stable convolution then gives $\|z_i\| = O(i^{-2/3})$ and $a_i = a_\infty + O(i^{-1/3})$. At residue-zero endpoints, $z_i(0)/\psi_i(0) = O(i^{-1/6})$. Finally,

$$\prod_{m=I+1}^i t_m = \text{const} \cdot k^i e^{3\lambda i^{1/3}} i^\beta (1 + O_k(i^{-1/3})), \quad F_i(0) \sim B \text{Ai}'(a) i^{-1/3}.$$

Norm factors telescope. Conversion (31.5) and Stirling give the scale (30.2). The published positive lower bound for this exact relaxed recurrence forces its finite limiting amplitude to be positive.

34 Recursion and transfer to all finite orders

Set $\tau = (1 - \varepsilon^3)^{-1/3}$ and

$$U(\varepsilon, x) = \frac{q^2[1 - x\varepsilon^2 + (k+1)\varepsilon^3]}{q + x\varepsilon^2 - \varepsilon^3}.$$

For $\Phi = \sum_{m \geq 0} \varepsilon^m \phi_m$ and scalar step series $\sigma(\varepsilon) = k + \sum_{r \geq 2} \sigma_r \varepsilon^r$, the exact formal evolving equation is

$$U\Phi(\varepsilon\tau, (x - \varepsilon)\tau) + \Phi(\varepsilon\tau, (x + q\varepsilon)\tau) = \sigma(\varepsilon)\Phi(\varepsilon, x). \quad (34.1)$$

Take $\phi_0 = f$, $\sigma_2 = k\lambda$, and require every $\phi_m(0) = 0$. At order ε^{m+2} the only new unknowns are ϕ_m and σ_{m+2} , with equation

$$kL_q\phi_m = \sigma_{m+2}f + P_m f + Q_m f', \quad L_q = (q/2)D^2 - x - \lambda.$$

All known forcing coefficients are polynomials. Writing $\phi_m = A_m f + D_m f'$, elimination reduces polynomial inversion to

$$\left[-\frac{q}{4}D^3 + 2(x + \lambda)D + 1\right]D_m = P - Q'/2, \quad A'_m = Q/q - D''_m/2. \quad (34.2)$$

The operator has diagonal $2d+1$ in polynomial degree d . Adding a constant to its right side changes only D_m 's constant term, so the new scalar uniquely enforces $D_m(0) = 0$. Choose $A_m(0) = 0$ to fix the amplitude gauge. This proves an obstruction-free recursion to every finite order.

For scalar matching write

$$H_i = k^i e^{3\lambda i^{1/3}} i^\beta \exp\left(\sum_{r=1}^J h_r i^{-r/3}\right)$$

with leading constant one. The new h_r first enters $\log(H_i/H_{i-1})$ at order ε^{r+3} with coefficient $-r/3$; all later coefficients can therefore be matched. The first two additional scalar stages are

$$\sigma_4 = -\frac{\lambda^2 k(6q^2 - 81q + 46)}{270q},$$

$$\sigma_5 = -\frac{\lambda k(8q^3 - 327q^2 + 657q - 92)}{810q}.$$

They give (30.6) and

$$h_2 = \frac{\lambda(4q^3 + 309q^2 + 531q - 46)}{270q}.$$

The endpoint profile begins

$$\Phi_i(0) = \varepsilon f'(0)(1 + e_2 \varepsilon^2 + \cdots), \quad e_2 = \frac{\lambda(6q^2 - 51q + 86)}{135q}.$$

There is no relative ε endpoint term. Stirling starts at ε^3 , so $c_1 = k^{-1/3}h_1$ and $c_2 = k^{-2/3}(h_1^2/2 + h_2 + e_2)$, as stated.

We give the analytic transfer, since formal matching alone is insufficient. For each $p > 1$, take sufficiently many complete stages and the logarithmic Airy cutoff. All denominators are bounded away from zero on its support; every fixed Taylor remainder is a power of ε times a square-integrable polynomial Airy envelope. Squared mesh sums cost $O(\varepsilon^{-1})$, exactly canceled by $N_i^2 \asymp \varepsilon^{-1}$. Every

bottom input is exactly zero and every top cutoff has a macroscopic margin. Cutoff errors decay faster than every power. Hence

$$\|T_i(H_{i-1}\Phi_{i-1}) - H_i\Phi_i\| \leq C_{k,p}H_iN_i i^{-p}.$$

Choose a fixed starting time L , let $P_i = \prod_{m=L}^i t_m$, $S_i = P_i N_i / N_{L-1}$, and $P_i / [k^i e^{3\lambda i^{1/3}} i^\beta] \rightarrow \kappa > 0$. Then

$$Z_i = \frac{\kappa}{N_{L-1}} \frac{H_i \Phi_i}{S_i}$$

has central projection tending to one for every truncation, and normalized defect $O(i^{-p})$. This fixes a common amplitude.

Subtract $a_\infty Z_i$ from the exact normalized solution and write the error as $\alpha_i \psi_i + w_i$, with $\alpha_i \rightarrow 0$. The leading blocks give

$$\begin{aligned} |\alpha_i - \alpha_{i-1}| &\leq C i^{-4/3} |\alpha_{i-1}| + C i^{-1} \|w_{i-1}\| + C i^{-p}, \\ \|w_i\| &\leq (1 - c i^{-2/3}) \|w_{i-1}\| + C i^{-4/3} |\alpha_{i-1}| + C i^{-p}. \end{aligned}$$

If $\alpha_i = O(i^{-\nu})$, stable convolution gives $\|w_i\| = O(i^{-\nu-2/3} + i^{2/3-p})$. Backward summation from the zero central limit then improves this to $\alpha_i = O(i^{-\nu-1/3} + i^{1-p})$. Starting at $\nu = 0$ and iterating finitely many times proves total norm error $O(i^{1-p})$. Endpoint evaluation loses at most $i^{1/2}$, so arbitrary p gives every finite asymptotic order with the same amplitude.

35 Compacted trees and automata as finite memory perturbations

For compacted trees use $a(x, m) = c(x, m)$, and for DFAs use $a(x, m) = b(x, m)/2^m$. Transform a by the same factorial and exponential weights. Through time $i = k$, the diagonal entries are $d_i(i) = q^{2i}/i!$; before k these are the only entries. At time k the extra bottom entry is

$$d_k(0) = \frac{q^{2q}}{q!} \quad \text{for compacted trees,} \quad d_k(0) = \frac{q^{2q}}{2q!} \quad \text{for DFAs.}$$

The second value must not be replaced by the relaxed seed. For $i \geq k+1$ the exact recurrence is

$$d_i = T_i d_{i-1} - D_i d_{i-k-1}, \quad (D_i u)(j) = \frac{\delta_m q^{2k}}{(X)_k} u(j-1), \quad (35.1)$$

where $X = (qi + j)/k$, $m = (i - j)/k$, and $\delta_m = m - 1$ for compacted trees or $m/2$ for normalized DFAs. Define $\delta_0 = 0$. Missing delayed inputs are zero. Every denominator is nonzero in this time range; the auxiliary negative source has already been handled in the seed and is never assigned a factorial transform.

35.1 A positive comparison despite the signed recurrence

Write ℓ_m for the horizontal-run length at level m before its rise. The relaxed run weight is $V_m(\ell) = (m+1)^\ell$. The exact compacted completed-run weight is

$$V_m(\ell) \left(1 - \frac{m}{(m+1)^k} \mathbf{1}_{\ell \geq k} \right).$$

For normalized DFAs the first allowed run has weight $V_0(\ell)/2$ for every $\ell \geq q$, and later runs have weight

$$V_m(\ell) \left(1 - \frac{1}{2(m+1)^q} \mathbf{1}_{\ell \geq k} \right), \quad m \geq 1.$$

To verify these factors, remove the last k horizontals of a marked completed run before a rise. Its removed relaxed weight is $(m+1)^k$; the negative source coefficient at the next level is m or $(m+1)/2$. Removal preserves every preceding barrier constraint and gives exactly the signed source term. Expanding the product over marked runs reproduces the recurrence. For the shortest initial DFA run, $\ell = q < k$, the auxiliary seed produces the same half factor as the ordinary subtraction does when $\ell \geq k$. At the final diagonal endpoint there is no open terminal horizontal run.

Consequently,

$$\begin{aligned} 0 < P_C &:= \prod_{m \geq 1} \left(1 - \frac{m}{(m+1)^k} \right) \leq \frac{C_n}{R_n} \leq 1, \\ 0 < P_B &:= \prod_{m \geq 1} \left(1 - \frac{1}{2(m+1)^q} \right) \leq \frac{B_n}{2^{n-1}R_n} \leq 1. \end{aligned} \tag{35.2}$$

These products are positive because the defects are summable for fixed $k \geq 3$. They will force strictly positive limiting amplitudes.

[write, 2 October 2026] For automata, this comparison, with the finite- n constant $P_{k,n} = \prod_{j=2}^n (1 - 1/(2j^{k-1})) \geq P_B$ in place of P_B , is Theorem 47.1 of the comparison section (source 29), its earliest form; at $k = 3$ it is (44.1) of the ternary section. Here it is a step of the proof, and the run weights are verified combinatorially; source 29 verifies them through the completed-run generating series.

35.2 Global control and the delayed bootstrap

For all sufficiently large i , every physical $X - r$, $0 \leq r < k$, is bounded below by a positive k -dependent multiple of i , while $|\delta_m| = O_k(i)$. Bounded weight ratios and the injective height shift give

$$\|D_i\| = O_k(i^{1-k}) = O_k(i^{-q})$$

globally, not merely in the Airy window. After division by the relaxed S_i , the delayed scalar product over $k+1$ steps stays bounded. The central/stable inequalities therefore acquire terms

$$C_k i^{-q} (|a_{i-k-1}| + \|z_{i-k-1}\|).$$

For $Q_i = |a_i| + K i^{-1/3} \|z_i\|$, these are bounded by $C_k i^{1/3-q} Q_{i-k-1}$. The running maximum of Q is controlled by the convergent product of $1 + C i^{-4/3} + C i^{1/3-q}$. Thus a_i is bounded and initially $z_i = O(i^{1/3})$. Reinsertion gives a delayed source no larger than $O(i^{-5/3})$, and hence the same $z_i = O(i^{-2/3})$ and convergent central projection as before. The bounds (35.2) force positive amplitudes.

At arbitrary order the delayed normalized error inequalities add $C i^{-q} \|w_{i-k-1}\|$ and $C i^{-q} |\alpha_{i-k-1}|$. For each power supersolution, its relative one-step change is $O(i^{-1})$, and the delayed coefficient is $O(i^{-q})$; both are absorbed by the stable decrement of order $i^{-2/3}$. Thus the same bootstrap gives

$$\|w_i\| = O(i^{-\nu-2/3} + i^{2/3-p}), \quad \alpha_i = O(i^{-\nu-1/3} + i^{1-p}).$$

This proves the all-finite-orders transfer for signed solutions without invoking positivity of their propagators or a spectral theorem for a companion matrix.

35.3 The delayed formal equation and first ratio terms

Let $\tau_l = (1 - l\varepsilon^3)^{-1/3}$ and

$$\mathcal{D}_k(\varepsilon, x) = \prod_{r=0}^{k-1} [q + x\varepsilon^2 - (1 + kr)\varepsilon^3].$$

The exact scaled delay coefficients in the Airy window are

$$\begin{aligned}\beta^B(\varepsilon, x) &= \frac{q^{2k} k^{k-1} \varepsilon^{3q} (1 - x\varepsilon^2 + \varepsilon^3)}{2 \mathcal{D}_k(\varepsilon, x)}, \\ \beta^C(\varepsilon, x) &= q^{2k} k^{k-1} \frac{\varepsilon^{3q} (1 - x\varepsilon^2 - q\varepsilon^3)}{\mathcal{D}_k(\varepsilon, x)}.\end{aligned}$$

Dividing by H_{i-1} gives the delayed formal equation

$$\begin{aligned}U\Phi(\varepsilon\tau_1, (x - \varepsilon)\tau_1) + \Phi(\varepsilon\tau_1, (x + q\varepsilon)\tau_1) \\ - \beta^X(\varepsilon, x) \left(\prod_{l=1}^k \sigma(\varepsilon\tau_l)^{-1} \right) \Phi(\varepsilon\tau_{k+1}, (x - \varepsilon)\tau_{k+1}) = \sigma(\varepsilon)\Phi(\varepsilon, x).\end{aligned}\quad (35.3)$$

There are k inverse scalar factors here, versus $k + 1$ after division by H_i . The delay begins at ε^{3q} , so at each new stage it adds only known lower-order forcing. The polynomial inverse (34.2) remains unchanged. Rational denominators stay bounded away from zero on the cutoff; there are only finitely many additional fixed shifts. The previous arbitrary-order remainder proof therefore applies in full, with the same exact bottom zeros and common normalization.

At first altered order the DFA subtraction is $q^k/(2k)\varepsilon^{3q}f$, while the compacted value is twice this. It is absorbed solely into the scalar, giving

$$\sigma_{3q}^B - \sigma_{3q}^R = -\frac{q^k}{2k}, \quad \sigma_{3q}^C - \sigma_{3q}^R = -\frac{q^k}{k}.$$

Earlier profiles agree, and the new profile at this first altered stage agrees too. Since h_{3q-3} enters the scalar logarithm with coefficient $-(q - 1)$, the first DFA change in it is $q^k/[2k^2(q - 1)]$, with twice this for compacted trees. At $i = kn$ this proves (30.9)–(30.10).

The positive first correction implies approach from above. For strict decrease, use enough finite terms to leave a remainder $o(n^{-(k-1)})$ in the logarithmic ratio. Subtract its values at $n + 1$ and n . The leading difference is $-(k - 2)An^{-(k-1)}$ with $A > 0$; all higher retained powers and the difference of the two remainders are smaller. This avoids differentiating an uncontrolled asymptotic remainder.

35.4 Cancellation between the two ratios

Let $u_n = C_n/R_n$ and $v_n = B_n/(2^{n-1}R_n)$, with positive limits $u_\infty = \rho_C$ and $v_\infty = \rho_B$. A stronger cancellation holds:

$$\log \frac{u_n}{u_\infty} - 2 \log \frac{v_n}{v_\infty} = -\frac{q^q}{k^k} n^{-q} + O_k(n^{-q-1/3}). \quad (35.4)$$

Every preceding fractional power vanishes. In particular the coefficient is $-4/27$ for $k = 3$ and $-27/256$ for $k = 4$.

Here is the order argument that controls nonlinear feedback. Put $d = 3q \geq 6$ and compare each formal solution to the relaxed one in the same boundary gauge. The first scalar change has order ε^d

and is pure f forcing, so its profile correction is zero. Thus the individual profile changes begin at order ε^{d-1} , and the inverse scalar-product changes begin at order ε^d . Every nonlinear response product has order at least $2d-1 > d+3$. Through order $d+3$ the response is therefore genuinely linear.

The exact coefficient identity is

$$\beta^C - 2\beta^B = -\frac{q^{2k}}{(X)_k} = -q^k k^k \varepsilon^{d+3} + O(\varepsilon^{d+5}).$$

After the k inverse scalar factors, the combined forcing is $+q^k \varepsilon^{d+3} f$. The polynomial inverse therefore gives

$$\sigma^C - 2\sigma^B + \sigma^R = q^k \varepsilon^{d+3} + O(\varepsilon^{d+4}),$$

with zero combined profiles through index $d+1$. Taking scalar logarithms introduces nonlinear terms only at order $2d$, still too late. Scalar matching then gives the combined coefficient $h_d^C - 2h_d^B + h_d^R = -q^{k-1}/k$. The endpoint logarithms contribute nothing through order ε^d , since all profiles vanish at zero and their combined correction starts at index $d+2$. Evaluation at $i = kn$ gives (35.4); the all-finite-orders theorem supplies its remainder with the same amplitudes.

36 Inverse counting scales and integer thresholds

[**write, 2 October 2026**] The Lambert anchor and the threshold rule are instances of theorems `p0:thm:lambert-core`, `p0:thm:perturbed-inversion` and `p0:thm:staircase` of the repository's transseries volume (Part 0). No novelty is claimed for the inversion mechanics.

This section applies separately to any of the three models. Write its expansion as

$$T_n = C(n!)^q A^n e^{\gamma n^{1/3}} n^\eta \left(1 + \sum_{r=1}^M c_r n^{-r/3} + O(n^{-(M+1)/3}) \right),$$

where $A = k^k/q^q$ for relaxed or compacted trees and $A = 2k^k/q^q$ for DFAs. The amplitude C is the corresponding positive model constant. Set

$$\nu = \frac{q}{2} + \eta = \frac{7k-5}{6}, \quad d = \log C + \frac{q}{2} \log(2\pi), \quad L = \log y.$$

Let $x(y)$ denote the inverse of the *log-linearly interpolated exact counts*: between consecutive integers interpolate $\log T_n$, then invert that eventually increasing function. With the principal Lambert function W_0 , define

$$w = W_0 \left(\frac{A^{1/q} L}{qe} \right), \quad n_0 = \frac{L}{qw}, \quad D = q(w+1).$$

Then

$$x(y) = n_0 - \frac{\gamma n_0^{1/3} + \nu \log n_0 + d}{D} + o\left(\frac{1}{\log n_0}\right). \quad (36.1)$$

An equivalent form displaying the constant displacement is

$$x(y) = n_0 - \frac{\gamma n_0^{1/3}}{D} - \frac{\nu}{q} - \frac{d - (\nu/q) \log A}{D} + o\left(\frac{1}{\log n_0}\right).$$

Indeed n_0 solves $qn_0 \log n_0 + (\log A - q)n_0 = L$, and the derivative of this leading expression is D . Inserting the displayed correction leaves logarithmic residual $O(n_0^{-1/3}) = o(1)$, including the omitted first finite-order term; the actual chord slopes are asymptotic to $q \log n_0$. Division by the slope proves (36.1).

For arbitrary finite inverse accuracy, take a sufficiently long smooth logarithmic truncation $H_M(t)$ of the proven expansion, interpolate its integer values linearly, and invert that interpolation. At nearby integer sites the logarithmic error is $O(n^{-(M+1)/3})$, which interpolation preserves. The inverse error is therefore

$$O\left(\frac{n_0^{-(M+1)/3}}{\log n_0}\right).$$

This is an effective all-finite-orders inversion procedure once the amplitude is specified; it does not numerically determine the amplitude.

A smooth inverse alone omits an oscillatory lattice correction. If $H(z) = \log y$ for a sufficiently accurate smooth logarithmic truncation and $\theta = \{z\}$, the first smooth-to-chord inverse correction is

$$-\frac{q\theta(1-\theta)}{2zH'(z)} + o\left(\frac{1}{z \log z}\right).$$

This follows from the chord excess $H''(z)\theta(1-\theta)/2$, with $H''(z) \sim q/z$. Finally, for large y , the exact threshold

$$N(y) = \min\{n : T_n \geq y\} = \lceil x(y) \rceil.$$

It need not equal the ceiling of a finite approximation to $x(y)$ when that approximation is too close to an integer.

37 Checks scope and further research

The independent verification chain separates analytic proofs from finite calculations. The exact fixed-weight and phase identities pass 2,095 rational checks for $k = 3, \dots, 11$. The source seeds, completed-run sums, and signed transforms pass 2,403 rational checks for $k = 3, \dots, 8$ and $n \leq 7$. A separate signed-model replay passes 11,016 independent checks. The first two relaxed coefficients were independently recomputed by a separate symbolic direct-linear-system method through order ε^5 . These checks guard algebra and indexing; compactness, tracking, and error transfer have separate proofs.

The only external asymptotic growth input needed for positivity is the relaxed lower bound in Dastidar–Wallner [3, Section 3.4, equation (9)]. It occurs before the paper’s subsequent high-path upper-bound argument. In particular, the present signed-delay route does not use a pointwise bound $U_i \leq q$, which is false at some active bottom edges, and needs no bridge-tail repair. The valid bound used here is $U_i(j) \leq q\alpha_i$ for active $j \geq 1$.

[**write, 2 October 2026**] That the published proof uses a false bound $U_i \leq q$ is source 18’s statement (and source 66’s, Section 44.1); this report has not checked it against the paper. See Part 0, “A correction of the literature that this report has not verified”.

The following remain distinct questions rather than consequences of this report.

1. **Explicit amplitude evaluation.** Construct a certified convergent numerical method with rigorous tail bounds for A_k^R, A_k^C, A_k^B , or obtain a structural representation that evaluates them. Positivity alone does not provide their values.
2. **Large-order control.** Determine the growth of the recursively generated coefficients, possible Gevrey bounds, optimal truncation, and exponentially small remainders. All finite orders do not by themselves imply an exponentially accurate or transseries completion.
3. **Growing arity.** Study whether a simultaneous k, n regime exists. The fixed-weight roots, spectral constants, and remainder bounds here may deteriorate with k ; no uniformity has been proved.
4. **Monotonicity at every size.** Are C_n/R_n and $B_n/(2^{n-1}R_n)$ nonincreasing for every n and fixed $k \geq 3$? The theorem proves only eventual decrease. The positive completed-run formula alone does not couple the bridge measures at different terminal sizes.
5. **Further model comparisons.** Beyond the cancellation in (35.4), determine additional universal combinations and their higher corrections. At sufficiently high orders the response becomes nonlinear, so further cancellation cannot be assumed from the first-order defect alone.
6. **Sharper endpoint errors.** Boundary-adapted resolvent or pointwise estimates may reduce the $i^{1/2}$ loss in norm-to-endpoint extraction. This could improve efficient rigorous numerical evaluation even though arbitrary finite orders are already available.

The ternary case (source 66)

Source. Batch-77 manuscript 66, archive `ternary-airy-amplitudes-result.zip`, main file `report/ternary-airy-amplitudes.tex` (656 lines; 14-page PDF), titled “Airy amplitudes for ternary trees and finite language automata”, author line “Research report”, dated 2 October 2026. Printed in full as the ternary section of Part III, label prefix `air:t3:`.

Abstract of source 66.

We prove positive limiting amplitudes for relaxed ternary trees and for minimal deterministic finite automata accepting finite languages over a three-letter alphabet. For relaxed trees we also construct an asymptotic expansion to every fixed finite order. The proof uses a fixed weighted Hilbert norm that matches the forward and adjoint boundary harmonics, a compact singular Airy limit, and evolving polynomial–Airy profiles. A corrected uniform bridge-tail estimate transfers the relaxed amplitude to the automata. The leading scales were established or motivated in earlier work; the results here strengthen the relaxed Theta estimate to an equivalent and transfer an equivalent to the ternary automata. No all-orders automaton expansion or general-alphabet theorem is asserted.

[**write, 2 October 2026**] Source 66 is the case $k = 3$ of source 18 and the base proof that source 18 generalizes (see the note at the head of this Part). Its Theorems 38.1 and 38.2 are the case $k = 3$ of Theorems 30.1 and 30.2: R_n and B_n are Part III’s counts at $k = 3$, $C_R = A_3^R$, $C_B = A_3^B$, $\rho = \rho_B$, and P_3 is the product P_B of (35.2) at $q = 2$, by Euler’s sine product. They are kept in source 66’s normalization because the proof that follows is that base proof; nothing in them is a separate result. Source 66’s own additions are the third coefficient c_3 and the scalar coefficients $\mathfrak{s}_3, \dots, \mathfrak{s}_6$; the explicit ternary weight of Lemma 39.1, with $1 \leq \omega \leq 3/2$ (Lemma 31.1 gives $1 \leq \omega \leq q$); and the bridge-tail route to the automaton amplitude of Section 44, a second

route beside source 18's signed-delay route. Its statements "no general-alphabet theorem" and "no all-orders automaton expansion" are about source 66 alone: source 18 proves both for every $k \geq 3$.

38 Statements and normalizations

Let R_n count relaxed ternary trees with n internal nodes, including $R_0 = 1$. This is the sequence A082162, beginning

$$1, 1, 7, 139, 5711, 408354, 45605881, \dots$$

Let B_n count minimal DFAs accepting finite languages on a fixed three-letter alphabet, with n transient states and one rejecting sink. Thus $B_0 = 1$ and

$$B_1 = 1, \quad B_2 = 14, \quad B_3 = 532, \quad B_4 = 42644.$$

These are different counting sequences. In particular, A082162 is not the minimal finite-language DFA sequence.

Write $a = a_1 < 0$ for the largest real zero of Ai, and put

$$\gamma = 3 \cdot 3^{1/3} a.$$

The published relaxed-tree theorem of Dastidar and Wallner [3] gives the Theta scale used below, including its positive lower bound. Our amplitude and finite-order conclusions are additional arguments, not statements attributed to that paper.

Theorem 38.1 (Positive amplitudes; the case $k = 3$). *There are constants $C_R, C_B > 0$ such that*

$$R_n \sim C_R (n!)^2 \left(\frac{27}{4}\right)^n e^{\gamma n^{1/3}} n^{5/3}, \quad (38.1)$$

$$B_n \sim C_B (n!)^2 \left(\frac{27}{2}\right)^n e^{\gamma n^{1/3}} n^{5/3}. \quad (38.2)$$

Moreover,

$$\frac{B_n}{2^{n-1} R_n} \longrightarrow \rho \in [P_3, 1], \quad P_3 = \frac{2\sqrt{2}}{\pi} \sin \frac{\pi}{\sqrt{2}} > 0, \quad C_B = \frac{\rho C_R}{2}.$$

Theorem 38.2 (Every fixed finite order for relaxed trees; the case $k = 3$). *For every integer $M \geq 0$ there are computable constants $c_1, \dots, c_M$ such that*

$$R_n = C_R (n!)^2 \left(\frac{27}{4}\right)^n e^{\gamma n^{1/3}} n^{5/3} \left(1 + \sum_{r=1}^M c_r n^{-r/3} + O(n^{-(M+1)/3})\right).$$

The same positive C_R works for every M . The assertion is an asymptotic expansion to each fixed finite order, not convergence of an infinite series.

The first coefficients, in exactly this normalization, are

$$\begin{aligned} c_1 &= 3^{-1/3} \frac{89a^2}{90}, \\ c_2 &= 3^{-2/3} \left(\frac{7921a^4}{16200} + \frac{115a}{27} \right), \\ c_3 &= \frac{704969a^6}{13122000} + \frac{115931a^3}{85050} + \frac{1514}{945}. \end{aligned}$$

The recursion and remainder argument, rather than finite symbolic checks alone, justify these coefficients. Neither theorem determines the leading constants in closed form.

[write, 2 October 2026] c_1 and c_2 are the $k = 3$ values displayed after Theorem 30.2; c_3 is source 66's own. A numerical check made at intake (exact R_n for $n \leq 640$; not shipped) is consistent with c_3 : with it the successive differences of the corrected amplitude shrink at the rate $n^{-4/3}$ predicts, without it at the rate n^{-1} .

39 Exact transfer and the boundary weight

The relaxed array has support $x \geq 2m \geq 0$ and satisfies

$$r_{x,m} = r_{x,m-1} + (m+1)r_{x-1,m}, \quad r_{x,0} = 1.$$

Set $i = x + m$, $j = x - 2m$, and use the exact normalization

$$d_i(j) = \frac{4^x}{x!} r_{x,m}.$$

This gauge multiplies every path to a fixed endpoint by the same positive factor, so normalized bridge measures are unchanged. The resulting array satisfies

$$d_i(j) = U_i(j)d_{i-1}(j-1) + d_{i-1}(j+2), \quad (39.1)$$

$$U_i(j) = \frac{4(i-j+3)}{2i+j} = 2 - v_i(j), \quad v_i(j) = \frac{6(j-2)}{2i+j}, \quad (39.2)$$

$$R_n = \frac{(2n)!}{16^n} d_{3n}(0). \quad (39.3)$$

Here $d_0(0) = 1$ and all missing coordinates are zero. At time i the physical space is

$$J_i = \{j : 0 \leq j \leq i, j \equiv i \pmod{3}\}.$$

The rectangular transfer T_i maps the space on J_{i-1} to that on J_i .

The critical infinite-phase transfer is

$$(Tu)(j) = 2u(j-1) + u(j+2), \quad j \geq 0.$$

It is not diagonally symmetrizable: its two directed jumps have unequal lengths. The following fixed weight is the useful replacement for a self-adjoint transfer.

Lemma 39.1 (Compatible forward and adjoint harmonics). *Define*

$$h(j) = j+1, \quad \ell(j) = j + \frac{4}{3} + \frac{1}{6} \left(-\frac{1}{2}\right)^j, \quad \omega(j) = \frac{\ell(j)}{h(j)}, \quad \pi(j) = \ell(j)h(j).$$

Then $1 \leq \omega(j) \leq 3/2$ and

$$Th = 3h, \quad T^*\ell = 3\ell,$$

where $\ell(-1) = \ell(-2) = 0$. In the fixed phase norms $\|u\|_\omega^2 = \sum_j \omega(j)|u(j)|^2$, one has $\|T\| \leq 3$. If $g(j) = u(j)/h(j)$ and the input is finitely supported, then

$$9\|u\|_\omega^2 - \|Tu\|_\omega^2 = \sum_{\substack{j \geq 1 \\ j \text{ in output phase}}} 2\omega(j)j(j+3)|g(j-1) - g(j+2)|^2. \quad (39.4)$$

Proof. Substitute the formulas into the two recurrences. In particular the adjoint bottom equations are $2\ell(1) = 3\ell(0)$ and $2\ell(2) = 3\ell(1)$; the geometric term is essential there. The h -conjugate of $T/3$ has row probabilities

$$p_j = \frac{2j}{3(j+1)}, \quad q_j = \frac{j+3}{3(j+1)}.$$

At $j = 0$, only the second term remains and equals one. The adjoint identity makes π invariant between successive residue phases. Jensen's inequality proves the contraction. Its exact two-point variance identity gives (39.4), including the last edge from the input support to its zero extension. $\square$

The weighted adjoint $T^\dagger = \omega^{-1}T^*\omega$ also sends h to $3h$. Thus both sides have the same boundary harmonic profile. The unweighted singular profiles, by contrast, have residue-dependent extrapolation lengths; ignoring that mismatch does not prove an amplitude theorem.

[**write, 2 October 2026**] This weight is Lemma 31.1 at $k = 3$: there $\sum_j l_j z^j = 3/(2 - 3z + z^3)$, whose coefficients are $l_j = j + \frac{4}{3} + \frac{1}{6}(-\frac{1}{2})^j$. It is the “matched forward and adjoint boundary profiles” that source 29 asks for in Section 51.

40 The singular Airy limit

Put $\alpha_i = 1 + 3/(2i + 1)$ and conjugate the finite transfer by h :

$$Q_i(j, k) = \frac{T_i(j, k)h(k)}{3\alpha_i h(j)}, \quad r_i(j) = \sum_k Q_i(j, k), \quad c_i(k) = \frac{1}{\pi(k)} \sum_j \pi(j) Q_i(j, k).$$

Since $U_i(j) \leq 2\alpha_i$ on active up edges, both r_i and c_i are at most one. The input column loss is exactly

$$1 - c_i(k) = \frac{\alpha_i - 1}{\alpha_i} + \frac{v_i(k+1)\ell(k+1)}{3\alpha_i \ell(k)}. \quad (40.1)$$

It vanishes at $k = 0$ and is at least ck/i for $k \geq 1$, with an absolute $c > 0$. One may take $c = 4/9$ for all physical input sites. There is no missing upper column correction: both possible nonnegative output rows are present.

For each row write $p = Q_i(j, j-1)$, $q = Q_i(j, j+2)$, assigning zero to a missing input. The following decomposition is exact:

$$\begin{aligned} 9\alpha_i^2 \|u\|_\omega^2 - \|T_i u\|_\omega^2 = 9\alpha_i^2 \Big\{ & \sum_k \pi(k)(1 - c_i(k))|g(k)|^2 \\ & + \sum_j \pi(j)pq|g(j-1) - g(j+2)|^2 \\ & + \sum_j \pi(j)(1 - r_i(j))[p|g(j-1)|^2 + q|g(j+2)|^2] \Big\}. \end{aligned} \quad (40.2)$$

It follows by summing the column masses and applying

$$p|a|^2 + q|b|^2 - |pa + qb|^2 = pq|a - b|^2 + (1 - p - q)(p|a|^2 + q|b|^2).$$

Every term in (40.2) is nonnegative.

Lemma 40.1 (Compact Airy limit). *For each fixed $m \geq 1$, the m th largest singular value of T_i in the fixed weighted norms satisfies*

$$\sigma_{i,m} = 3 \left(1 + a_m i^{-2/3} + o(i^{-2/3}) \right),$$

where $a_1 > a_2 > \dots$ are the negative Airy zeros. The normalized first right singular vector tends, under mesh- $3i^{-1/3}$ interpolation, to the positive normalized $\text{Ai}(a_1 + x)$ on $\mathbb{R}_+$.

Proof. Let $\varepsilon = i^{-1/3}$ and consider the self-adjoint singular forms

$$\mathcal{E}_i(u) = \frac{9 \|u\|_\omega^2 - \|T_i u\|_\omega^2}{18\varepsilon^2}.$$

Replacing 9 by $9\alpha_i^2$ changes this form by $O(\varepsilon) \|u\|_\omega^2$. In particular the forms are bounded below uniformly. We give the compactness and limiting constants explicitly.

On an input phase r , let $H_k = r + 3k + 1$ and write $u_k = u(r + 3k)$. Direct telescoping gives

$$\sum_{k \geq 0} H_k H_{k+1} \left| \frac{u_{k+1}}{H_{k+1}} - \frac{u_k}{H_k} \right|^2 = \sum_{k \geq 0} |u_{k+1} - u_k|^2 + \frac{3|u_0|^2}{r+1}. \quad (40.3)$$

The input is extended by zero at infinity. If F_i is linear between the nodes

$$x_k = \varepsilon H_k, \quad F_i(x_k) = \frac{u_k}{\sqrt{3\varepsilon}}, \quad F_i(0) = 0,$$

then the right side of (40.3), divided by $9\varepsilon^2$, is exactly $\int |F'_i|^2$.

The moving variance sum omits the critical top edge to zero. Therefore localize first: multiply u_k by a cutoff equal to one for $x_k \leq R$ and zero for $x_k \geq 2R$. For fixed R and large i , these edges are far below the physical top and U_i is bounded below. Writing $\mathcal{Q}$ for the radial sum in (40.3), product differences and $H_{k+1}/H_k \leq 4$ give

$$\mathcal{Q}(\chi_R u) \leq C \mathcal{Q}(u; x \leq 2R + 3\varepsilon) + C\varepsilon^2 R^{-2} \|u\|^2.$$

Hence bounded norms and bounded $\mathcal{E}_i$ imply uniform local H^1 bounds for the zero-anchored interpolants. In particular

$$\int_0^\delta |F_i|^2 \leq \frac{\delta^2}{2} \int_0^\delta |F'_i|^2,$$

and the corresponding nodal mass is $O(\delta^2 + \varepsilon^2)$. The lower endpoint cannot carry concentrated mass.

The column loss (40.1) also yields

$$\sum_{\varepsilon k \geq R} \omega(k) |u(k)|^2 \leq C/R \quad (40.4)$$

for bounded form energy. Local compactness, the lower-end estimate and (40.4) give global strong L^2 compactness. Weighted nodal norms, unweighted nodal norms and interpolated norms have the same limit: away from zero $\omega \rightarrow 1$, and the two tails are uniformly negligible. The same holds for inner products.

On any compact interval $0 < \delta < x < R$, the variance term in (40.2) converges to $\int x^2 |(F/x)'|^2$. Each loss term contributes one half of $\int x |F|^2$. Indeed $v_i(j) = 3x\varepsilon^2 + O(\varepsilon^3)$ there, and both

normalized row and column losses equal $x\varepsilon^2 + O(\varepsilon^3)$. Local strong L^2 convergence and the gradient bound justify translated sampling. Discarding the nonnegative terms outside $[\delta, R]$, then sending $\delta \downarrow 0$ and $R \uparrow \infty$, gives the liminf form

$$\mathcal{E}(F) = \int_0^\infty (|F'|^2 + x|F|^2) dx, \quad F \in H_0^1(\mathbb{R}_+).$$

Here $\int x^2 |(F/x)'|^2 = \int |F'|^2$ by Hardy's inequality and integration by parts. The zero trace was supplied by (40.3); no Neumann or Robin boundary has been introduced.

For recovery sequences sample smooth compactly supported functions inside $(0, \infty)$ as $u_k = \sqrt{3\varepsilon} F(x_k)$. Direct Riemann sums give the norm and form limits. Such functions form a core of the displayed form. Finite-dimensional recovery spaces give the min-max upper bounds. For the lower bounds, compactness of the first m discrete eigenvectors preserves their orthonormality and gives an m -dimensional limiting subspace; apply the liminf to its linear combinations. The limiting operator is $-\partial_x^2 + x$ with Dirichlet boundary, with eigenvalues $-a_m$. Thus

$$\frac{9 - \sigma_{i,m}^2}{18\varepsilon^2} \longrightarrow -a_m.$$

Simplicity of the first limit eigenvalue gives ground-vector convergence. The first discrete vector can be chosen positive because $T_i^\dagger T_i$ is irreducible on its input residue chain. $\square$

Remark 40.2. The lemma concerns one occupied residue channel. On the direct sum of all three channels, the first three singular values have the same leading Airy level. It would be incorrect to use a_2 as the second level on that full space.

41 An evolving Airy profile

Let $f(x) = \text{Ai}(a+x)$ and

$$g(x) = \left(-\frac{5x^2}{12} - \frac{ax}{6} \right) f(x).$$

Choose a fixed smooth cutoff shape, equal to one up to one and zero past two, and apply it at $x/\log i$. With $\varepsilon = i^{-1/3}$ define

$$\begin{aligned} F_i(j) &= \chi_i(\varepsilon(j+1)) [f(\varepsilon(j+1)) + \varepsilon g(\varepsilon(j+1))], \\ N_i &= \|F_i\|_\omega, \quad \psi_i = F_i/N_i, \\ t_i &= 3 \left(1 + ai^{-2/3} + \frac{5}{2i} \right), \quad s_i = t_i \frac{N_i}{N_{i-1}}. \end{aligned}$$

All estimates from this point begin at one fixed sufficiently large time I . Earlier evolution is absorbed into its finite initial vector.

Lemma 41.1 (Two-sided evolving residual). *Uniformly through all three phases,*

$$\|T_i \psi_{i-1}/s_i - \psi_i\|_\omega = O(i^{-4/3}), \quad (41.1)$$

$$\|T_i^\dagger \psi_i/s_i - \psi_{i-1}\|_\omega = O(i^{-1}). \quad (41.2)$$

Also $N_i \asymp i^{1/6}$, $N_i/N_{i-1} = 1 + O(i^{-1})$, and, when $i \equiv 0 \pmod{3}$, $\psi_i(0) \asymp i^{-1/2}$.

Proof. The profile is evolving, not a frozen eigenvector. In particular $(i-1)^{-1/3} = \varepsilon(1 + \varepsilon^3/3 + O(\varepsilon^6))$ must be retained. At $x = \varepsilon(j+1)$, Taylor expansion of the uncut $T_i F_{i-1}$ gives

$$3f + 3\varepsilon g + 3\varepsilon^2(f'' - xf) + \varepsilon^3\{f''' + 4xf' + 9f + 3(g'' - xg)\} + O(\varepsilon^4).$$

The $4xf'$ includes xf' from the time dilation. Using $f'' = (x+a)f'$,

$$f''' + 4xf' + 9f = 10f + (5x+a)f',$$

while

$$3\{g'' - (x+a)g\} = -\frac{5}{2}f - (5x+a)f'.$$

The result is $t_i(f + \varepsilon g) + O(\varepsilon^4)$. The remainder has a square-integrable polynomial–Airy envelope, uniformly on the moving support. A mesh sum costs $O(\varepsilon^{-1})$ in squared norm, canceled by $N_i^2 \asymp \varepsilon^{-1}$. The lower missing input corresponds exactly to $x = 0$, where $f = g = 0$. This proves (41.1) after normalization.

We spell out the adjoint boundary estimate, where an unweighted argument would fail. The exact weighted adjoint is

$$(T_i^\dagger u)(t) = \frac{U_i(t+1)\omega(t+1)u(t+1) + \omega(t-2)u(t-2)}{\omega(t)}.$$

Let $B = \omega^{-1}T^*\omega$ be the critical weighted adjoint, and let T^* here denote its unweighted counterpart. For $t \geq 2$ put

$$a_t = \frac{2(\omega(t+1) - \omega(t))}{\omega(t)}.$$

Both B and T^* send h to $3h$ at those sites. Hence, exactly,

$$[(B - T^*)F](t) = a_t h(t+1) \left[\frac{F(t+1)}{h(t+1)} - \frac{F(t-2)}{h(t-2)} \right]. \quad (41.3)$$

Its coefficient is bounded by $C((t+1)^{-1} + 2^{-t})$. Write $F(t)/h(t) = \varepsilon\Phi_\varepsilon(\varepsilon(t+1))$, where $\Phi_\varepsilon(x) = (f(x) + \varepsilon g(x))/x$. For $0 \leq x \leq 1$, $|\Phi'_\varepsilon(x)| \leq C(x + \varepsilon)$ because $f(0) = f''(0) = 0$. Thus (41.3) is $O(\varepsilon^3)$ per site near the bottom. Above the Airy scale, the same estimate has an integrable polynomial–Airy envelope. The operator $T^* - T$ is a third finite difference and has the same order.

At the exceptional sites check directly:

$$(BF)(0) = \frac{3}{2}F(1), \quad (BF)(1) = 2F(2), \quad (TF)(0) = F(2), \quad (TF)(1) = 2F(0) + F(3).$$

The linear terms cancel, and the quadratic term of f vanishes. The remaining terms are $O(\varepsilon^3)$. The difference of the adjoint and forward inhomogeneous perturbations is

$$v_i(t)F(t-1) - v_i(t+1)\frac{\omega(t+1)}{\omega(t)}F(t+1).$$

A first-difference estimate gives normalized norm $O(\varepsilon^3)$; at fixed t it is $O(\varepsilon^4)$. Replacing F_i by F_{i-1} has normalized cost $O(\varepsilon^3)$ by time dilation. These facts prove (41.2).

All cutoff discrepancies occur at $x \geq \log i - O(\varepsilon)$, where the Airy envelope is bounded by a polynomial times $e^{-c(\log i)^{3/2}}$. They are smaller than every fixed power of i .

Finally, Euler–Maclaurin on a residue phase r gives

$$N_{r,\varepsilon}^2 = C_{-1}\varepsilon^{-1} + C_0 + C_1\varepsilon + O(\varepsilon^2), \quad C_{-1} = \frac{1}{3} \int_0^\infty f^2 > 0, \quad (41.4)$$

with the first three constants independent of r . To see the phase order, set $H_\varepsilon = (f + \varepsilon g)^2$. Then $H_\varepsilon(0) = H'_\varepsilon(0) = 0$; ordinary phase-sum endpoint corrections begin at $O(\varepsilon^2)$. The algebraic weight contributes $\varepsilon H_\varepsilon(x)/(3x)$, whose quotient vanishes at zero, and its first phase-dependent correction is again $O(\varepsilon^2)$. The geometric part of ω is $O(\varepsilon^2)$ in the sum. Formula (41.4) proves the adjacent-time ratio, including the change of phase. The endpoint estimate follows from $f(\varepsilon) = \text{Ai}'(a)\varepsilon + O(\varepsilon^3)$. $\square$

42 Tracking and the relaxed amplitude

Normalize the evolution by $A_i = T_i/s_i$ and decompose an exact solution as

$$y_i = A_i y_{i-1} = b_i \psi_i + z_i, \quad z_i \perp \psi_i.$$

The two residuals give the block estimates

$$b_i = (1 + O(i^{-4/3}))b_{i-1} + \beta_i z_{i-1}, \quad \|\beta_i\| = O(i^{-1}), \quad (42.1)$$

$$z_i = v_i b_{i-1} + D_i z_{i-1}, \quad \|v_i\| = O(i^{-4/3}). \quad (42.2)$$

The vector symbol v_i in this section is unrelated to the scalar up-weight loss in (39.2).

Lemma 42.1 (Compressed contraction). *For some $c > 0$ and all sufficiently large i , $\|D_i\| \leq 1 - ci^{-2/3}$.*

Proof. The explicit input profile converges to the same Airy ground function as the first right singular vector, so its squared overlap is $1 - o(1)$. If $x \perp \psi_{i-1}$, singular decomposition gives

$$\|T_i x\|^2 \leq [\sigma_{i,2}^2 + (\sigma_{i,1}^2 - \sigma_{i,2}^2)o(1)] \|x\|^2.$$

By Lemma 40.1 and $s_i = 3(1 + ai^{-2/3} + O(i^{-1}))$, division by s_i^2 leaves a strict loss of order $i^{-2/3}$. Output projection cannot increase the norm. This argument uses singular values of one channel and does not assume self-adjointness of T_i or a frozen eigenvector equation for ψ_i . $\square$

Let $q_i = |b_i|$ and $Z_i = \|z_i\|$. The quantity $q_i + Ki^{-1/3}Z_i$ is bounded for sufficiently large fixed K . Indeed the negative term $-Kci^{-1}Z_{i-1}$ absorbs the $O(i^{-1})Z_{i-1}$ cross term, while the remaining multiplicative errors are summable $O(i^{-4/3})$. Iterating the stable block gives

$$Z_i = O(i^{-2/3}), \quad b_i = b_\infty + O(i^{-1/3}).$$

For the convolution estimate, sources older than $i/2$ have stretched-exponential suppression; the remaining kernel is bounded by $\exp[-c(i-m)i^{-2/3}]$ and has total mass $O(i^{2/3})$.

Endpoint evaluation is bounded in the fixed norm. Since $\psi_i(0) \asymp i^{-1/2}$ at residue-zero times,

$$\frac{z_i(0)}{\psi_i(0)} = O(i^{-1/6}) \longrightarrow 0.$$

The scalar norm factors telescope, while

$$\prod_{m=I}^i t_m = K_I 3^i e^{3ai^{1/3}} i^{5/2} (1 + O(i^{-1/3})), \quad F_i(0) \sim \text{Ai}'(a) i^{-1/3}.$$

Consequently

$$d_{3n}(0) \sim C_d 27^n e^{\gamma n^{1/3}} n^{13/6}. \quad (42.3)$$

The constant is finite. It is strictly positive because the published relaxed Theta lower bound in [3] rules out a zero limit for this exact solution. Thus the proof does not assume positivity of a nonnormal spectral projection. Equation (39.3) and Stirling prove (38.1). The same argument applies to every fixed-time initial vector, giving a finite limiting endpoint ratio to the relaxed reference solution; that limit need not be known to be positive.

43 Polynomial Airy recursion to every finite order

Put $\tau = (1 - \varepsilon^3)^{-1/3}$ and seek formal series

$$\Phi(\varepsilon, x) = \sum_{m \geq 0} \varepsilon^m \phi_m(x), \quad \mathfrak{s}(\varepsilon) = 3 + \sum_{m \geq 2} \mathfrak{s}_m \varepsilon^m, \quad \phi_0 = f, \quad \mathfrak{s}_2 = 3a.$$

The exact formal evolving equation is

$$\mathcal{U}(\varepsilon, x) \Phi(\varepsilon \tau, (x - \varepsilon) \tau) + \Phi(\varepsilon \tau, (x + 2\varepsilon) \tau) = \mathfrak{s}(\varepsilon) \Phi(\varepsilon, x), \quad (43.1)$$

where

$$\mathcal{U}(\varepsilon, x) = 2 - \frac{6(x\varepsilon^2 - 3\varepsilon^3)}{2 + x\varepsilon^2 - \varepsilon^3}.$$

Require $\phi_m(0) = 0$ at every order. At order ε^{m+2} the only new profile is ϕ_m , and the only new scalar is $\mathfrak{s}_{m+2}$. Zero drift cancels the intervening order. After reducing derivatives by $f'' = (x + a)f$, the equation is

$$3(\partial_x^2 - x - a)\phi_m = \mathfrak{s}_{m+2}f + P_m f + Q_m f', \quad (43.2)$$

with known polynomials P_m, Q_m .

For a polynomial–Airy function $Af + Bf'$,

$$(\partial_x^2 - x - a)(Af + Bf') = [A'' + 2(x + a)B' + B]f + [2A' + B'']f'.$$

After dividing (43.2) by three, polynomial forcing $Pf + Qf'$ is solved by

$$\mathcal{K}B := -\frac{1}{2}B''' + 2(x + a)B' + B = P - \frac{1}{2}Q', \quad A' = \frac{1}{2}(Q - B''). \quad (43.3)$$

On polynomials of degree at most d , $\mathcal{K}$ has diagonal entries $1, 3, \dots, 2d + 1$ and lower-degree remaining terms. It is invertible. Adding a constant to the right side adds precisely that constant to B . The new scalar therefore uniquely enforces $B(0) = 0$, which is the boundary condition $\phi_m(0) = 0$. Fix $A(0) = 0$ to remove the remaining amplitude gauge. This constructs every finite stage with no Fredholm obstruction or logarithmic profile.

For example, the first scalar coefficients in this gauge are

$$\mathfrak{s}_3 = \frac{15}{2}, \quad \mathfrak{s}_4 = \frac{23a^2}{45}, \quad \mathfrak{s}_5 = \frac{11a}{270}, \quad \mathfrak{s}_6 = -\frac{1714a^3}{14175} - \frac{5293}{1890}.$$

Two independent exact symbolic recursions verify these values. The infinite-stage existence statement follows from (43.3), not from those finite computations.

43.1 Uniform remainders and a common amplitude

Fix $p > 1$, take an integer $K \geq \max(4, \lceil 3p \rceil)$, construct profiles through $K - 2$ and scalars through K , and allow one further order if desired. Let

$$G_i = 3^i e^{3ai^{1/3}} i^{5/2}, \quad H_i = G_i \exp \left(\sum_{r=1}^{K-3} h_r i^{-r/3} \right)$$

with leading constant exactly one. Match $\log(H_i/H_{i-1})$ to $\log \mathfrak{s}(i^{-1/3})$. The first two nonconstant orders already agree. The new h_r first appears at order ε^{r+3} with coefficient $-r/3$, so every further scalar equation is uniquely soluble.

Apply the same cutoff to the finite profile sum Φ_i . Each fixed derivative of each polynomial–Airy correction has a bound $C(1+x)^D e^{-cx^{3/2}}$ for $x \geq 1$ and is bounded near zero. On $x \leq 2 \log i$, the rational denominators in (43.1) stay uniformly away from zero. Taylor remainders are powers of ε times a common integrable polynomial–Airy envelope; spatial displacements are $O(\varepsilon + \varepsilon^3 x)$. Thus weighted mesh sums divide by $N_i^2 \asymp \varepsilon^{-1}$ without logarithmic loss. Bottom values vanish exactly, the physical top lies beyond the cutoff, and cutoff discrepancies are superpolynomially small. These estimates prove

$$\|T_i[H_{i-1}\Phi_{i-1}] - H_i\Phi_i\|_\omega \leq C_p H_i N_i i^{-p}. \quad (43.4)$$

Constants and starting indices may depend on the requested finite order.

Fix the same lower index I in every construction and write

$$P_i = \prod_{m=I}^i t_m \sim \kappa G_i, \quad S_i = \prod_{m=I}^i s_m = P_i \frac{N_i}{N_{I-1}}.$$

Then

$$Z_i = \frac{\kappa}{N_{I-1}} \frac{H_i \Phi_i}{S_i}$$

has projection onto ψ_i tending to one, independently of truncation. Indeed $H_i/G_i \rightarrow 1$ and $\Phi_i/N_i - \psi_i \rightarrow 0$. Dividing (43.4) by S_i yields $\|Z_i - A_i Z_{i-1}\| = O(i^{-p})$. The reciprocal constants in this normalization are essential; one cannot independently absorb a different constant into each finite truncation.

43.2 The zero-limit bootstrap

Let the exact normalized solution have limiting projection C , and set $E_i = y_i - CZ_i = \eta_i \psi_i + w_i$, where $\eta_i \rightarrow 0$. The already proved block estimates imply

$$\begin{aligned} |\eta_i - \eta_{i-1}| &\leq C i^{-4/3} |\eta_{i-1}| + C i^{-1} \|w_{i-1}\| + C i^{-p}, \\ \|w_i\| &\leq (1 - c i^{-2/3}) \|w_{i-1}\| + C i^{-4/3} |\eta_{i-1}| + C i^{-p}. \end{aligned}$$

If $\eta_i = O(i^{-q})$, the contraction convolution gives

$$\|w_i\| = O(i^{-q-2/3} + i^{2/3-p}).$$

Summing the central increment *backward*, using $\eta_\infty = 0$ and $p > 1$, gives

$$\eta_i = O(i^{-q-1/3} + i^{1-p}).$$

Starting at bounded η and iterating finitely many times yields $\|E_i\| = O(i^{1-p})$. Endpoint extraction costs at most $i^{1/2}$, so the relative endpoint error is $O(i^{3/2-p})$. For the M -term theorem choose $p > 3/2 + (M+1)/3$ and sufficiently many formal terms. The sufficiently large starting index of these estimates may depend on M , while the scalar normalization retains its fixed lower limit I . This transfers every finite order with the same amplitude.

Finally, every ϕ_m vanishes at zero, so $\Phi_i(0)$ has an ordinary expansion in ε after factoring out $\text{Ai}'(a)\varepsilon$. Independent calculations give

$$h_1 = \frac{89a^2}{90}, \quad h_2 = \frac{571a}{135}, \quad \frac{\Phi_i(0)}{\text{Ai}'(a)\varepsilon} = 1 + \frac{4a}{135}\varepsilon^2 - \frac{103}{810}\varepsilon^3 + O(\varepsilon^4).$$

Combining the scalar factor, endpoint series, $i = 3n$, and Stirling proves Theorem 38.2, including its displayed coefficients.

44 Positive defects and the automaton amplitude

The ternary DFA recurrence is

$$b_{x,m} = 2b_{x,m-1} + (m+1)b_{x-1,m} - mb_{x-3,m-1}, \quad x \geq 2m.$$

The base row is $b_{x,0} = 1$ for $x \geq 0$. The necessary auxiliary boundary value is $b_{-1,0} = 1$; all other out-of-domain values vanish. Equivalently the first row is $b_{x,1} = 2^{x-1} - 1$ for $x \geq 2$. This boundary convention reproduces the source's tabulated initial counts.

We use a positive completed-run representation rather than the signed recurrence. At level $m \geq 1$, put

$$V_m(\ell) = (m+1)^\ell, \quad W_m(\ell) = \begin{cases} 2(m+1)^\ell, & 0 \leq \ell < 3, \\ 2(m+1)^\ell - (m+1)^{\ell-2}, & \ell \geq 3. \end{cases}$$

Start both completed-run arrays at $a(x,1) = A(x,1) = 1$ for $x \geq 2$, with triangular support. Convolution by V_m gives $a(x, m+1)$; convolution by W_m gives $A(x, m+1)$. The endpoints are $R_n = a(2n, n)$ and $B_n = A(2n, n)$. For verification,

$$\sum_{\ell \geq 0} W_m(\ell) t^\ell = \frac{2 - (m+1)t^3}{1 - (m+1)t},$$

and appending the final horizontal run recovers the signed recurrence. In particular all weights in this representation are nonnegative.

Pointwise,

$$2 \left(1 - \frac{1}{2(m+1)^2} \right) V_m(\ell) \leq W_m(\ell) \leq 2V_m(\ell).$$

Induction gives

$$\prod_{j=2}^n \left(1 - \frac{1}{2j^2} \right) \leq \frac{B_n}{2^{n-1}R_n} \leq 1. \quad (44.1)$$

Euler's sine product evaluates the infinite lower product as P_3 in Theorem 38.1.

[**write, 2 October 2026**] This is the case $k = 3$ of Theorem 47.1 (source 29), whose proof of the run weights goes through the completed-run generating series as here.

Under the relaxed bridge measure to $(2n, n)$, let L_m be the horizontal-run length at vertical level m before rising to $m + 1$. Then

$$\rho_n := \frac{B_n}{2^{n-1}R_n} = \mathbb{E}_n \prod_{m=1}^{n-1} \left(1 - \frac{\mathbf{1}_{\{L_m \geq 3\}}}{2(m+1)^2}\right). \quad (44.2)$$

The factor 2^{n-1} is constant over the colored relaxed paths, so coloring does not change this bridge probability.

44.1 A corrected uniform high-path bound

[**write, 2 October 2026**] That the published lemma asserts $U_i(j) \leq 2$ is source 66's statement, not checked against the paper by this report (Part 0). The repair below, with the monotonicity of u_r and the bridge comparison, is a second route to the automaton amplitude at $k = 3$. Source 18 needs none of it: its signed-delay route (Section 35.1 and the following subsections) uses only $U_i(j) \leq q\alpha_i$.

The proof of the high-path lemma in [3] (published Lemma 15, page 15:12; Lemma 14 in arXiv v1) contains the pointwise assertion $U_i(j) \leq 2$. That assertion is false near the boundary, for example $U_4(1) = 8/3$. The following repair proves the required conclusion.

At active up destinations $j \geq 1$,

$$U_i(j) \leq U_i(1) = 2 \left(1 + \frac{3}{2i+1}\right) \leq 2 \left(1 + \frac{3}{2i}\right).$$

The product of the extra factors along any path of length t is at most $Ct^{3/2}$. Thus

$$d_{3x}(3y) \leq Cx^{3/2}2^{2x+y} \binom{3x}{x-y}. \quad (44.3)$$

For continuations to $(T, 0)$, write $p(r, s; T)$ for their total weight and $u_r(s) = p(r, s; T)/(s+1)$. Its backward coefficients are

$$A_r(s) = \frac{4(r-s+3)(s+2)}{(2r+s+3)(s+1)}, \quad B_r(s) = \frac{(s-1)_+}{s+1}.$$

For $s \geq 1$ their row sum is

$$V_r(s) = 3 - 6 \frac{s-1}{s+1} \frac{s+2}{2r+s+3},$$

which decreases because both nonnegative fractions increase. At $s = 0$ the sole coefficient is at least four, whereas $V_r(1) = 3$, so the boundary comparison also holds. Adjacent same-phase starts $s, s+3$ have next supports $\{s-2, s+1\}$ and $\{s+1, s+4\}$. If the next normalized continuation decreases, all values on the first support are at least the common middle value and all on the second are at most it. The row-sum comparison preserves monotonicity. The terminal sequence is one at zero and zero at other physical sites, so reverse induction proves that u_r decreases on each phase. Consequently

$$p(3x, 3y; T) \leq (3y+1)p(3x, 0; T),$$

and comparison through height zero yields

$$\mathbb{P}_T\{\text{path visits } (3x, 3y)\} \leq (3y+1) \frac{d_{3x}(3y)}{d_{3x}(0)}. \quad (44.4)$$

The relaxed lower bound gives $d_{3x}(0) \geq c27^x e^{-Cx^{1/3}} x^{13/6}$. Furthermore,

$$\frac{2^{2x+y}}{27^x} \binom{3x}{x-y} = \mathbb{P}\{\text{Bin}(3x, 1/3) = x-y\} \leq e^{-2y^2/(3x)}.$$

The last inequality follows from the binomial lower-tail Chernoff bound. Combining (44.3) and (44.4), summing over $y > x^{3/4}$ and $y \leq x$, and then over $x > I$, bounds the forbidden bridge mass by

$$\sum_{x>I} Cx^{4/3} \exp\left(-\frac{2}{3}\sqrt{x} + Cx^{1/3}\right) \longrightarrow 0. \quad (44.5)$$

This estimate is uniform in the terminal size. The correction to the false critical-weight domination is only polynomial and is harmless here.

44.2 From fixed height to fixed time

In (44.2), first retain only factors with $m < M$, obtaining $\rho_n^{[M]}$. Pointwise product comparison gives, uniformly in n ,

$$|\rho_n - \rho_n^{[M]}| \leq \frac{1}{2} \sum_{j>M} j^{-2} \leq \frac{1}{2M}.$$

For fixed M , additionally ignore a retained factor whose terminating rise occurs after path time L , obtaining $\rho_n^{[M,L]}$. This depends only on a fixed finite prefix. Summing the reweighted prefixes gives a finite initial vector at time L ; afterward the transfer is exactly relaxed. The tracking theorem therefore proves convergence of $\rho_n^{[M,L]}$ as $n \rightarrow \infty$.

If a retained rise occurs at time $i > L$, then its level is $m+1 \leq M$ and its transformed height is $j = i - 3(m+1) \geq i - 3M$. At the preceding block time $p = 3\lfloor i/3 \rfloor$, the height is at least $p - 3M$, because a forward step raises height by at most one. For fixed M and sufficiently large p , this exceeds $3(p/3)^{3/4}$. The uniform tail bound (44.5) therefore gives

$$\sup_n |\rho_n^{[M]} - \rho_n^{[M,L]}| \longrightarrow 0 \quad (L \rightarrow \infty).$$

Thus $\rho_n^{[M]}$ converges as a uniform limit of convergent sequences; then the $1/(2M)$ estimate proves convergence of ρ_n . Its limit lies in $[P_3, 1]$ by (44.1). Combining with the relaxed equivalent proves the DFA part of Theorem 38.1, including $C_B = \rho C_R/2$.

This transfer is qualitative in its truncation levels. It does not yield a finite-order DFA expansion.

45 Inverse asymptotics and integer thresholds

[write, 2 October 2026] An instance of theorems `p0:thm:lambert-core`, `p0:thm:perturbed-inversion` and `p0:thm:staircase` of the repository's transseries volume (Part 0); no novelty is claimed for the inversion mechanics.

Let T_n denote either R_n or B_n , and set respectively

$$(A, C) = (27/4, C_R) \quad \text{or} \quad (27/2, C_B).$$

For $y \rightarrow \infty$, define

$$L = \log y, \quad w = W_0 \left(\frac{\sqrt{A} L}{2e} \right), \quad n_0 = \frac{L}{2w}, \quad D = 2(w + 1),$$

where W_0 is the principal real Lambert branch. Let $x(y)$ be the inverse of the linear interpolation of $\log T_n$ between consecutive integers, with target value $\log y$.

Corollary 45.1 (Inverse equivalent and corrections). *For both counting sequences,*

$$\begin{aligned} x(y) &= n_0 - \frac{\gamma n_0^{1/3} + (8/3) \log n_0 + \log(2\pi C)}{D} + o(1/\log n_0) \\ &= n_0 - \frac{\gamma n_0^{1/3}}{D} - \frac{4}{3} - \frac{\log(2\pi C) - (4/3) \log A}{D} + o(1/\log n_0). \end{aligned} \quad (45.1)$$

The least integer threshold $\nu(y) = \min\{n : T_n \geq y\}$ equals $\lceil x(y) \rceil$ eventually. In particular (45.1) gives its value with an $O(1)$ final error. Rounding the displayed approximation itself is not an exact uniform formula near integer thresholds.

Proof. Stirling and the leading equivalent give

$$\log T_n = H_0(n) + g(n) + o(1),$$

where

$$H_0(t) = 2t \log t + (\log A - 2)t, \quad g(t) = \gamma t^{1/3} + \frac{8}{3} \log t + \log(2\pi C).$$

The coefficient $8/3$ includes the logarithm from the two factorials. Consecutive log increments are $2 \log n + \log A + o(1) > 0$, so both inverses are eventually well defined. Linear interpolation of the smooth model adds $O(1/n)$, and interpolation of the sequence's $o(1)$ error remains uniformly $o(1)$.

The equation $H_0(n_0) = L$ has the stated exact Lambert solution, with $H'_0(n_0) = D$. Put $\delta = -g(n_0)/D = O(n_0^{1/3}/\log n_0)$. Taylor's formula gives

$$H_0(n_0 + \delta) + g(n_0 + \delta) - L = O(\delta^2/n_0) + O(n_0^{-2/3}|\delta|) = o(1).$$

The actual interpolant has slopes asymptotic to $2 \log n_0$, proving the error in (45.1). The second expression follows from $(8/3) \log n_0 = (4/3)D - (4/3) \log A$. Exact rounding applies to the actual inverse by monotonicity, not automatically to its approximation. $\square$

In particular $x(y) \sim L/(2 \log L)$. Since $\gamma < 0$, the stretched correction increases the inverse by

$$-\frac{\gamma n_0^{1/3}}{D} \sim \frac{-\gamma}{2^{4/3}} \frac{L^{1/3}}{(\log L)^{4/3}}.$$

For relaxed trees, Theorem 38.2 also gives constructive inverses to every finite order, provided the interpolation convention is retained. Taking logarithms and Stirling yields a smooth truncation

$$H_M(t) = H_0(t) + g(t) + \sum_{r=1}^M \lambda_r t^{-r/3},$$

with error $O(n^{-(M+1)/3})$ at integers. Let $\mathcal{I}$ denote linear interpolation and define x_M by $(\mathcal{I}H_M)(x_M) = L$. Then

$$x(y) - x_M = O\left(\frac{n_0^{-(M+1)/3}}{\log n_0}\right).$$

This statement requires no derivative bounds on the enumeration remainder. A smooth-model inverse instead solves $H_M(z_M) = L$ and can be obtained by Newton iteration from n_0 ; its error squares with factor $O((n_0 \log n_0)^{-1})$ at each step. These two inverses differ at high order. In fact, with $\theta = \{z_M\}$,

$$x_M - z_M = -\frac{\theta(1-\theta)}{z_M H'_M(z_M)} + O\left(\frac{z_M^{-5/3}}{\log z_M}\right).$$

The fractional-part term comes from the chord error of the convex leading logarithm. A pure smooth inverse series must not silently be identified with the actual log-linear inverse beyond this order. No analogous all-orders DFA conclusion is needed or asserted here.

46 Verification and scope

The accompanying reproducibility package contains two independent exact polynomial–Airy recursion implementations, rational checks of the critical and moving boundary identities, exact completed-run and backward-monotonicity checks, and diagnostic numerical residual and inverse tests. The diagnostic floating checks verify implementations and signs; the proofs above supply the infinite-range conclusions.

Independent reviews cover the singular compactness and quasimode argument, the every-fixed-order transfer, the DFA amplitude transfer with the repaired tail, and the inverse conventions. The report keeps the inherited relaxed Theta theorem separate from its new proof steps. It includes no full third-party paper and makes no claim of an exhaustive novelty search. All results here are for the ternary case. The leading amplitudes are positive but are not numerically or algebraically evaluated.

Orders of growth by the completed-run comparison (source 29)

Source. Batch-77 manuscript 29, archive `larger-alphabet-automata-report.zip`, main file `article/larger-alphabet-automata.tex` (344 lines; 6-page PDF), titled “Finite languages over larger alphabets: Summable exclusion and sharp order estimates”, dated 2 October 2026, empty author line. Handed over by cluster P4 and printed in full as the comparison section of Part III, label prefix `air:ka:al:`.

Abstract of source 29.

For each fixed alphabet size $k \geq 3$, we transfer the known sharp order of relaxed k -ary trees to minimal deterministic automata recognizing finite languages. The proof is a positive completed-run representation: exclusion reduces each run weight by at most $1/[2(m+1)^{k-1}]$, whose sum converges. We obtain an explicit uniform comparison constant, the factorial, exponential, Airy and polynomial scales, a uniform finite-level approximation, and a logarithmic inverse enclosure. We do not prove a limiting amplitude or an all-orders expansion. A boundary calculation explains why the binary one-step singular-profile argument does not extend without modification.

[**write, 2 October 2026**] Source 29 is the earliest form of the positive completed-run comparison that sources 18 and 66 use as a proof step ((35.2), (44.1)); Theorem 47.1 is where this report states it. Its Θ conclusion (47.4) is superseded by Theorem 30.1, which proves a positive amplitude and every fixed order. What remains source 29's own: the explicit finite- n constant $P_{k,n}$, uniform in $k \geq 3$ and $n \geq 1$; the explicit lower bound $P_k \geq (2\sqrt{2}/\pi) \sin(\pi/\sqrt{2})$; the uniform finite-level approximation of Proposition 49.1; the Θ -only inverse enclosure of Theorem 50.1, with the injection $B_{n+1} \geq kB_n$; the count by total states; and the boundary obstruction of Section 51, which the ternary weight (Lemma 39.1) and Lemma 31.1 resolve. Source 66's transfer audit lists source 29's research proof among its audited inputs.

47 Statement and state convention

Fix an integer $k \geq 3$, and write $q = k - 1$. Let B_n count finite languages over a fixed ordered k -letter alphabet whose minimal complete deterministic automaton has exactly n transient states and one rejecting sink, up to state isomorphism. Thus $B_0 = 1$ counts the empty language, and $B_1 = 1$ counts $\{\varepsilon\}$. Let R_n count relaxed k -ary trees with n internal nodes, in the convention of Ghosh Dastidar–Wallner [3].

Set

$$C_k = \frac{2k^k}{q^q}, \quad \beta_k = 3z \left(\frac{kq}{2} \right)^{1/3}, \quad \alpha_k = \frac{2k-1}{3}, \quad (47.1)$$

where $z = -2.338107410\dots$ is the largest negative zero of Ai.

Theorem 47.1. *For every $n \geq 1$,*

$$P_{k,n} \leq \frac{B_n}{2^{n-1}R_n} \leq 1, \quad P_{k,n} = \prod_{j=2}^n \left(1 - \frac{1}{2j^{k-1}} \right). \quad (47.2)$$

Furthermore,

$$P_k := \prod_{j=2}^{\infty} \left(1 - \frac{1}{2j^{k-1}} \right) \geq \frac{2\sqrt{2}}{\pi} \sin \frac{\pi}{\sqrt{2}} > 0. \quad (47.3)$$

Consequently, for each fixed $k \geq 3$,

$$B_n = \Theta \left((n!)^{k-1} C_k^n \exp(\beta_k n^{1/3}) n^{\alpha_k} \right). \quad (47.4)$$

The only asymptotic input is the published relaxed-tree theorem [3, Theorem 1]. The comparison (47.2) is exact and elementary. Its lower constant is uniform in both $k \geq 3$ and $n \geq 1$; the unspecified constants in (47.4) may depend on k . For ternary alphabets the scales are

$$(n!)^2 (27/2)^n \exp(3 \cdot 3^{1/3} z n^{1/3}) n^{5/3}.$$

Remark 47.2 (Total states). The exponent $\alpha_k = (2k-1)/3$ uses *transient* states. If $N = n+1$ is the total number of states and $D_N = B_{N-1}$, then

$$D_N = \Theta \left((N!)^{k-1} C_k^N \exp(\beta_k N^{1/3}) N^{(2-k)/3} \right).$$

Indeed $(N-1)! = N!/N$, and the remaining shifted factors change only by a bounded factor tending to a positive constant.

48 Exact recurrence and positive completed runs

The arrays in [3, Propositions 8 and 12] satisfy, on $x \geq qm \geq q$,

$$r_{x,m} = r_{x,m-1} + (m+1)r_{x-1,m}, \quad (48.1)$$

$$b_{x,m} = 2b_{x,m-1} + (m+1)b_{x-1,m} - mb_{x-k,m-1}. \quad (48.2)$$

Their base row is $r_{x,0} = b_{x,0} = 1$ for $x \geq 0$, and they vanish when $x < qm$ for $m \geq 1$. Retain the auxiliary value $b_{-1,0} = 1$; all other negative-coordinate values are zero. Then

$$R_n = r_{qn,n}, \quad B_n = b_{qn,n}.$$

The auxiliary boundary is essential: omitting it would give $B_1 = 2$, in conflict with the combinatorial interpretation and Table 3 of [3]. Equivalently, prescribe $b_{x,1} = 2^{x-k+2} - 1$ for $x \geq k-1$ and use (48.2) above that row. This makes explicit the first-state convention implicit in the terse published recurrence and consistent with the binary case [2].

[write, 2 October 2026] The binary recurrence with this auxiliary value is Part IV's (53.1).

For $m \geq 1$ and $\ell \geq 0$ define

$$V_m(\ell) = (m+1)^\ell, \quad W_m(\ell) = 2(m+1)^\ell - \mathbf{1}_{\{\ell \geq k\}}(m+1)^{\ell-k+1}. \quad (48.3)$$

Let $a(x,1) = A(x,1) = 1$ for $x \geq q$. Both arrays vanish for $x < qm$. For $x \geq q(m+1)$ set

$$a(x, m+1) = \sum_{\ell=0}^{x-qm} V_m(\ell) a(x-\ell, m), \quad (48.4)$$

$$A(x, m+1) = \sum_{\ell=0}^{x-qm} W_m(\ell) A(x-\ell, m). \quad (48.5)$$

Proposition 48.1. *All these weights are nonnegative. For $x \geq qm$ and $m \geq 1$,*

$$r_{x,m} = \sum_{\ell=0}^{x-qm} (m+1)^\ell a(x-\ell, m), \quad (48.6)$$

$$b_{x,m} = \sum_{\ell=0}^{x-qm} (m+1)^\ell A(x-\ell, m). \quad (48.7)$$

In particular $R_n = a(qn, n)$ and $B_n = A(qn, n)$ for $n \geq 1$.

Proof. The completed-run generating series is

$$\sum_{\ell \geq 0} W_m(\ell) t^\ell = \frac{2 - (m+1)t^k}{1 - (m+1)t}.$$

Thus (48.5) and (48.7) give $A(x, m+1) = 2b_{x,m} - (m+1)b_{x-k,m}$, while adding the trailing horizontal steps gives $b_{x,m+1} = A(x, m+1) + (m+2)b_{x-1,m+1}$. Their combination is (48.2). The first row in (48.7) is $\sum_{\ell=0}^{x-q} 2^\ell = 2^{x-k+2} - 1$. Uniqueness of the triangular recurrence proves the identity. Replacing the numerator by 1 gives (48.1) and (48.6). At $x = qn, m = n$ only the term $\ell = 0$ survives. $\square$

49 Summable exclusion and its consequences

Proof of Theorem 47.1. For every $m \geq 1$ and $\ell \geq 0$,

$$2 \left(1 - \frac{1}{2(m+1)^{k-1}} \right) V_m(\ell) \leq W_m(\ell) \leq 2V_m(\ell).$$

Induction in the nonnegative convolutions (48.4)–(48.5) therefore proves

$$2^{m-1} P_{k,m} a(x, m) \leq A(x, m) \leq 2^{m-1} a(x, m).$$

The common trailing-run convolution preserves these inequalities for r, b . Taking the endpoint proves (47.2). Since $k \geq 3$, the sum of the deficits $1/(2j^{k-1})$ converges and all factors are positive; hence $P_k > 0$. The factors increase with k . Euler’s sine product gives

$$P_3 = \frac{\prod_{j=1}^{\infty} (1 - 1/(2j^2))}{1 - 1/2} = \frac{2\sqrt{2}}{\pi} \sin \frac{\pi}{\sqrt{2}}.$$

Finally, the published theorem [3] is

$$R_n = \Theta \left((n!)^{k-1} \left(\frac{k^k}{(k-1)^{k-1}} \right)^n \exp(\beta_k n^{1/3}) n^{(2k-1)/3} \right).$$

Combining it with (47.2) proves (47.4). □

The same comparison holds when $k = 2$, but its lower product then tends to zero like $n^{-1/2}$. It cannot recover the binary-DFA power $7/8$. The summability threshold at $k = 3$ is the reason this larger-alphabet leading-order extension is simpler than the binary amplitude problem.

[write, 2 October 2026] The binary amplitude problem is solved in Part IV (source 48): $b_n \sim \gamma n! 8^n e^{3zn^{1/3}} n^{7/8}$ with $\gamma > 0$, by cancelling a non-summable drift rather than by a summable comparison.

Proposition 49.1 (Uniform finite-level approximation). *For an integer $M \geq 1$, retain W_m at levels $m < M$, replace it by $2V_m$ for $m \geq M$, and keep the first row unchanged. Write $B_n^{[M]}$ for the resulting endpoint count. For every $n \geq 1$ and $k \geq 3$,*

$$0 \leq 1 - \frac{B_n}{B_n^{[M]}} \leq \frac{1}{2} \sum_{j>M} j^{-(k-1)} \leq \frac{1}{2(k-2)M^{k-2}}. \quad (49.1)$$

Proof. The exact ratio lies between 1 and $\prod_{j=M+1}^{\max(M,n)} (1 - 1/(2j^{k-1}))$, by the same convolution comparison. Use $1 - \prod_j (1 - u_j) \leq \sum_j u_j$ and the integral bound for the decreasing tail. □

If $B_n^{[M]}/(2^{n-1}R_n)$ converges for every fixed M , then (49.1) proves convergence of $B_n/(2^{n-1}R_n)$ by a uniform approximation argument. This reduces, but does not solve, the ratio-limit problem.

[write, 2 October 2026] The ratio-limit problem is solved in Part III: for $k = 3$ by exactly this route in Section 44.2 (source 66), and for every $k \geq 3$, with the first defect and eventual monotonicity, by Theorem 30.2 (source 18).

50 A logarithmic inverse enclosure

[write, 2 October 2026] An instance of theorems `p0:thm:lambert-core` and `p0:thm:staircase` of the repository's transseries volume (Part 0); no novelty is claimed for the inversion mechanics. Theorem 30.1 now gives the amplitude term that this enclosure leaves open, and Section 36 the corresponding finer inverse.

This section uses only (47.4); it does not posit an unknown amplitude. Put

$$\lambda_k = C_k^{1/q}/e, \quad \eta_k = \alpha_k + q/2 = \frac{7k-5}{6},$$

and define for large positive t

$$F_k(t) = qt \log(\lambda_k t) + \beta_k t^{1/3} + \eta_k \log t. \quad (50.1)$$

Stirling's formula and (47.4) give the precise bounded-error statement

$$\log B_n = F_k(n) + O_k(1). \quad (50.2)$$

In particular the coefficient of $\log n$ is η_k , not α_k .

For $y \rightarrow \infty$, let W be the positive real Lambert function and set

$$t = \frac{y}{qW(\lambda_k y/q)}, \quad L = 1 + \log(\lambda_k t), \quad (50.3)$$

$$\nu_k(y) = t - \frac{\beta_k t^{1/3} + \eta_k \log t}{qL}. \quad (50.4)$$

Theorem 50.1. *There are constants $K_k, y_k > 0$ such that, for $y \geq y_k$,*

$$\left| \nu_k(y) - \frac{K_k}{\log \nu_k(y)} \right| \leq I_k(e^y) \leq \left| \nu_k(y) + \frac{K_k}{\log \nu_k(y)} \right|, \quad (50.5)$$

where $I_k(X) = \min\{n \geq 1 : B_n \geq X\}$. Moreover

$$n = \nu_k(\log B_n) + O_k(1/\log n).$$

The constants here are existence bounds, not effective numerical certificates.

Proof. Write $f(t) = qt \log(\lambda_k t)$ and $g(t) = \beta_k t^{1/3} + \eta_k \log t$. Formula (50.3) gives $f(t) = y$ and $f'(t) = qL$. With $\delta = -g(t)/(qL) = O(t^{1/3}/\log t)$, Taylor's formula yields

$$F_k(t + \delta) - y = O(\delta^2/t) + O((t^{-2/3} + t^{-1})|\delta|) = o(1).$$

Since $F'_k(t) \sim q \log t > 0$, (50.2) implies the asserted $O(1/\log n)$ inversion at sequence values. Choose a constant bounding the error in (50.2). Increasing K_k if necessary makes every sufficiently large integer below $\nu_k(y) - K_k/\log \nu_k(y)$ have $\log B_n < y$, and every sufficiently large integer at or above the upper endpoint have $\log B_n \geq y$. Taking ceilings gives (50.5).

The threshold is well defined: for every nonempty finite language L , each language aL , with a a letter, has one more transient state in its minimal complete automaton. The old initial residual L is reached after a ; all its residuals remain reachable, and the new initial residual has strictly larger maximum word length. The k constructions are disjoint and injective, so $B_{n+1} \geq kB_n$ for $n \geq 1$. $\square$

The enclosure respects integer rounding. It does not assert a smooth interpolation canonically determined by the sequence, nor an everywhere unambiguous rounded estimate without the displayed error interval.

51 What remains open and a boundary obstruction

Neither the relaxed theorem nor the comparison proves that the quotient in (47.4) has a limit. A positive limiting amplitude and higher-order endpoint corrections for this family remain unproved here. A relaxed amplitude theorem alone would still require control of the finite-level perturbed ratios in Proposition 49.1.

[**write, 2 October 2026**] A positive limiting amplitude, and the endpoint corrections to every fixed order, are proved for every fixed $k \geq 3$ in Theorem 30.1 (source 18; for $k = 3$ the amplitude also in Theorem 38.1, source 66). The obstruction below is resolved there by the fixed weight $\omega = \ell/h$, which matches the forward and adjoint boundary harmonics on every residue phase (Lemmas 39.1 and 31.1); the paragraph after Lemma 39.1 names the residue-dependent extrapolation lengths computed here.

The critically tilted relaxed transfer has two shifts:

$$(T_i f)(j) = U(i, j)f(j-1) + f(j+k-1), \quad U(i, j) = \frac{(k-1)^2(i-j+k)}{(k-1)i+j}.$$

For $k \geq 3$, its directed zero pattern has a forward edge of length one without its reverse. Positive diagonal conjugation cannot make it symmetric. Although $T_i^* T_i$ is Jacobi on residue classes modulo k , the following exact frozen calculation prevents an automatic binary singular-profile proof.

Let $T = q \text{ up} + \text{down}_q$ on the half-line, and let S be the unilateral lower shift in residue coordinate $\ell \geq 0$. From input residue r to output residue $r+1 \pmod k$, the matrix is

$$T_r = qI + S^* \quad (0 \leq r < q), \quad T_q = I + qS.$$

At the bulk squared edge k^2 , a formal linear solution $\ell + c$ of the right and left Jacobi boundary equations has constants

input residue	c_{right}	c_{left}	physical extrapolated zero
$0 \leq r < q$	q/k	1	$r - q$
$r = q$	1	$1/k$	-1

For example the first right equation is $q^2 c + q(c+1) = k^2 c$, giving $c = q/k$; the first left equation has bottom diagonal $q^2 + 1$ and gives $c = 1$. In physical coordinates the right and left zeros agree within one step, but change between successive residues. The boundary offsets cycle as $q, q-1, \dots, 1, 1$, so only the binary case has a constant offset. Thus an $O(i^{-1/3})$ boundary-scale mismatch may occur between successive singular profiles, and an $O(i^{-1})$ tracking estimate requires an additional cancellation argument. The finite operator's Airy asymptotics are not asserted by this frozen calculation.

A whole k -step analysis or matched forward and adjoint boundary profiles could address this issue. Also $U(i, j)$ can exceed q near the boundary, so global contraction requires its own normalization. These analytic questions are separate from the positive proof of Theorem 47.1.

52 Verification and literature scope

The attached standard-library Python verifier checks signed and positive recurrences, all-cell comparisons, truncation bounds, and published terms for $2 \leq k \leq 8$ through 16 transient states.

Independent small-automaton checks and mathematical review are recorded separately. In particular,

$$\begin{aligned} k = 3 : & \quad 1, 1, 14, 532, 42644, 6011320, 1330452032, \dots, \\ k = 4 : & \quad 1, 1, 30, 3900, 1460700, 1220162880, \dots \end{aligned}$$

A331120 is the binary minimal-DFA sequence. A082162 counts relaxed ternary trees, not minimal ternary DFAs. No larger-arity minimal-DFA OEIS identifier was verified in the bounded search.

The 2024 primary paper’s conclusion leaves compacted and minimal-DFA asymptotics to a projected long version. The current author bibliography, focused later-literature searches, and authorized ProveIt semantic searches were checked on 2 October 2026; no matching later result was located. Wallner’s 2026 decompression paper [5] concerns a different statistic on chains. These are bounded overlap checks, not a universal novelty guarantee. The attached receipts specify the queries and limitations.

[**write, 2 October 2026**] The repository searches were right when made (the receipt refers to commit 4b874cea0). The repository’s material on these automata is now this report, which the same batch delivered. The raw search receipt `sources/overlap.json` is not shipped; see the README.

Part IV

Binary finite-language automata

Source. Batch-77 manuscript 48, archive `oeis-dfa-finite-language-report.zip`, main file `article/dfa-finite-languages.tex` (967 lines; 16-page PDF), titled “Positive amplitude and all orders for finite binary languages”, dated 2 October 2026, empty author line, placed by commit aa7345800 (cluster P2). Label prefix `air:dfa:`. Its passages that are word for word Part II’s are replaced by pointers, and its appendix is Appendix A; every passage in which the automaton differs is printed.

Abstract of the source.

A331120 counts finite binary languages by the number of transient states in their minimal deterministic automata. The known estimate has order $n!8^n \exp(3zn^{1/3})n^{7/8}$, where z is the largest negative Airy zero. We strengthen it to a positive limiting amplitude and a relative expansion at every finite order in $n^{-1/3}$. An exact nonnegative renewal model establishes positivity and domination, while a half-strength signed delay of a contractive Jacobi evolution identifies the polynomial correction. A finite-memory projection argument and an arbitrary-order residual bootstrap yield endpoint accuracy without assuming that the signed propagator is positive. We give three explicit logarithmic corrections, a constructive coefficient recursion, a selected smooth inverse, and the proper distinction between inverse interpolation and discrete thresholds. The shared spectral estimates are proved in a self-contained appendix. The amplitude exists and is positive; its decimal digits are not certified.

53 Statement and mathematical scope

Let $b_{n,m}$ be the array

$$\begin{aligned} b_{n,m} &= 2b_{n,m-1} + (m+1)b_{n-1,m} - mb_{n-2,m-1} \quad (n \geq m \geq 1), \\ b_{n,0} &= 1 \quad (n \geq -1), \quad b_{n,m} = 0 \quad (m > n). \end{aligned} \quad (53.1)$$

The auxiliary value $b_{-1,0} = 1$ is retained. Put $b_n = b_{n,n}$. This is A331120, beginning

$$1, 1, 6, 60, 900, 18480, 487560, 15824880, 612504240, \dots$$

It counts minimal complete deterministic automata over a fixed binary alphabet, up to isomorphism of states, with n transient states and one rejecting sink. Equivalently it counts the corresponding finite languages by minimal-state complexity. Proposition 5 of [2] proves the exact recurrence. The same paper's Theorem 1 proves

$$b_n = \Theta\left(n!8^n e^{3zn^{1/3}} n^{7/8}\right), \quad (53.2)$$

where $z = -2.338107410459767\dots$ is the largest real zero of Ai. The Airy exponential and exponent $7/8$ are therefore established inputs. The additional result here is a convergent positive amplitude and its complete finite algebraic correction hierarchy.

Set $a = 2^{-1/3}z$ and $F(x) = \text{Ai}(z + 2^{1/3}x)$, so $F'' = 2(x+a)F$, $F(0) = 0$, and $F'(0) > 0$.

Theorem 53.1. *There is a constant $\gamma > 0$ such that, for every fixed integer $M \geq 0$,*

$$b_n = \gamma n!8^n e^{3zn^{1/3}} n^{7/8} \left(1 + \sum_{k=1}^M \mathbf{c}_k n^{-k/3} + O_M(n^{-(M+1)/3})\right). \quad (53.3)$$

All $\mathbf{c}_k$ are constructible elements of $\mathbb{Q}[z]$. In logarithmic form the first three corrections are

$$\begin{aligned} \log \frac{b_n}{\gamma n!8^n e^{3zn^{1/3}} n^{7/8}} &= \frac{53z^2}{90} n^{-1/3} + \frac{623z}{432} n^{-2/3} \\ &+ \left(\frac{3497}{4480} - \frac{1304z^3}{42525}\right) n^{-1} + O(n^{-4/3}). \end{aligned} \quad (53.4)$$

This is a Poincaré expansion to every fixed finite order. It is not a claim of an exponentially improved transseries or a canonical analytic continuation of an integer sequence. Strict positivity of γ uses the published lower bound in (53.2) only after the convergence and all-orders error estimates have been proved.

54 Exact positivity and the initial boundary

Normalize uniformly by $c_{n,m} = b_{n,m}/2^m$. Then

$$c_{n,m} = c_{n,m-1} + (m+1)c_{n-1,m} - \frac{m}{2}c_{n-2,m-1}. \quad (54.1)$$

In particular $c_{n,0} = 1$, $c_{-1,0} = 1$, and $c_{n,1} = (2^n - 1)/2$. A normalization by 2^{m-1} only for $m \geq 1$ requires a separate boundary row; (54.1) avoids that exception.

Proposition 54.1 (Positive renewal representation). *Set $A(n, 1) = 1$ for every $n \geq 1$ and $A(n, m) = 0$ for $n < m$. For $m \geq 1$ put*

$$W_m(k) = \begin{cases} 2, & k = 0, \\ 2(m+1), & k = 1, \\ (2(m+1)^2 - (m+1))(m+1)^{k-2}, & k \geq 2. \end{cases} \quad (54.2)$$

For $n \geq m+1$, define

$$A(n, m+1) = \sum_{k=0}^{n-m} W_m(k) A(n-k, m).$$

Then

$$b_{n,m} = \sum_{k=0}^{n-m} (m+1)^k A(n-k, m) \quad (m \geq 1). \quad (54.3)$$

All weights are nonnegative, and $b_n = A(n, n)$ for $n \geq 1$.

Proof. The completed-run generating series is

$$\sum_{k \geq 0} W_m(k) t^k = \frac{2 - (m+1)t^2}{1 - (m+1)t}.$$

Consequently completing a run followed by a vertical step gives $A(n, m) = 2b_{n,m-1} - mb_{n-2,m-1}$ for $m \geq 2$. Adding possible trailing horizontal steps in (54.3) recovers (53.1). For $m = 1$, the initial renewal values give $b_{n,1} = \sum_{k=0}^{n-1} 2^k = 2^n - 1$, which agrees with (53.1), including its initial value $b_{-1,0}$. The triangular cutoff ensures that only $k = 0$ survives on $n = m$. Uniqueness of the triangular recurrence proves the assertion. $\square$

Combinatorially A counts prefixes ending in a vertical step. At height m , a horizontal run has $m+1$ choices per edge and two vertical colors; the subtracted completed-run decorations are exactly the forbidden repeated transition signature. The exceptional first row includes the mandatory initial horizontal edge and the requirement that the first non-sink state be accepting. The algebraic proof above makes the initial convention exact.

Let $r_{n,m}$ denote the relaxed array $r_{n,0} = 1$, $r_{n,m} = r_{n,m-1} + (m+1)r_{n-1,m}$, with invalid entries zero. Since $c_{n,m} \geq 0$, induction in (54.1) yields

$$0 \leq c_{n,m} \leq r_{n,m}. \quad (54.4)$$

This is a domination of exact counts; no positivity of the signed linear propagator will be assumed.

55 The shared Jacobi evolution

[write, 2 October 2026] Sections 55–60 and Appendix A of source 48 follow the compacted-tree argument of Part II line by line; only the renewal model, the delay coefficient (with its $m = 1$ term), the drift $1/(8N)$, the seeds and the coefficients differ. Source 48 does not say so. Its Section 3 up to the paragraph below is, word for word, the beginning of Part II's Section 18: the gauge (18.1), the Jacobi matrix S_N with edge weights $b_{N,j}$ and the contraction Q_N (18.2), the vectors g_N, h_N and the operators P_N, M_N, Π_N (18.3), Lemma 18.1 with (18.4) and the finite profile (18.5), and the remark that the gap is parity-restricted. It is not reprinted, and the

references below to these displays and to Lemma 18.1 are to Part II. Source 48 adds only the conventions $D_0(0) = 1$, $P_0 = 1$ and $v_0^{(r)} = \delta_0$. Its appendix “Self-contained Jacobi estimates” is Appendix A with “DFA theorem” in place of “compacted theorem”, and is not reprinted either.

The relaxed counts $r_{n,m}$ satisfy

$$r_{n,0} = 1, \quad r_{n,m} = r_{n,m-1} + (m+1)r_{n-1,m},$$

with invalid cells zero. By (54.4), $0 \leq c_{n,m} \leq r_{n,m}$. The common gauge turns the relaxed recurrence into $v_N^{(r)} = S_N Q_N v_{N-1}^{(r)}$; consequently $\|v_N^{(r)} / P_N\| \leq 1$. Only this domination and the explicit Jacobi estimates, not an automatic extension of a relaxed amplitude theorem, will be used to prove Theorem 53.1.

56 A globally bounded signed delay

Set

$$e_{N,j} = \frac{c_{(N+j)/2, (N-j)/2}}{((N+j)/2)!}, \quad v_N(j) = \frac{e_{N,j}}{D_N(j)}$$

on $0 \leq j \leq N$, $j \equiv N \pmod{2}$, and zero otherwise. For $N \geq 3$, direct substitution in (54.1) gives

$$e_{N,j} = \frac{N-j+2}{N+j} e_{N-1,j-1} + e_{N-1,j+1} - \frac{N-j}{(N+j)(N+j-2)} e_{N-3,j-1}. \quad (56.1)$$

The physical recurrence starts at $N = 3$, using the exact initial vectors

$$e_0(0) = 1, \quad e_1(1) = 1, \quad e_2(0) = e_2(2) = \frac{1}{2},$$

with all other entries zero. At $N = 2, j = 0$ the fictitious $c_{-1,0} = 1$ contributes to (54.1); omitting it would change the seed. For $N \geq 3$ the delay is used only on $1 \leq j \leq N-2$ with the correct parity. It is zero at $j = 0$ and absent on $m = 0$. It is *nonzero* at $m = 1, j = N-2$, unlike the compacted-tree delay.

[**write, 2 October 2026**] In Part II the delay coefficient $2(N-j-2)/((N+j)(N+j-2))$ vanishes at $j = N-2$ (Section 19); here it is $(N-j)/((N+j)(N+j-2))$, nonzero there.

For all sufficiently large N , the exact gauge recurrence is

$$v_N = S_N Q_N v_{N-1} - L_N v_{N-3}, \quad (L_N f)(j) = \ell_{N,j} f(j-1), \quad (56.2)$$

where on $1 \leq j \leq N-2$

$$\ell_{N,j} = \frac{N-j}{(N+j)(N+j-2)} \frac{D_{N-3}(j-1)}{D_N(j)}, \quad (56.3)$$

and it is zero elsewhere. Cancellation of factorials gives

$$\left(\frac{D_{N-3}(j-1)}{D_N(j)} \right)^2 = \frac{(N-j+1)(N-j)(N+j)(N+j-1)(N+j-2)(N+j-3)}{(N+1)N^2(N-1)^2(N-2)}. \quad (56.4)$$

For $N \geq 6$ this ratio is uniformly bounded on the entire physical support; the rational prefactor in (56.3) is $O(N^{-1})$. Hence

$$\|L_N\| \leq C N^{-1}. \quad (56.5)$$

This is a global operator bound, not an estimate limited to the Airy window.

For $1 \leq j \leq 2N^{1/2}$ on the physical support, expanding (56.3)–(56.4) gives

$$\ell_{N,j} = \frac{1}{N} + O\left(\frac{j+1}{N^2}\right).$$

Let U be the unilateral upward shift: $(Uf)(0) = 0$ and $(Uf)(j) = f(j-1)$. The finite profile in Lemma 18.1, its Airy moments, and (56.5) imply

$$\left\|L_N g_{N-3} - \frac{1}{N} U g_{N-3}\right\| = O(N^{-5/3}), \quad \|U g_{N-3} - g_N\| = O(N^{-1/3}). \quad (56.6)$$

For the first estimate the typical height is $O(N^{1/3})$, and the $O(N^{-4/3})$ quasimode error is multiplied by $O(N^{-1})$. Cutoff tails are exponentially small. The second estimate uses a one-site sample shift and an $O(N^{-1})$ time change. The shifted parity is exactly the parity occupied at time N .

57 Cancellation of the scalar drift

Put

$$R_N = P_N N^{-1/8}, \quad u_N = v_N / R_N, \quad (57.1)$$

and begin the following recurrences at any fixed sufficiently large N . Then

$$u_N = \eta_N M_N u_{N-1} - B_N u_{N-3}, \quad \eta_N = \left(\frac{N}{N-1}\right)^{1/8}, \quad B_N = \frac{R_{N-3}}{R_N} L_N. \quad (57.2)$$

Since $R_{N-3}/R_N = 1/8 + O(N^{-2/3})$, Lemma 18.1 and (56.6) yield

$$\begin{aligned} \eta_N &= 1 + \frac{1}{8N} + O(N^{-2}), \quad \|B_N\| = O(N^{-1}), \\ \|B_N g_{N-3} - \frac{1}{8N} g_N\| &= O(N^{-4/3}), \\ b_N := \langle g_N, B_N g_{N-3} \rangle &= \frac{1}{8N} + O(N^{-4/3}), \\ \eta_N t_N - 1 - b_N &= O(N^{-4/3}). \end{aligned} \quad (57.3)$$

The last line cancels the nonsummable order- N^{-1} drift. It explains the additional factor $N^{-1/8}$ relative to the relaxed spectral normalization.

[**write, 2 October 2026**] Source 48 states here the finite-memory Lemma 20.1, with (20.4), (20.5) and its proof, word for word as in Part II; it is not reprinted, and the references to it below are to Part II.

58 Convergence of the scalar amplitude

The positive gauge preserves $0 \leq v_N \leq v_N^{(r)}$ entrywise. The relaxed contraction therefore gives the initial bound

$$\|u_N\| \leq N^{1/8}. \quad (58.1)$$

Decompose

$$u_N = A_N g_N + w_N, \quad w_N \perp g_N.$$

Applying Lemma 18.1 to the transverse part of (57.2), and bounding the delayed whole vector using (58.1), gives

$$\|w_N\| \leq (1 - c' N^{-2/3}) \|w_{N-1}\| + O(N^{-7/8}).$$

The factor $\eta_N = 1 + O(N^{-1})$ is absorbed in the gap. Lemma 20.1 then gives

$$\|w_N\| = O(N^{-5/24}). \quad (58.2)$$

No cancellation or positivity of the delayed evolution is needed here.

Projecting the recurrence gives

$$A_N = \eta_N t_N A_{N-1} - b_N A_{N-3} + \xi_N, \quad |\xi_N| \leq C N^{-1} (\|w_{N-1}\| + \|w_{N-3}\|). \quad (58.3)$$

The one-step cross term uses $\|M_N^* g_N - g_{N-1}\| = O(N^{-1})$ and orthogonality; the delayed cross term uses only $\|B_N\| = O(N^{-1})$. Set $\delta_N = A_N - A_{N-1}$. Then

$$\delta_N = b_N (\delta_{N-1} + \delta_{N-2}) + (\eta_N t_N - 1 - b_N) A_{N-1} + \xi_N. \quad (58.4)$$

Equations (57.3), (58.1) and (58.2) give (20.5) with $r = 29/24$. Thus $\delta_N = O(N^{-29/24})$ is summable, and A_N has a finite limit A_∞ . Since u_N and g_N are nonnegative, $A_\infty \geq 0$.

Now A_N is bounded and $w_N \rightarrow 0$, so u_N is bounded. Repeating the transverse and scalar estimates improves them to

$$A_N = A_\infty + O(N^{-1/3}), \quad \|w_N\| = O(N^{-1/3}), \quad \|u_N\| = O(1). \quad (58.5)$$

There is not yet an endpoint conclusion: the normalized ground endpoint itself is only of order $N^{-1/2}$.

59 Formal profiles at every finite order

Let $\varepsilon = N^{-1/3}$ and $x = (j+1)\varepsilon$. Seek an approximate solution to the ungauged recurrence in the form

$$Z_{N,j} = H_N \Phi_N(x), \quad \Phi_N(x) = \sum_{k=0}^K \varepsilon^k G_k(x), \quad G_0 = F, \quad (59.1)$$

with $G_k(0) = G'_k(0) = 0$ for $k \geq 1$, and formal ratio

$$\frac{H_N}{H_{N-1}} \sim 2 \left(1 + \sum_{m \geq 2} s_m \varepsilon^m \right). \quad (59.2)$$

The one-step arguments are $(x \pm \varepsilon)(1 - \varepsilon^3)^{-1/3}$, and the delayed argument is $(x - \varepsilon)(1 - 3\varepsilon^3)^{-1/3}$. The horizontal and delayed coefficients in (56.1) become

$$\begin{aligned} \mathcal{A}(x, \varepsilon) &= \frac{1 - x\varepsilon^2 + 3\varepsilon^3}{1 + x\varepsilon^2 - \varepsilon^3}, \\ \beta(x, \varepsilon) &= \frac{\varepsilon^3(1 - x\varepsilon^2 + \varepsilon^3)}{(1 + x\varepsilon^2 - \varepsilon^3)(1 + x\varepsilon^2 - 3\varepsilon^3)}. \end{aligned} \quad (59.3)$$

After division by H_{N-1} , the delayed term has the extra factor

$$\frac{H_{N-3}}{H_{N-1}} = \left(\frac{H_{N-1}}{H_{N-2}} \frac{H_{N-2}}{H_{N-3}} \right)^{-1},$$

whose leading constant is $1/4$. As β starts at degree three, neither the new profile nor the new scalar in those previous ratios changes the leading Airy inversion at the current degree.

[**write, 2 October 2026**] Source 48 continues with the paragraph of Part II that gives the formal equation at degree m , (22.4), in which $R_m, S_m \in \mathbb{Q}[a][x]$ are already known and $s_2 = a$, and its triangular polynomial reduction (22.5), with existence and uniqueness under $Q(0) = 0$ and $P(0) + Q'(0) = 0$. The paragraph is word for word Part II's and is not reprinted; here the delayed coefficient is this Part's (59.3).

The initial scalar coefficients are

$$\begin{aligned} s_2 &= a, & s_3 &= \frac{29}{24}, & s_4 &= \frac{29a^2}{270}, \\ s_5 &= -\frac{103a}{648}, & s_6 &= \frac{5341815 - 562112a^3}{5443200}. \end{aligned} \quad (59.4)$$

A genuine finite logarithmic amplitude may consequently be chosen as

$$\log H_N = N \log 2 + 3aN^{1/3} + \frac{29}{24} \log N + \sum_{k=1}^J h_k N^{-k/3}. \quad (59.5)$$

At degree $k+3$ in the one-step ratio, the new h_k has coefficient $-k/3$, so any finite list of ratio coefficients can be matched uniquely. For every truncation the leading multiplicative constant in H_N is one. Initially,

$$h_1 = \frac{53a^2}{45}, \quad h_2 = \frac{551a}{216}, \quad h_3 = \frac{2371}{6720} - \frac{5216a^3}{42525}. \quad (59.6)$$

60 Global residuals and the zero-limit bootstrap

60.1 Approximate solutions in the normalized norm

Fix any $p > 1$. Take sufficiently many terms of (59.1) and (59.5), multiply by the cutoff $\chi(j/N^{1/2})$, restrict to occupied parity, and define

$$z_N = D_N^{-1} Z_N / R_N.$$

Then

$$\|z_N\| = O(1), \quad r_N := z_N - \eta_N M_N z_{N-1} + B_N z_{N-3}, \quad \|r_N\| = O(N^{-p}), \quad (60.1)$$

and

$$\langle g_N, z_N \rangle \longrightarrow q_0 = \kappa^{-1} \sqrt{I/2} > 0, \quad I = \int_0^\infty F(x)^2 dx. \quad (60.2)$$

Here is the required global justification. On $j \leq 2N^{1/2}$ the factorial product for D_N has $\log D_N^{-1} = O(j^2/N)$, so D_N^{-1} is uniformly bounded there. Taylor remainders in (59.3), the time dilations and

the finite amplitude ratios are bounded by an arbitrarily prescribed power of $N^{-1/3}$ times a fixed polynomial–Airy envelope. All such envelopes are square-integrable. The coefficient denominators stay uniformly away from zero. Cutoff derivative errors, and differences between one- and three-step cutoff supports, are exponentially small in $N^{1/8}$, up to powers of N . The bounded gauge preserves this property. At the left boundary all fictitious samples vanish exactly, because every $G_k(0) = 0$; the upper moving boundary lies outside the cutoff for large N . These facts give the norm residual after summation.

[**write, 2 October 2026**] Part II has $N^{1/4}$ here. The cutoff $\chi(j/N^{1/2})$ is the same, so both statements hold; the one with $N^{1/4}$ is the stronger.

For the normalization, $H_N/R_N \sim \kappa^{-1}N^{-1/6}$, and a parity-restricted Airy sample has norm asymptotic to $N^{1/6}\sqrt{I/2}$. Its direction tends to g_N , proving (60.2). Only the leading asymptotic of P_N has been used: the original approximate recurrence is divided by the exact R_N , so an all-orders expansion of the relaxed spectral product is unnecessary.

60.2 A bootstrap with vanishing limiting projection

[**write, 2 October 2026**] Source 48’s Section 8.2 is, word for word, Part II’s subsection of the same title (Section 23.2), equations (23.3)–(23.7), read with this Part’s η_N , B_N , r_N , z_N and Lemma 20.1. It is not reprinted. It sets

$$C_* = A_\infty/q_0, \quad E_N = u_N - C_*z_N = \alpha_N g_N + W_N, \quad W_N \perp g_N, \quad (60.3)$$

and concludes

$$\|E_N\| = O(N^{1-p}), \quad (60.4)$$

assuming neither positivity nor contractivity of the signed delayed propagator.

61 The endpoint and the common positive amplitude

Point evaluation has norm one on ℓ^2 , so $|E_N(0)| \leq \|E_N\| = O(N^{1-p})$. For even N , the leading normalized approximate endpoint is of order $N^{-1/2}$, because

$$H_N/R_N \sim \kappa^{-1}N^{-1/6}, \quad \Phi_N(\varepsilon) \sim F'(0)\varepsilon.$$

We may choose p arbitrarily large. For the relative remainder in (53.3), it is enough to take $p \geq 3/2 + (M+1)/3$ and retain enough formal terms. Thus (60.4) supplies every fixed finite relative endpoint order; no endpoint smoothing estimate for the signed kernel is needed.

If $A_\infty = 0$, then $C_* = 0$. Equation (60.4) with $p > 3/2$ would imply $u_N(0) = o(N^{-1/2})$. On the other hand, the known lower bound (53.2), divided by $2^n n! R_{2n}$, implies $u_{2n}(0) \geq cN^{-1/2}$ for large even $N = 2n$. This contradiction proves $A_\infty > 0$.

Since $D_N(0) = 1$ and $b_n = 2^n n! e_{2n,0}$, conversion from N to $2n$ gives

$$\begin{aligned} \gamma &= 2^{7/8} C_* F'(0) = 2^{15/8} \kappa A_\infty > 0, \\ A_\infty &= \lim_{N \rightarrow \infty} \left\langle g_N, v_N / (P_N N^{-1/8}) \right\rangle. \end{aligned} \quad (61.1)$$

We used the Airy identity $F'(0)/\sqrt{I} = \sqrt{2}$, proved in the appendix. Every truncation has the same q_0 and the same leading normalization of H_N . Hence C_* and γ are independent of the expansion order. This completes the analytic part of Theorem 53.1.

62 Explicit coefficients and their rational field

The normalized endpoint factor begins

$$\frac{\Phi_N(\varepsilon)}{F'(0)\varepsilon} = 1 + \frac{a}{3}\varepsilon^2 + \frac{29}{24}\varepsilon^3 + O(\varepsilon^4).$$

Combining this with (59.6), then setting $N = 2n$, gives (53.4). Exponentiation gives

$$\begin{aligned} \mathfrak{c}_1 &= \frac{53z^2}{90}, \\ \mathfrak{c}_2 &= \frac{z(5618z^3 + 46725)}{32400}, \\ \mathfrak{c}_3 &= \frac{8337112z^6 + 200508660z^3 + 191198475}{244944000}. \end{aligned} \tag{62.1}$$

These coefficients were independently checked by a direct polynomial linear-system solve, separate from the triangular recursion.

[**write, 2 October 2026**] The rest of source 48’s Section 10, the proof that every coefficient lies in $\mathbb{Q}[z]$ by the grading under $(a, \varepsilon, x) \mapsto (\omega a, \omega \varepsilon, \omega x)$, is word for word the second half of Part II’s Section 25, its invariance statement covering “the full formal recurrence, including its delayed dilation and both previous ratios”. It is not reprinted.

63 Selected smooth inverse and discrete thresholds

[**write, 2 October 2026**] The anchor $X = Y/W((8/e)Y)$ and the corrections are instances of theorems `p0:thm:lambert-core` and `p0:thm:perturbed-inversion` of the repository’s transseries volume, and the threshold remark is the separation step of `p0:thm:staircase` (Part 0). No novelty is claimed for the inversion mechanics.

Write ℓ_k for the logarithmic correction coefficients in Theorem 53.1. For each finite M define a positive-real model by exponentiating

$$\mathcal{F}_M(x) = \log \gamma + \log \Gamma(x+1) + x \log 8 + 3zx^{1/3} + \frac{7}{8} \log x + \sum_{k=1}^M \ell_k x^{-k/3}. \tag{63.1}$$

Its derivative is $\log(8x) + O(x^{-2/3})$, positive for sufficiently large x . Therefore $\mathcal{F}_M$ has a unique eventual inverse $x_M(Y)$. No values of b_n at nonintegers are asserted.

Put

$$Y = \log y, \quad w = W((8/e)Y), \quad X = Y/w.$$

The principal real Lambert function is used. Then X solves $X(\log X + \log 8 - 1) = Y$. Stirling expansion gives

$$\mathcal{F}_M(x) = x(\log x + \log 8 - 1) + 3zx^{1/3} + \frac{11}{8} \log x + \log(\gamma\sqrt{2\pi}) + O(x^{-1/3}).$$

Consequently

$$x_M(Y) = X - \frac{3zX^{1/3} + \frac{11}{8}\log X + \log(\gamma\sqrt{2\pi})}{1+w} + O\left(\frac{X^{-1/3}}{\log X}\right). \quad (63.2)$$

Indeed the correction is $O(X^{1/3}/\log X)$; the change in the stretched term's derivative contributes $O(X^{-1/3}/(\log X)^2)$ to the index, and the quadratic contribution from the factorial core is $O(X^{-1/3}/(\log X)^3)$. The first omitted forward coefficient is at most the remainder displayed in (63.2).

Higher corrections can be generated by reversion of (63.1), or by Newton iteration

$$x^{(r+1)} = x^{(r)} - \frac{\mathcal{F}_M(x^{(r)}) - Y}{\mathcal{F}'_M(x^{(r)})}.$$

On $x \asymp X$, $\mathcal{F}''_M(x) = O(1/x)$ and $\mathcal{F}'_M(x) \asymp \log X$, so each sufficiently accurate Newton step squares the index error up to the factor $O((X \log X)^{-1})$. Finite forward orders and finite Newton depth therefore give any desired finite algebraic inverse accuracy. They remain distinct errors: solving a fixed model more accurately does not improve its agreement with b_n . Precisely, Theorem 53.1 and the mean-value theorem imply

$$x_M(\log b_n) - n = O_M\left(\frac{n^{-(M+1)/3}}{\log n}\right). \quad (63.3)$$

For arbitrary y , the first integer index with $b_n \geq y$ is a different object. The asymptotics show eventual strict increase, since $b_{n+1}/b_n \sim 8n$, but a threshold still requires checking the adjacent integers or a certified bracket. An unspecified big- O constant is not a certified finite threshold algorithm.

64 Verification and numerical limitations

The accompanying code separates analytic proof from finite evidence. The exact script checks 5,050 triangular cells through $n = 100$ against the positive renewal construction, verifies 5,049 normalized recurrence instances, includes the nonzero $m = 1$ delay, and reproduces all 20 displayed OEIS terms. The symbolic coefficient generator implements the triangular polynomial inversion. A separate direct-linear-system computation checks its coefficients without importing the generator. The analytic estimates above, rather than these finite checks, establish the theorem.

For orientation only, direct floating evaluation of the normalized amplitude gives approximately 178.317, 164.945, 150.038, 136.080, and 131.985 at $n = 10, 20, 40, 80, 100$, respectively. These slowly varying values are *not* enclosures of γ , do not give certified digits, and are not used in the proof. Inverse numerical tests, when a trial amplitude is chosen, verify only the specified model with that trial parameter.

The package contains this standalone article, its source and build script, exact and symbolic verification programs and JSON outputs, a separate mathematical review, bounded literature and overlap receipts, a clean replay script, and a file manifest. The PDF build and all pages were visually checked separately from mathematical review.

[**write, 2 October 2026**] This paragraph describes source 48's delivered archive (commit 096ee7b87); its manuscript, PDF, build script and checksum manifest are not shipped. The README of this report lists the shipped files.

65 Literature boundary and further questions

The earlier binary-DFA result [2] provides the two-sided order, not the positive limiting amplitude. The later general-arity paper [3] gives a broader exact recurrence and a relaxed-tree asymptotic theorem. Its general-arity setting does not supply the present endpoint amplitude. Ordinary minimal automata for arbitrary languages have different asymptotics and should not be confused with this finite-language class. The official entry [7], the cited papers, the author’s current project bibliography, and focused 2025–2026 searches were checked on 2 October 2026. No later amplitude theorem was located; this bounded search is not a guarantee of universal novelty.

[write, 2 October 2026] The repository searches recorded in `48-dfa-literature-status.md` (at commit `a68b926cf`) were right when made. The same batch delivered the relaxed, compacted and fixed-arity companions, printed as Parts I–III of this report; source 48 cites neither its binary companions nor their shared text.

Several extensions deserve separate investigation.

1. Obtain certified decimal enclosures of γ by effective error bounds or positive interval propagation. Existence alone does not do this.
2. For a normalized exclusion coefficient $(\alpha m + \beta)$, the same formal drift is $\alpha/(4N)$, suggesting power $1 - \alpha/4$. The relaxed, compacted, and present DFA cases have parameters $(0, 0)$, $(1, -1)$, and $(1/2, 0)$, respectively. Nonnegative renewal weights and upper domination can hold on a larger parameter set, but a matching lower bound or another positivity proof is still required. No general parameter theorem is claimed here.
3. Extend the method to the non-nearest-neighbor general-arity recurrences, controlling the non-selfadjoint modes rather than assuming that the binary Jacobi argument automatically applies.

[write, 2 October 2026] For every fixed alphabet size $k \geq 3$ this is done in Part III (source 18), with a fixed weighted norm and a summable delay instead of the binary Jacobi argument, as this item anticipates.

4. Analyze exponentially small corrections beyond the present finite-order Poincaré expansion.

Appendix

A Self-contained Jacobi estimates

[write, 2 October 2026] This appendix is source 15’s (Part II); source 48 ships the same text with “DFA theorem” in its first sentence. “Section 3” of source 15 is Section 18 here. The appendix serves Parts II and IV, and is a second, self-contained proof of the estimates of Part I, Sections 2–5.

This appendix proves all shared gauge, gap, profile, projection and spectral product estimates used in the compacted theorem. Its objects are the explicit matrices in Section 18. The relaxed all-orders theorem is not assumed here.

A.1 Exact symmetrization and norm contraction

For the relaxed recurrence define

$$d_{N,j} = \frac{r_{(N+j)/2, (N-j)/2}}{((N+j)/2)!}$$

on the physical support and zero otherwise. Direct substitution gives

$$d_{0,j} = \mathbf{1}_{j=0}, \quad d_{N,-1} = 0, \quad d_{N,j} = \frac{N-j+2}{N+j} d_{N-1,j-1} + d_{N-1,j+1}. \quad (\text{A.1})$$

For $v_N^{(r)}(j) = d_{N,j}/D_N(j)$ the factorial identities

$$\frac{D_N(j)}{D_N(j-1)} = b_{N,j}, \quad \frac{D_{N-1}(j)}{D_N(j)} = Q_N(j)$$

on their required supports prove $v_N^{(r)} = S_N Q_N v_{N-1}^{(r)}$. All edge and diagonal coefficients lie in $[0, 1]$. The finite Jacobi matrix is irreducible, so its largest eigenvalue has a positive unit eigenvector. The matrix exchanges the two parity subspaces, so its spectrum is symmetric about zero and $\|S_N\| = \lambda_N$. Consequently $\|S_N Q_N\| \leq \lambda_N$ and $\|v_N^{(r)}/P_N\| \leq 1$.

A.2 The coarse Airy limit and its gap

Write $\varepsilon = N^{-1/3}$ and $x_j = (j+1)\varepsilon$. For vectors supported on $0, \dots, N+1$, the exact quadratic form is

$$\begin{aligned} \langle v, (2I - S_N)v \rangle &= \sum_{j=0}^N b_{N,j+1} |v_{j+1} - v_j|^2 + |v_0|^2 \\ &\quad + \sum_{j=1}^N (2 - b_{N,j} - b_{N,j+1}) |v_j|^2 + (2 - b_{N,N+1}) |v_{N+1}|^2. \end{aligned} \quad (\text{A.2})$$

The term $|v_0|^2$ is the first Dirichlet-gradient interval, with fictitious value $v_{-1} = 0$. No nonphysical edge coefficient $b_{N,0}$ is needed. For $1 \leq j \leq N+1$,

$$1 - b_{N,j} = \frac{2(j-1)}{(N+j)(1+b_{N,j})} \geq \frac{j-1}{N+j}.$$

Thus for universal $c, C > 0$,

$$\varepsilon^{-2} \langle v, (2I - S_N)v \rangle \geq c \sum_{j=0}^{N+1} x_j |v_j|^2 - C\varepsilon \|v\|^2. \quad (\text{A.3})$$

On every fixed x window the edge weights tend uniformly to one and

$$\varepsilon^{-2}(2 - b_{N,j} - b_{N,j+1}) \longrightarrow 2x_j$$

uniformly there. Interpolate linearly with value $\varepsilon^{-1/2}v_j$ at x_j and zero at the origin; add a zero node after the last finite node and extend by zero. The terminal term of (A.2) controls the final interpolation interval. Bounded scaled energy gives local H^1 bounds. Equation (A.3) gives uniformly small L^2 tails outside large fixed windows. Local compactness followed by this tail estimate gives strong L^2 compactness; the boundary interval enforces zero trace at zero. The limiting lower form is

$$\int_0^\infty (|f'(x)|^2 + 2x|f(x)|^2) dx.$$

Norm convergence follows on a fixed window from the gradient bound and Riemann sums, then globally from (A.3). Sampling smooth, compactly supported functions vanishing at zero supplies the reverse upper form bound. Applying the min-max principle in both directions proves convergence of every fixed low eigenvalue of $\varepsilon^{-2}(2I - S_N)$ to the corresponding eigenvalue of

$$\mathcal{H} = -\frac{d^2}{dx^2} + 2x, \quad f(0) = 0.$$

These eigenvalues are $\mu_k = -2^{2/3}z_k$, where $z_1 = z > z_2 > \dots$ are the negative Airy zeros. Hence

$$\lambda_N = 2 + 2a\varepsilon^2 + o(\varepsilon^2), \quad \lambda_N - \lambda_{2,N} \sim 2^{2/3}(z - z_2)\varepsilon^2, \quad (\text{A.4})$$

where $\lambda_{2,N}$ is the second-largest positive eigenvalue for large N . Between opposite parity subspaces, the singular values of S_N are its positive eigenvalues, possibly with additional zeros. Removing the top singular direction therefore gives norm at most $\lambda_N(1 - cN^{-2/3})$. The negative eigenvalue $-\lambda_N$ is not a second slow direction within occupied parity.

A.3 A finite quantitative quasimode

The Airy equation is $F'' = 2(x + a)F$. Define

$$\begin{aligned} F_2(x) &= \frac{2(-a + 2x)}{15}F(x) + \frac{x(2a + x)}{15}F'(x), \\ F_3(x) &= \frac{F(x) - xF'(x)}{3}, \\ f_N(x) &= F(x) + \varepsilon^2 F_2(x) + \varepsilon^3 F_3(x), \\ \hat{\lambda}_N &= 2 + 2a\varepsilon^2 + 3\varepsilon^3 + \frac{13a^2}{15}\varepsilon^4 + \frac{5a}{3}\varepsilon^5. \end{aligned} \quad (\text{A.5})$$

Choose a fixed smooth cutoff χ , equal to one on $[0, 1]$ and zero on $[2, \infty)$, and let $\hat{f}_N$ be the ℓ^2 normalization of $\chi(j/N^{1/2})f_N(x_j)$. Changing or omitting the cutoff in local calculations gives an error smaller than every inverse power of N .

For a direct residual check, the two edge coefficients at a sample point are

$$b_-(x, \varepsilon) = \sqrt{\frac{1 - x\varepsilon^2 + 3\varepsilon^3}{1 + x\varepsilon^2 - \varepsilon^3}}, \quad b_+(x, \varepsilon) = \sqrt{\frac{1 - x\varepsilon^2 + 2\varepsilon^3}{1 + x\varepsilon^2}}.$$

Expand

$$b_-f_N(x - \varepsilon) + b_+f_N(x + \varepsilon) - \hat{\lambda}_N f_N(x)$$

through degree five in ε , reducing derivatives by $F'' = 2(x + a)F$. Every coefficient vanishes. The omitted left sample is zero because $F(0) = F_2(0) = F_3(0) = 0$. On the cutoff support, $x\varepsilon^2 = O(N^{-1/2})$, and Taylor remainders are bounded by $C\varepsilon^6$ times a fixed polynomial times Airy decay. Such envelopes are square-integrable. Cutoff derivatives contribute only $\exp(-cN^{1/4})$ times powers of N . The unnormalized sampled norm is asymptotic to $\varepsilon^{-1/2} \|F\|_{L^2}$, so

$$\|(S_N - \hat{\lambda}_N)\hat{f}_N\| = O(\varepsilon^6). \quad (\text{A.6})$$

The coarse gap identifies the nearby eigenvalue as λ_N ; projecting onto the complementary eigenspaces gives

$$\lambda_N = \hat{\lambda}_N + O(\varepsilon^6), \quad \|\psi_N - \hat{f}_N\| = O(\varepsilon^4). \quad (\text{A.7})$$

The sign follows from the positive leading Airy profile.

The normalized samples have ℓ^2 derivative $O(N^{-1})$ with respect to continuous N , and one-site shift difference $O(N^{-1/3})$. They have bounded polynomial moments on the Airy scale. Together with (A.7), these facts give

$$\|\psi_N - \psi_{N-1}\| = O(N^{-1}), \quad \|(I - Q_N)\psi_N\| = O(N^{-4/3}), \quad \psi_N(0) = \sqrt{2}N^{-1/2}(1 + O(N^{-1/3})). \quad (\text{A.8})$$

For the diagonal estimate use $1 - \sqrt{1-t} \leq t$ on $[0, 1]$, the sampled moments, and (A.7). For the endpoint use

$$I = \int_0^\infty F(x)^2 dx = 2^{-1/3} \text{Ai}'(z)^2, \quad \frac{F'(0)}{\sqrt{I}} = \sqrt{2}. \quad (\text{A.9})$$

The integral identity follows by differentiating $u \text{Ai}(u)^2 - \text{Ai}'(u)^2$ and integrating from z to infinity. The derivative $\text{Ai}'(z)$ is positive for the largest zero. A Riemann-sum estimate with integrable derivative gives relative normalization error $O(\varepsilon)$. The endpoint Taylor error is $O(\varepsilon^2)$, and the normalized quasimode error divided by the leading endpoint is $O(\varepsilon^{5/2})$. This proves the endpoint assertion. The needed Airy decay and polynomially weighted integrability are standard consequences of its decaying ODE solution; see [6].

A.4 Projection and adjoint estimates

Writing $S_N\psi_N = \lambda_N\psi_N$ on its two parity components and taking the corresponding scalar products shows that both components have equal squared norm. Thus g_N, h_N in (18.3) are unit vectors and $S_N h_N = \lambda_N g_N$, $S_N g_N = \lambda_N h_N$. The estimates above imply

$$\|h_N - g_{N-1}\| = O(N^{-1}), \quad \|(I - Q_N)h_N\| + \|(I - Q_N)g_{N-1}\| = O(N^{-4/3}). \quad (\text{A.10})$$

For the second term use the explicit sampled approximation at $N - 1$, rather than the weaker $O(N^{-1})$ profile difference. Its Airy moments with Q_N give the same bound, and its norm error is already $O(N^{-4/3})$.

On the preceding parity,

$$\Pi_N S_N = S_N(I - |h_N\rangle\langle h_N|).$$

The singular gap therefore gives, for every such y ,

$$\|\Pi_N M_N y\| \leq (1 - cN^{-2/3}) \|Q_N y\| \leq (1 - cN^{-2/3}) \|y\|.$$

It is unnecessary to assume $y \perp h_N$. Furthermore

$$\|M_N g_{N-1} - g_N\| = O(N^{-1}), \quad M_N^* g_N = Q_N h_N,$$

so (A.10) also gives the adjoint-ground bound. Finally,

$$t_N = \langle Q_N h_N, g_{N-1} \rangle = 1 + O(N^{-4/3}).$$

Indeed the diagonal defect is $O(N^{-4/3})$, and the unit-vector identity $\langle h, g \rangle = 1 - \|h - g\|^2/2$ makes the profile overlap defect $O(N^{-2})$. These prove every projection estimate in Lemma 18.1.

A.5 The positive spectral product constant

From (A.7),

$$\log \lambda_N = \log 2 + aN^{-2/3} + \frac{3}{2}N^{-1} + O(N^{-4/3}).$$

Integral or Euler summation for the first two powers and absolute convergence of the residual series give

$$P_N = \kappa 2^N \exp(3aN^{1/3})N^{3/2}(1 + O(N^{-1/3})), \quad \kappa > 0.$$

For completeness, κ is the positive limit of the quotient in this formula. Each factor λ_N is positive, and the logarithm of the quotient converges to a finite real number. This completes the proof of Lemma 18.1.

References

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- [2] Andrew Elvey Price, Wenjie Fang and Michael Wallner. *Asymptotics of Minimal Deterministic Finite Automata Recognizing a Finite Binary Language*. LIPIcs AofA **159** (2020), 11:1–11:13. doi:10.4230/LIPIcs.AofA.2020.11. Source 48: the recurrence (Proposition 5) and the two-sided estimate (Theorem 1) of Part IV. Cited by source 29 as **binary**.
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